A SHARP DENOMINATOR THEOREM FOR THE CELLULAR RATIONAL APPROXIMATIONS TO $\zeta(5)$

Abstract. Brown and Zudilin analyse a family of cellular integrals on the moduli space $\mathcal{M}_{0,8}$ whose totally symmetric member produces simultaneous rational approximations $I'_n = Q_n\zeta(5) - P_n$ and $I''_n = Q_n\zeta(3) - \hat{P}_n$. On the basis of extensive computation they record the inclusions $Q_n,\ d_n^2 d_{2n}\hat{P}_n,\ d_n^5 P_n \in \mathbb{Z}$, where $d_n = \operatorname{lcm}(1,\dots,n)$. We complete this picture. The inclusions hold after multiplication by the single global factor $12 = 2^2 \cdot 3$ — as their own printed value $P_2 = 1190161/384$ already shows, since $d_2^5 = 32$ and $384 = 12 \cdot 32$ — and the factor 12 is sharp.

Our main theorem is the $p \geq 5$ half of the resulting law: $\operatorname{ord}_p(P_n) \geq -5\lfloor \log_p n \rfloor = -\operatorname{ord}_p(d_n^5)$ for every prime $p \geq 5$ and every $n \geq 1$. The proof is a digit-scaled induction on the number of base-p digits of n, applied to the graded remainder $W_n = P_n - H_n^{(5)} Q_n$, and it rests on exactly two inputs that are not proved here. The first is the finite decomposition identity $P_n = \sum_{k,l} T(n,k,l)\, w_5(n,k,l)$ over the Brown–Zudilin summand $T(n,k,l) = \binom{n+k}{n}\binom{n}{k}^2\binom{n+l}{n}\binom{n}{l}^2\binom{n+k+l}{n}$ of $Q_n = \sum_{k,l} T(n,k,l)$, with w_5 an explicit weight-5 form in harmonic letters; it is verified exactly over $\mathbb{Q}$ for $n \leq 34$ (for the representative used in the base case, $n \leq 20$), agrees with P_n at every $n \leq 360$ and satisfies the Brown–Zudilin operator at 748 consecutive values modulo two 25-bit primes, but is not yet machine-certified. The second is the linear-algebra certificate (DEPTH) cutting out the depth-conditioned family in which w_5 is chosen: an exact rank computation, reproduced at two auxiliary primes with identical pivot sets, but not a human proof. We state both statuses precisely and isolate what a certificate for the first must contain.

Everything else in that chain is proved: an exact Lucas congruence $Q_{ap+r} \equiv Q_a Q_r \pmod{p}$ for the bottom row (formalised in Lean 4 with no axioms beyond `propext`, `Classical.choice`, `Quot.sound`), a graded $H^{(5)}$-layer reduction, a level-independent depth calculus for w_5, an exact fibre block factorisation of T whose only external input is Wolstenholme's congruence, a descent Kummer lemma, and a base case that dissolves at the midpoint $n = (p+1)/2$ into a three-cell corner sum equal to $-14\sum_{j<p} j^{-2} \equiv 0 \pmod{p}$. Those two classical facts — Wolstenholme, and $\sum_{j<p} j^{-2} \equiv 0$ — are the only arithmetic inputs of the chain that constrain the prime (Wilson's theorem is used as well, but holds at every prime and so costs nothing), and they are exactly why $p \geq 5$ is the right hypothesis. We also prove two structural obstruction theorems that explain why several natural routes cannot work; and we prove the weight-3 companion statement for $\hat{P}_n$ modulo a third unproved input, the weight-3 decomposition identity $\hat{P}_n = \sum_{k,l} T\,\hat{w}_3$ (Theorem B, likewise verified but not certified), which the $p \geq 5$ law for P_n does *not* use. The residual factor 12 at $p \in \{2,3\}$ is of a different nature; we record its status honestly, including the fact that the geometric mechanism proposed for it provably cannot produce the factor 3.

Contents

1. Introduction 2
2. Structure of the proof 6
3. The bottom row 7
4. The graded H-layer 9

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5. Carry lemmas for the Brown–Zudilin summand 10
6. The decomposition identities 11
7. The pole calculus and the depth conditions 14
8. The fibre congruence 19
9. The digit-scaled induction 25
10. The descent term off-regime 26
11. The nucleus: the base case 29
12. Two structural theorems 34
13. The middle row 36
14. Certification status of the decomposition identities 38
15. The primes 2 and 3 41
16. Summary of the dependency structure 43
Appendix A. Reproduction 43
Acknowledgments 44
References 44

1. Introduction

1.1. **The Brown–Zudilin family.** In [BZ] Brown and Zudilin study the family of five-fold cellular integrals

$$I(\boldsymbol{a}) = I(a_1, \dots, a_8)$$

of [Br16, Example 7.5], period integrals on the moduli space $\mathcal{M}_{0,8}$ of genus zero curves with eight marked points. General results on periods of these moduli spaces [Br09] force $I(\boldsymbol{a})$ to be a $\mathbb{Q}$-linear combination of multiple zeta values of weight ≤ 5; the cellular nature of the integral kills the subleading weight and pins the leading weight to the single period $\zeta(5) + 2\zeta(3)\zeta(2)$, and the coefficient of $\zeta(3)$ vanishes identically. The upshot [BZ, (deco)] is a decomposition

$$I = Q \cdot \big(2\zeta(5) + 4\zeta(3)\zeta(2)\big) - 4\hat{P} \cdot \zeta(2) - 2P, \qquad Q \in \mathbb{Z},\ P, \hat{P} \in \mathbb{Q},$$

and, after "setting $\zeta(2)$ to zero", a *pair* of simultaneous rational approximations

$$I'(\boldsymbol{a}) = Q\zeta(5) - P, \qquad I''(\boldsymbol{a}) = Q\zeta(3) - \hat{P}.$$

A large transformation group $\mathfrak{G}$ of order 7! acts on the normalised integrals and on $Q, P, \hat{P}$, and it is by exploiting that action that Brown and Zudilin compute a sharp denominator; such Rhin–Viola groups are a recurring and arithmetically decisive feature of rational approximations to $\zeta(2)$, $\zeta(3)$ and $\zeta(4)$ [RV96, RV01, MZ20], as they are of the hypergeometric constructions for odd zeta values generally [Zu04, Zu02]. The whole subject descends from Apéry [Ap79] and Beukers' re-reading of it [Be79].

We work throughout with the *totally symmetric* member $a_1 = \cdots = a_8 = n$ of the family, and write Q_n, P_n, $\hat{P}_n$ for the resulting coefficients, normalised as in [BZ]:

$$Q_0 = 1, \quad Q_1 = 21, \quad Q_2 = 2989; \qquad P_1 = \tfrac{87}{4}, \quad P_2 = \tfrac{1190161}{384}; \qquad \hat{P}_1 = \tfrac{101}{4}, \quad \hat{P}_2 = \tfrac{344923}{96}. \tag{1.1}$$

All three sequences satisfy one and the same Apéry-type recursion of order 3 [Zu02b], which we call L_{BZ}:

$$c_0(\nu)Y_\nu + c_1(\nu)Y_{\nu+1} + c_2(\nu)Y_{\nu+2} + c_3(\nu)Y_{\nu+3} = 0, \tag{1.2}$$

$$\begin{aligned} c_0(\nu) &= (\nu+1)^5(\nu+2)\,a_0(\nu+1), & c_1(\nu) &= -2(\nu+2)B_8(\nu), \\ c_2(\nu) &= -2B_9(\nu), & c_3(\nu) &= 2(\nu+3)^5(2\nu+5)\,a_0(\nu), \end{aligned} \tag{1.3}$$

with $a_0(x) = 41218x^3 + 198849x^2 + 320790x + 173057$ and B_8, B_9 explicit polynomials making all four c_i of degree 9.

The bottom row has a closed form [BZ, (Q_n)]: with

$$T(n,k,l) := \binom{n+k}{n}\binom{n}{k}^2\binom{n+l}{n}\binom{n}{l}^2\binom{n+k+l}{n}, \tag{1.4}$$

one has

$$Q_n = \sum_{k,l=0}^{n} T(n,k,l) \in \mathbb{Z}. \tag{1.5}$$

We refer to T as the *Brown–Zudilin summand*, and to the triple $(Q_n, \hat{P}_n, P_n)$ as the *bottom*, *middle* and *top row* of the ladder.

1.2. **The denominator law, and the factor** 12.

On the basis of extensive computation, Brown and Zudilin record [BZ, (dn-totsym)]

$$Q_n, \qquad d_n^2 d_{2n}\hat{P}_n, \qquad d_n^5 P_n \;\in\; \mathbb{Z} \qquad \text{for } n = 0, 1, 2, \ldots, \tag{1.6}$$

where $d_n = \operatorname{lcm}(1, 2, \ldots, n)$; the general-parameter analogues are their inclusions [BZ, (incl), (sharp_incl)], which they describe as an experimental observation that "it is in principle possible to prove, with considerable effort".

The purpose of this paper is to complete (1.6). The inclusions hold after multiplication by one global constant, and that constant is $12 = 2^2 \cdot 3$. This is already visible in [BZ]'s own printed value: $P_2 = 1190161/384$ and $d_2^5 = 2^5 = 32$, so $d_2^5 P_2 = 1190161/12 \notin \mathbb{Z}$, while $12\, d_2^5 P_2 \in \mathbb{Z}$. The corrected law is sharp in both directions.

Evidence 1.1 (the empirical law). [VERIFIED] Over the 361 exact ladder values $n = 0, \ldots, 360$, the number of n with $\operatorname{den}(P_n) \mid c\, d_n^5$ is

c	1	2	3	4	6	12	24
$\#\{n \le 360\}$	1	1	1	330	1	361	361

So the constant is exactly 12: it suffices, and neither of its maximal proper divisors 6 and 4 does. The two halves fail very differently. Dropping the 2^2 to 6 (hence also 3, 2, 1) fails at all 360 values $n \ge 1$; dropping the 3 to 4 fails at only 31 values, the first being $n = 2, 6, 8, 18, 20, 24, 26, 54$ — so the factor 3 is real but rare, which is consistent with its provenance being the least understood part of the constant (§15). Moreover $\log \operatorname{den}(P_n)/\log(d_n^5) \to 1$ (values $0.9953, 0.9928, 0.9947, 0.9936$ at $n = 80, 200, 320, 355$), so the exponent 5 is itself tight.

Remark 1.2. We stress the framing. Brown and Zudilin state (1.6) explicitly as an experimental observation, not as a theorem, and their paper is careful to flag which of its assertions are proved and which are computational. What we do here is supply the missing constant and prove the arithmetic half of the resulting statement; we regard this as completing their analysis rather than correcting it. The 2-adic and 3-adic content of the constant 12 is, as we explain in §15, of a genuinely different flavour from the $p \ge 5$ part, and Brown and Zudilin's own heuristic — that the construction obtains I' by motivically setting $\zeta(2) := 0$, a row operation against the period $(2\pi i)^2$ whose integral normalisation carries a Bernoulli denominator — is precisely the mechanism one would expect to leave a universal $2^2 \cdot 3$ behind.

1.3. **The main theorem.** Throughout, p denotes a prime, v_p (equivalently ord_p) the p-adic valuation on $\mathbb{Q}$, and

$$L = L(n) := \lfloor \log_p n \rfloor = v_p(d_n), \qquad H_N^{(m)} := \sum_{i=1}^{N} i^{-m}.$$

Theorem 1.3 (Sharp-12, $p \geq 5$ part)**.** *Assume the decomposition identity* (T1-top) *of §6 and the linear-algebra certificate* (DEPTH) *of §7. Then for every prime $p \geq 5$ and every $n \geq 1$,*

$$\mathrm{ord}_p(P_n) \;\geq\; -5\lfloor \log_p n \rfloor \;=\; -\,\mathrm{ord}_p(d_n^5).$$

Equivalently $d_n^5 P_n \in \mathbb{Z}[1/6]$, which is the $p \geq 5$ half of $\mathrm{den}(P_n) \mid 12\, d_n^5$.

The two hypotheses are both finite, and isolated as sharply as we know how:

- **(T1-top).** $P_n = \sum_{k,l=0}^{n} T(n,k,l)\, w_5(n,k,l)$, where w_5 is an explicit $\mathbb{Q}$-combination of weight-5 monomials in the harmonic letters A_m, B_m, C_m, N_m of §6. Evidence class [VERIFIED].
- **(DEPTH).** The linear system of §7 cutting out the depth-conditioned family in which w_5 is chosen is consistent, with the ranks recorded in Theorem 7.8. Evidence class [CERT]: an exact rank computation reproduced at two auxiliary primes with identical pivot sets. It is what Proposition 7.12 and hence the depth bound (DEPTH-gen) rest on, and it is inherited by every statement below that quotes a depth bound.

Neither is [PROVED]: see §14 for exactly what has been checked for (T1-top), what a creative-telescoping certificate would have to contain, and a structural obstruction (Theorem 14.3) which shows why the cheap route that works at weight 3 has no weight-5 analogue. Section 14 is written so that a single sentence flips when that certificate lands. We stress that the weight-3 decomposition identity (Theorem 6.1) is *not* among the hypotheses of Theorem 1.3: nothing in the P_n chain of §§3–11 consumes $\hat{w}_3$. Theorem 6.1 is a third unproved input of this paper, used only for the middle row (Theorem 1.7).

Everything else in the chain is proved. The statements below record the main intermediate results together with the hypotheses each one carries; of them only Theorem 1.4 is unconditional.

Theorem 1.4 (Bottom row)**.** *For every prime p, every $a \geq 0$ and every $0 \leq r < p$,*

$$Q_{ap+r} \equiv Q_a\, Q_r \pmod{p},$$

hence $Q_n \equiv \prod_i Q_{n_i} \pmod{p}$ over the base-p digits of n. Moreover

$$Q_{ap+r} \equiv Q_a\big(Q_r + p\, a\, \Psi_a\big) \pmod{p^2} \qquad (p \geq 5,\ 1 \leq a < p,\ p \nmid Q_a)$$

with Ψ_a the explicit level-r weight-1 harmonic functional of Corollary 8.9.

Theorem 1.5 (Base case)**.** *Assume* (T1-top) *for the representative w_5^{I} of §6 and the linear-algebra certificate* (DEPTH)*. Then $\mathrm{ord}_p(P_n) \geq 0$ for every prime $p \geq 5$ and every $n < p$.*

Theorem 1.6 (One radical short, unconditionally in n)**.** *Assume* (T1-top) *and* (DEPTH)*. For every prime $p \geq 5$ and every $n \geq 1$,*

$$\mathrm{ord}_p(P_n) \;\geq\; -5\lfloor \log_p n \rfloor - 1,$$

i.e. $d_n^5 \cdot \prod_{5 \leq p \leq 3n} p \cdot P_n \in \mathbb{Z}[1/6]$; and $\mathrm{ord}_p(P_n) \geq 0$ outright whenever $p > 3n$.

Theorem 1.6 does not use the base case or the off-regime descent, and it is recorded separately because it is what the depth calculus alone buys.

Theorem 1.7 (Middle row)**.** *Assume the weight-*3 *decomposition identity* (Theorem B) *of §6. Let $p \geq 5$, $1 \leq a < p$, $0 \leq r < p$, $n = ap + r$ and $p \nmid Q_a$. Then*

$$v_p\big(p^3 \hat{P}_n - \hat{P}_a Q_r\big) \;\geq\; 1 + \min\big(0, v_p(\hat{P}_a)\big).$$

In particular $p^3 \hat{P}_{ap+r} \equiv \hat{P}_a Q_r \pmod p$ if and only if $\hat{P}_a \in \mathbb{Z}_p$.

Theorem 1.7 corrects a natural expectation: the naïve "Frobenius $\mathrm{diag}(1, p^3, p^5)$" product congruence is *false* in the middle slot, and §13 explains exactly why — the letter $A_3(k)$ has upper argument $n + k$ reaching $2n$, which is the source of the d_{2n} in (1.6).

Finally, two obstruction theorems delimit the landscape; they are stated in §12 and are, we think, of independent interest.

Theorem 1.8 (The θ wall, informally)**.** *The entire content of the base case is a linear functional $\theta_{p,n}$ on the space of depth-conditioned weight-*5 *forms which* factors through the value map $w \mapsto \sum_{k,l} T(n,k,l)w(n,k,l)$. *Consequently no exact summation identity $\sum_{k,l} T \cdot M = 0$ — no member of the "Lemma Φ species", including the new weight-*2 *identities of Lemma 12.9 — can ever move θ; and $\theta \not\equiv 0$ on that space. The depth calculus plus arbitrarily much summand-side identity work therefore cannot prove the base case.*

1.4. **What is new.** The proof of Theorem 1.3 is a digit-scaled induction whose four ingredients are, we believe, all new:

(1) a *level-lifting* proposition (Proposition 7.12) showing that a finite list of 42 linear conditions on the coefficients of w_5, computed at a single base-p digit, controls the p-adic depth of w_5 at *every* digit level simultaneously and at every prime $p \geq 5$ at once;
(2) an exact block factorisation of T over a base-p fibre (Lemma 8.2) together with an exact residue identity (Lemma 8.4) that annihilates the entire first-order dependence of the fibre sum on the fibre coordinates, giving the sharp fibre congruence Lemma 8.8 and its multi-digit weakening Lemma 8.13; the only external input is Wolstenholme's congruence, used once;
(3) a descent Kummer lemma (Lemma 10.2) whose one-line mechanism is that *off-regime means a carry in base-p position* 0*, the depth-pattern indicators live in position $L \geq 1$, and Kummer's theorem counts both*;
(4) a base case that is not a cancellation argument at all above or below the midpoint, and at the midpoint reduces to three cells whose residues sum to $-14 \sum_{j<p} j^{-2}$ (Theorem 11.7).

Along the way we obtain: a mod p^2 supercongruence for the bottom row (Corollary 8.9); an explicit order-4 left multiple $\tilde{L}$ of L_{BZ} whose leading coefficient is $2D(\nu+3)^5(2\nu+5)$, i.e. with the factor a_0 removed (§11.4); a universal normalised midpoint row $(11907, -334374, -19292)$ (§11.5); new exact weight-2 residue identities on the Brown–Zudilin summand (Lemma 12.9); and two new harmonic closed forms for Apéry's $\zeta(3)$ numerators b_n (§6.4), one of which is cell-wise p-integral.

1.5. **Conventions on evidence.** Because part of the chain is machine-assisted, we are explicit about evidence classes and use them uniformly.

- [PROVED] — a complete human proof is given here or in a named predecessor.
- [CERTIFIED] — a machine proof object (a creative-telescoping operator together with its certificate) has been produced *and* independently re-checked by exact rational-function arithmetic in a session that does not trust the search.
- [CERT] — a linear-algebra certificate: an exact rank computation reproduced at two or three auxiliary primes with identical pivot sets.

- [VERIFIED] r — an exact finite check on the stated range r, 0 failures. This is evidence, never proof. Ranges are always given.
- [LEAN] — formalised in Lean 4 with 0 `sorry`s and no axioms beyond the three standard ones (`propext`, `Classical.choice` and `Quot.sound`).
- [OPEN] — not settled; stated precisely.

All numerical checks quoted below are exact (integer, `Fraction`, $\mathbb{F}_q$, or tracked-precision p-adic arithmetic); no floating point enters any arithmetic statement. Appendix A lists the scripts.

2. Structure of the proof

This section is a narrative map. Nothing is proved here; every statement is proved in the section named.

2.1. **The shape of the object.** The three rows of the ladder sit in a graded tower. The bottom row is integral and obeys an exact Lucas congruence (§3). The upper rows are not integral, and their denominators are exactly what we must control. The first move (§4) is to strip off the part of P_n that is visibly responsible for the pole: writing

$$W_n := P_n - H_n^{(5)} Q_n,$$

the "H-layer" $H_n^{(5)} Q_n$ contributes $-5L$ for free because $Q_n \in \mathbb{Z}$, and all remaining content sits in W_n. The second move is to give W_n a summand-level shape:

$$W_n = \sum_{k,l} T(n,k,l)\, v_5(n,k,l), \qquad v_5 := w_5 - H_n^{(5)},$$

which is exactly the decomposition identity (T1-top). Everything downstream is a *cell-by-cell ledger*: a comparison, at each summand, between the p-adic *depth* of v_5 (how negative its valuation can be) and the Kummer *carries* of T (how many powers of p the summand supplies).

2.2. **The three walls, and what they forced.** The route to Theorem 1.3 was not the expected one, and three hard negatives shaped it. We state them because each is a theorem, and each closes off a natural approach.

(W1) **Lemma F is sharp** (§8). The fibre congruence delivers relative precision $p^{2+\min(v_p T, 2)}$ and that value is *attained*; so no strengthening of the fibre analysis can pay for the weight-5 depth. The fix had to come from the w_5 side, and it is a *linear* fix: the depth conditions of §7.

(W2) **No p-independent representative makes the base case cell-wise** (§7, Theorem 7.16). The strengthened depth systems (DEPTH^{+}) and (DEPTH^{++}) are *inconsistent* with the fitting system, at two auxiliary primes, and the inconsistency localises to exactly the three pole patterns carrying the deficit. Enlarging the alphabet by Apéry-type letters $R^{(a)}$ or by genuinely nested depth-2 letters Y_{ab}, V_{ab} does not help (Theorem 7.17).

(W3) **The θ wall** (§12, Theorem 12.5). The obstruction functional factors through the value map, so no summand-side identity can move it. This *explains* (W2) rather than merely recording it: those linear systems were trying to buy, with p-independent linear conditions, something that is not p-independent-linear.

The walls all point the same way: the base case is a statement about the *number* P_n, so a proof must import a property of P_n. There are exactly two such properties available: the summand decomposition, and the recurrence L_{BZ}. The proof in §11 uses both.

2.3. **The nucleus dissolves at the midpoint.** The base case is $\mathrm{ord}_p(P_n) \geq 0$ for $n < p$. The cell-wise bound the depth calculus supplies is -1, and the deficit is *attained*, at $(n,k,l) = \left(\frac{p+1}{2}, 0, \frac{p-1}{2}\right)$ for every p. So a genuine cancellation is needed. The proof splits the range at the midpoint and finds that the cancellation is a three-term identity:

- For $n \leq (p-1)/2$ the region $\mathrm{III} = \{k, l \geq q := p-n,\ p \leq k+l < p+q\}$ where the chosen representative w_5^{I} can fail to be cell-wise integral is *empty*: it requires $2n \geq p$. Half the base case is free.
- At $n = (p+1)/2$ the region collapses to the *three corner cells* $(n-1,n)$, $(n,n-1)$, (n,n). There $T/p^2 \equiv (2,2,24)$ by Wilson's theorem, Lucas' theorem and Kummer's theorem, and the relevant coefficients of the pole expansion are $K_3 = (3 - s_2/2,\ 3 - s_2/2,\ -1/2 - s_2/2)$ where $s_2 = \sum_{j<p} j^{-2}$; the corner sum is
$$2\Big(3 - \tfrac{s_2}{2}\Big) + 2\Big(3 - \tfrac{s_2}{2}\Big) + 24\Big(-\tfrac{1}{2} - \tfrac{s_2}{2}\Big) \;=\; -14\, s_2 \;\equiv\; 0 \pmod{p},$$
because $\sum_{j<p} j^{-2} \equiv \sum_{j<p} j^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0$ for $p \geq 5$. *One classical input; no Fermat quotient, no Bernoulli number.*
- Above the midpoint the forward L_{BZ} induction has only one kind of exceptional step, the roots of a_0 modulo p, and those are *apparent singularities*: all three 2×2 minors of a certain 2×3 coefficient matrix are divisible by $a_0(\nu)$ in $\mathbb{Q}[\nu]$. Equivalently there is an explicit order-4 left multiple $\tilde{L}$ of L_{BZ} with a_0 removed from the leading coefficient (§11.4).

The two halves dovetail with no residue: *on the desingularised ladder the only singular step below* $p-3$ *is exactly the midpoint, which is exactly the single level the region computation supplies.* A congruence (REC-$\star$) that had been expected to be the *input* to the base case comes out as its *output*, and its universal normalised row $(11907, -334374, -19292)$ turns out to be the same row that governs the $a=1$ midpoint band one digit up in the prior campaign of this program [P-M].

2.4. **The induction.** With the base case in hand, the induction (§9) is a per-digit ledger. Writing $n = ap + r$ and grouping the cells of W_n by $(b,c) = (\lfloor k/p \rfloor, \lfloor l/p \rfloor)$,
$$p^5 W_n - Q_r W_a = \underbrace{\sum_{k,l} T(n,k,l)\, \mathcal{E}(n,k,l)}_{\text{(I) descent}} + \underbrace{\sum_{b,c} v_5(a,b,c)\big[\mathcal{T}(b,c) - Q_r T(a,b,c)\big]}_{\text{(II) fibre}},$$
$\mathcal{E} = p^5 v_5(n,k,l) - v_5(a,b,c)$ and $\mathcal{T}(b,c) = \sum_{s,t} T(n, bp+s, cp+t)$. Term (II) is paid for by the depth bound (DEPTH-gen) against the fibre congruence Lemma 8.13, with slack *exactly zero*. Term (I) splits by regime: in-regime the letter descent is exact and costs nothing; off-regime it is Lemma 10.2. Every line of the ledger is tight; there is no spare power of p anywhere in the chain.

Two remarks on that. First, the route the off-regime term was expected to take — a letter-wise carry ledger — is *refuted* (§10.3): at weight 5 the mismatch of the letter $A_m(k)$ costs $m\lambda$ against a total Kummer gain of $1+\lambda$, and at $p=5$, $n=19$, $(k,l) = (6,0)$ the letter-wise ledger is short by exactly one power. The pole is not in $\mathcal{E}$ at all; it is annihilated inside v_5 by the depth conditions at level n. Second, the fact that the two sharpnesses — Lemma F's and (DEPTH)'s — meet exactly is not a coincidence of bookkeeping but the reason the theorem is true at the stated exponent.

3. The bottom row

3.1. **The closed form is certified.**

Theorem 3.1. *[CERTIFIED] The double sum $\sum_{k,l} T(n,k,l)$ is annihilated by L_{BZ}; together with the natural boundaries and agreement at $n = 0, 1, 2$ this proves (1.5).*

The certificate was produced by two-step creative telescoping [Ze91, WZ92] using the package of [Ko10]. The single call `CreativeTelescoping`[ann, $\{S_k - 1, S_l - 1\}, \{S_n\}$] returns `$Failed` even here, where a telescoper provably exists; the two-step split is the fix:

$$\mathrm{ann} = \texttt{Annihilator}[T, \{S_n, S_k, S_l\}] \rightsquigarrow \mathrm{ct}_1 = \texttt{CreativeTelescoping}[\mathrm{ann}, S_k - 1, \{S_n, S_l\}] \\ \rightsquigarrow \texttt{OreGroebnerBasis} \rightsquigarrow \mathrm{ct}_2.$$

The resulting order-3 telescoper has coefficients *identically* those of (1.3): the coefficient ratio is $\{1, 1, 1, 1\}$ and `Expand`[coeffs $- L_{\mathrm{BZ}}$] $= \{0, 0, 0, 0\}$. Every check was redone by exact rational-function arithmetic on the explicit binomial expression for T — each annihilator generator satisfies `Together`[$(L \cdot T)/T$] $= 0$, and each k-step pair satisfies $\mathrm{tel} \cdot T + (S_k - 1)(\mathrm{cert} \cdot T) = 0$. Separately, a *single*-certificate (WZ-pair) form

$$L_{\mathrm{BZ}} \cdot T = \Delta_k(\rho T) + \Delta_l(\sigma T)$$

with explicit rational ρ, σ has been produced and checked to exactly 0 twice, the second time in a kernel that never loaded the package at all; this is the object every weight-lowering step of §14 consumes.

3.2. **Lucas for the bottom row.** The following is the "T-level" input used everywhere below. Its proof is uniform in p (no prime is excluded) and, importantly, uniform in a — it is already multi-digit.

Lemma 3.2 (Lucas step)**.** *For $a, b \geq 0$ and $0 \leq r, s < p$: $\binom{ap+r}{bp+s} \equiv \binom{a}{b}\binom{r}{s} \pmod{p}$.*

Proof. Lucas' theorem applied twice: first to the last digit s against r, then to the reassembled higher digits. □

Lemma 3.3 (Single carry annihilation)**.** *With $N = ap + r$, $k = bp + s$: if $r + s < p$ then $\binom{N+k}{N} \equiv \binom{a+b}{a}\binom{r+s}{r}$, and if $r + s \geq p$ then $\binom{N+k}{N} \equiv 0 \pmod{p}$.*

Proof. If $r + s < p$ then $N + k = (a + b)p + (r + s)$ and Lemma 3.2 applies. If $r + s \geq p$ then $N + k = (a + b + 1)p + (r + s - p)$ with $0 \leq r + s - p < p$, and Lemma 3.2 gives $\binom{N+k}{N} \equiv \binom{a+b+1}{a}\binom{r+s-p}{r}$; but $s < p$ forces $r + s - p < r$, so $\binom{r+s-p}{r} = 0$. □

Lemma 3.4 (Double carry annihilation)**.** *Assume $s \leq r$, $t \leq r$, $r + s < p$, $r + t < p$. Then $s + t < p$, and $\binom{N+k+l}{N} \equiv \binom{a+b+c}{a}\binom{r+s+t}{r} \pmod{p}$ if $r + s + t < p$, while $\binom{N+k+l}{N} \equiv 0 \pmod{p}$ otherwise.*

Proof. First, $s + t < p$. Suppose $s + t \geq p$. From $s \leq r$ and $t \leq r$ we get $p \leq s + t \leq 2r$, so $r \geq p/2$; from $r + s < p$ and $r + t < p$ we get $s, t \leq p - 1 - r$, so $s + t \leq 2(p - 1 - r) < 2(p - p/2) = p$, a contradiction. Hence $s + t < p$, so $r + s + t < 2p$ and $N + k + l = (a + b + c + \varepsilon)p + (r + s + t - \varepsilon p)$ with $\varepsilon \in \{0, 1\}$. If $\varepsilon = 0$, Lemma 3.2 gives the stated product. If $\varepsilon = 1$, the low digit is $\rho = r + s + t - p$, and $\rho < r$ precisely because $s + t < p$; Lemma 3.2 then produces a factor $\binom{\rho}{r} = 0$. □

The deduction "$s + t < p$" in Lemma 3.4 is the non-obvious step, and it is what upgrades a previously verified "carry-kill predicate" to a proof.

Lemma 3.5 (Summand factorisation)**.** *With $N = ap + r$, $k = bp + s$, $l = cp + t$ as above,*

$$T(N, bp + s, cp + t) \equiv [\![r + s + t < p]\!] \cdot T_{\mathrm{hi}}(a, b, c)\, T_{\mathrm{lo}}(r, s, t) \pmod{p},$$

$$T_{\mathrm{hi}}(a, b, c) = \binom{a+b}{a}\binom{a}{b}^2\binom{a+c}{a}\binom{a}{c}^2\binom{a+b+c}{a}, \qquad T_{\mathrm{lo}}(r, s, t) = \binom{r+s}{r}\binom{r}{s}^2\binom{r+t}{r}\binom{r}{t}^2\binom{r+s+t}{r}.$$

Proof. $\binom{N}{k}^2 \equiv \binom{a}{b}^2\binom{r}{s}^2$ and $\binom{N}{l}^2 \equiv \binom{a}{c}^2\binom{r}{t}^2$ by Lemma 3.2; these vanish unless $s \le r$ and $t \le r$. The factors $\binom{N+k}{N}$, $\binom{N+l}{N}$ are handled by Lemma 3.3 (they vanish unless $r+s<p$, $r+t<p$). On the surviving regime Lemma 3.4 applies to $\binom{N+k+l}{N}$. Multiplying the five congruences, and noting that when any factor is $\equiv 0$ both sides vanish, gives the statement. □

Proof of Theorem 1.4, Lucas part. Write every summation index in base p: $k = bp+s$, $l = cp+t$ with $0 \le s,t < p$; since $k,l \le N < (a+1)p$ we have $0 \le b,c \le a$. Summing Lemma 3.5 over $0 \le b,c \le a$ and $0 \le s,t < p$,

$$Q_N \equiv \sum_{b,c=0}^{a} \sum_{s,t:\, r+s+t<p} T_{\mathrm{hi}}(a,b,c)\, T_{\mathrm{lo}}(r,s,t) \pmod p.$$

Two boundary points. First, the true index region is $\{(b,s) : bp+s \le N\} \times \{(c,t) : cp+t \le N\}$ intersected with the surviving set; completing to the full box adds only terms with $b = a$, $s > r$ (for which $\binom{r}{s} = 0$), and identically for (c,t), so the completed sum is congruent to the true one. Second, the (b,c)-region and the (s,t)-region are independent, since $r+s+t < p$ constrains only s,t; so the double sum factors. The high factor is $\sum_{b,c=0}^{a} T_{\mathrm{hi}}(a,b,c) = Q_a$, an *integer identity* (it is (1.5) at $n = a$). For the low factor, $Q_r = \sum_{s,t=0}^{r} T_{\mathrm{lo}}(r,s,t)$, and every term with $r+s+t \ge p$ vanishes mod p: if $r+s \ge p$ or $r+t \ge p$ this is the computation of Lemma 3.3; otherwise Lemma 3.4 applies and $\binom{r+s+t}{r} \equiv 0$. Hence $\sum_{s,t:\, r+s+t<p} T_{\mathrm{lo}} \equiv Q_r$ and $Q_N \equiv Q_a Q_r$. The multi-digit product form follows by induction on the number of base-p digits, since the two-digit statement holds for *all* $a \ge 0$. □

Evidence 3.6. [PROVED], and [LEAN]: the statement $Q_{ap+r} \equiv Q_a Q_r \pmod p$ and its full digit-product form are formalised in Lean 4 with 0 `sorry`s, no `native_decide`, and axioms exactly {`propext`, `Classical.choice`, `Quot.sound`}. The same development formalises the Apéry-row analogue $a_{ap+r} \equiv a_a a_r$, an abstract χ-twisted weight-w Frobenius descent theorem, and its minimal-Apéry ($p^3 b_{ap+r} \equiv b_a a_r$) and Franel ($p^2 B_{ap+r} \equiv B_a A_r$) instances. In the minimal-Apéry instance the formalised object is $b_{\min}(n) = \sum_k \binom{n}{k}^2\binom{n+k}{k}^2(2H_n^{(3)} - H_k^{(3)})$; the identity $b_{\min} = b$ is proved on paper but is not formalised, and likewise for the Franel analogue. The mod p^2 supercongruence of Corollary 8.9 is not formalised either. Independently, [VERIFIED] 0 failures for $p \in \{5,\dots,31\}$, all single-digit cells and iterated digits, $n \le 360$.

4. The graded H-layer

Lemma 4.1 (Digit split of a harmonic sum)**.** *For $x \ge 0$ and $m \ge 1$, $H_x^{(m)} = U_m(x) + p^{-m} H_{\lfloor x/p\rfloor}^{(m)}$ with $U_m(x) := \sum_{j\le x,\ p\nmid j} j^{-m} \in \mathbb{Z}_p$.*

Proof. Split the summation range according to whether $p \mid j$. □

Theorem 4.2 (H-layer reduction)**.** *Let $p \ge 5$, $1 \le a < p$, $0 \le r < p$, $n = ap + r$ (so $n < p^2$). Put $W_n := P_n - H_n^{(5)} Q_n$ and $\hat W_n := \hat P_n - H_n^{(3)} Q_n$. Then*

$$p^5 P_n - P_a Q_r \equiv p^5 W_n - W_a Q_r \pmod p, \qquad p^3 \hat P_n - \hat P_a Q_r \equiv p^3 \hat W_n - \hat W_a Q_r \pmod p,$$

each congruence being the assertion that the difference of its two sides lies in $p\mathbb{Z}_p$.

Remark 4.3. The difference of the two sides is $p^5 H_n^{(5)} Q_n - H_a^{(5)} Q_a Q_r$ (respectively $p^3 H_n^{(3)} Q_n - H_a^{(3)} Q_a Q_r$), and both of *those* terms are p-integral for free, since $v_p(H_N^{(m)}) \ge -m\lfloor \log_p N\rfloor$ gives $p^5 H_n^{(5)}$, $p^3 H_n^{(3)} \in \mathbb{Z}_p$ for $n < p^2$, while $H_a^{(m)} \in \mathbb{Z}_p$ for $a < p$ and $Q_n, Q_a, Q_r \in \mathbb{Z}$. That is what the proof below establishes, and it is unconditional. The p-integrality of the individual sides is *not* available here: $\mathrm{ord}_p(P_a) \ge 0$ for $a < p$ is the base case (Theorem 1.5), $\mathrm{ord}_p(p^5 P_n) \ge 0$ for $n < p^2$ is Theorem 1.3 at $L = 1$, and the middle-row analogues are Proposition 13.3 and Theorem 1.7 — all of them later, and all of them conditional. Nothing below needs them.

Proof. For $n = ap + r < p^2$ the indices $m \le n$ divisible by p are exactly $m = jp$ with $1 \le j \le \lfloor n/p \rfloor = a$, and $p \nmid j$ since $j \le a < p$. Hence $H_n^{(5)} = S + p^{-5} H_a^{(5)}$ with $S := \sum_{m \le n,\ p \nmid m} m^{-5} \in \mathbb{Z}_p$, so $p^5 H_n^{(5)} = p^5 S + H_a^{(5)} \equiv H_a^{(5)} \pmod{p^5}$. Since $Q_n \in \mathbb{Z}$, we get $p^5 H_n^{(5)} Q_n \equiv H_a^{(5)} Q_n \pmod{p^5}$; reducing mod p and using Theorem 1.4 together with $H_a^{(5)} \in \mathbb{Z}_p$ (as $a < p$) gives $p^5 H_n^{(5)} Q_n \equiv H_a^{(5)} Q_a Q_r \pmod{p}$. On the other side $P_a Q_r = H_a^{(5)} Q_a Q_r + W_a Q_r$. Subtract. The weight-3 statement is identical with $5 \mapsto 3$. □

Theorem 4.2 is unconditional and weight-agnostic: it needs nothing but Theorem 1.4 and a two-line valuation computation, and it works verbatim at every weight. Its consequence is that the entire "p^5 pole" of the one-digit product congruence

$$(\mathrm{LB}_5) \qquad p^5 P_{ap+r} \equiv P_a\, Q_r \pmod{p} \qquad (p \ge 5,\ 1 \le a < p,\ 0 \le r < p)$$

is free, and (LB_5) is *equivalent* to the same statement for W:

$$(\mathrm{W5}) \qquad p^5 W_{ap+r} \equiv W_a\, Q_r \pmod{p}.$$

Evidence 4.4. [VERIFIED] 0 failures: (W5) holds for every $p \in \{5, 7, 11, 13, 17, 19, 23, 29, 31\}$ and every single-digit cell $n = ap + r \le 360$, with $\min v_p(p^5 W_n - W_a Q_r) = 1$ exactly at every one of those primes — so (W5) is sharp and carries the full content. Also $\min_{n<p^2} v_p(W_n) = -5$ at every one of those primes. The prime $p = 5$ is not exceptional.

Remark 4.5. The induction of §9 does *not* use (LB_5). As assembled, the last step of (LB_5) needs $v_p((\Lambda - Q_r)W_a) \ge 1$, i.e. it needs the base case; the inequality-shaped induction on $v_p(W_n)$ avoids this entirely and needs only Lemma 8.13, which is unconditional. We record (LB_5) because it is the sharp statement at $n < p^2$ and because the middle-row theorem is of that shape.

5. Carry lemmas for the Brown–Zudilin summand

In the $\zeta(3)$ template the summand $\binom{n}{k}^2\binom{n+k}{k}^2$ is a perfect square, so a single Kummer carry already gives $v_p \ge 2$. The Brown–Zudilin summand (1.4) has *three unsquared* binomials, so a single carry gives only $v_p \ge 1$. The replacement is that one digit overflow forces carries in two *different* unsquared binomials.

Lemma 5.1 (Triple-carry slack). *Let p be prime, $N = ap + r$ with $0 \le a, r < p$, and $k = bp + s$, $l = cp + t$ with $0 \le s, t < p$ and $0 \le b, c \le a$ (automatic when $k, l \le N$). If $a + b \ge p$ then $v_p(T(N, k, l)) \ge 2$.*

Proof. By Kummer's theorem $v_p\binom{N+k}{N}$ is the number of carries in the base-p addition $N + k$. The position-0 carry is $\varepsilon_0 = [\![r + s \ge p]\!]$; the position-1 sum is $a + b + \varepsilon_0 \ge a + b \ge p$, so there is a carry at position 1. Hence $v_p\binom{N+k}{N} \ge 1$, with ≥ 2 if $r + s \ge p$. So assume, else we are done: (i) $r + s < p$; (ii) $s \le r$ and $t \le r$ — otherwise $s > r$ (resp. $t > r$) forces a borrow in $N - k$ (resp. $N - l$), so $v_p\binom{N}{k} \ge 1$ and the *squared* factor already gives $v_p \ge 2$; (iii) $a + c < p$ — otherwise the position-1 argument applied to $\binom{N+l}{N}$ gives another power; (iv) $r + t < p$ — otherwise $\binom{N+l}{N}$ has a position-0 carry. From (i),(ii),(iv), $s + t < p$: if $s + t \ge p$ then $p \le s + t \le 2r$ forces $r \ge p/2$, while $r + s < p$, $r + t < p$ force $s, t \le p - 1 - r$, hence $s + t \le 2(p - 1 - r) < p$, a contradiction. From (iii) and $b \le a$, $b + c \le a + c < p$. Therefore $k + l = (b + c)p + (s + t)$ is the base-p

expansion, and the position-1 sum in $N + (k + l)$ is $a + (b + c) + [\![r + s + t \geq p]\!] \geq a + b \geq p$: a carry. So $v_p\binom{N+k+l}{N} \geq 1$ as well, and $v_pT \geq 2$. □

Lemma 5.2 (Refined triple carry)**.** *With N, k, l as above set $\beta := \lfloor (N+k)/p \rfloor = a+b+[\![r+s \geq p]\!]$. If $\beta \geq p$ then $v_p(T(N, k, l)) \geq 2$ for every l.*

Proof. If $a + b \geq p$ this is Lemma 5.1. Otherwise $\beta \geq p$ forces $[\![r + s \geq p]\!] = 1$ and $a + b = p - 1$; then $N + k$ carries at position 0 and at position 1 ($a + b + 1 = p$), so $v_p\binom{N+k}{N} \geq 2$ by Kummer alone. □

Lemma 5.3 (Boundary carry)**.** *Let p be prime, $N = ap + r$, $k = bp + s$, $l = cp + t$ with $0 \leq a, r, s, t < p$, $0 \leq b, c \leq a$, $k, l \leq N$, and put $\beta = \lfloor (N + k)/p \rfloor$. If $\beta \geq p$ and $a + b < p$ — equivalently $a + b = p - 1$ and $r + s \geq p$ — then $v_p(T(N, k, l)) \geq 4$.*

Proof. $a + b = p - 1$ and $r + s \geq p$ make $N + k = p^2 + (r + s - p)$: the addition carries at position 0 and at position 1, so by Kummer

$$v_p\binom{N+k}{N} = 2. \tag{5.1}$$

Two further powers are needed. Note $b \leq a$ and $a + b = p - 1$ force $2a + 1 \geq p$.

Case $s > r$. The subtraction $N - k$ borrows at position 0, so $v_p\binom{N}{k} \geq 1$ and the squared factor supplies 2; with (5.1), $v_pT \geq 4$.

Case $s \leq r$, $s+t < p$. Then $k + l = (b + c)p + (s + t)$ is the base-p expansion and $r + s + t \geq r + s \geq p$, so $N + (k + l)$ carries at position 0; its position-1 sum is $a + (b + c) + 1 \geq a + b + 1 = p$, so it carries there too. Hence $v_p\binom{N+k+l}{N} \geq 2$.

Case $s \leq r$, $s + t \geq p$. From $s \leq r$ and $s + t \geq p$ we get $t \geq p - s \geq p - r$, i.e.

$$r + t \geq p, \tag{5.2}$$

so $N + l$ carries at position 0 and $v_p\binom{N+l}{N} \geq 1$. Write $\beta' := b + c + 1$, the position-1 digit sum of $k + l$ (here $k + l = \beta' p + (s + t - p)$). If $\beta' < p$, the position-1 sum of $N + (k + l)$ is $a + \beta' + [\![r + s + t \geq 2p]\!] = p + c + [\![\cdot]\!] \geq p$ (using $a + b = p - 1$), a carry, so $v_p\binom{N+k+l}{N} \geq 1$; total $2 + 1 + 1 = 4$. If $\beta' \geq p$ then $c \geq p - 1 - b = a$, so $c = a$; by (5.2) the position-0 carry of $N + l$ exists and its position-1 sum is $a + c + 1 = 2a + 1 \geq p$, a second carry, so $v_p\binom{N+l}{N} \geq 2$ and with (5.1), $v_pT \geq 4$. □

Evidence 5.4. Lemma 5.1: [VERIFIED] 24 741 cases ($p \in \{5, 7\}$, all a, r, all $k, l \leq N$), 0 failures. The three-carry strengthening is *false*: "$a + b \geq p$ and $a + c \geq p$" does *not* give $v_p \geq 3$ (649 counterexamples in the same range), so Lemma 5.1 is exactly the available slack. Lemma 5.2: [VERIFIED] 28 128 cases, 0 failures. Lemma 5.3: [VERIFIED] 0 failures with $\min v_p(T) = 4$ exactly (hence sharp) at $p = 5, 7, 11, 13$ on 518/2 869/28 340/65 918 index triples, of which 174/892/8 175/18 620 lie in the last sub-case $s \leq r$, $s + t \geq p$. By contrast, $\beta \geq p$ *with* $a + b \geq p$ gives $\min v_p(T) = 2$ exactly over 1.3M triples — Lemma 5.1 is sharp there.

6. The decomposition identities

6.1. **The alphabet.** For $n \geq 0$, $0 \leq k, l \leq n$ and $1 \leq m \leq 5$ define the *letters*

$$A_m(x) := H^{(m)}_{n+x} - H^{(m)}_x, \qquad B_m(x) := H^{(m)}_{n-x} - H^{(m)}_x, \qquad C_m := H^{(m)}_{n+k+l} - H^{(m)}_{k+l}, \qquad N_m := H^{(m)}_n, \tag{6.1}$$

with $x \in \{k, l\}$; each has *weight* m. These are exactly the parameter log-derivatives of the Γ-quotient form of T, which is why they are the right alphabet: with

$$F(n,k,l;\alpha_k,\alpha_l,\beta_k,\beta_l,g) = \frac{\Gamma(n+k+1+\alpha_k)}{\Gamma(n+1)\Gamma(k+1+\alpha_k)}\Big(\frac{\Gamma(n+1)}{\Gamma(k+1+\beta_k)\Gamma(n-k+1-\beta_k)}\Big)^2$$
$$\times\,(\text{the same with } k \mapsto l) \times \frac{\Gamma(n+k+l+1+g)}{\Gamma(n+1)\Gamma(k+l+1+g)},$$

one has $\partial^m_{\alpha_k} \log F|_0 = (-1)^{m+1}(m-1)!\,A_m(k)$, $\partial^m_{\beta_k} \log F|_0 = 2(-1)^{m+1}(m-1)!\,B_m(k)$ and $\partial^m_g \log F|_0 = (-1)^{m+1}(m-1)!\,C_m$.

6.2. **The middle row: Theorem B.**

Theorem 6.1 (Theorem B)**.** *[VERIFIED]*

$$\hat{P}_n = \sum_{k,l=0}^{n} T(n,k,l)\,\hat{w}_3(n,k,l),$$

where

$$\hat{w}_3(n,k,l) = H_n^{(3)} + A_3(k) + A_3(l) - \tfrac{1}{4}\big[A_2(k)A_1(k) + A_2(l)A_1(l)\big]$$
$$- \tfrac{3}{4}\big[A_2(k)B_1(k) + A_2(l)B_1(l)\big] - \tfrac{3}{8}\big[A_2(k) + A_2(l)\big]C_1 - \tfrac{1}{8}\big[A_2(k)A_1(l) + A_2(l)A_1(k)\big]. \tag{6.2}$$

This is a decomposition of the middle row over the *same* Brown–Zudilin summand that gives Q_n, with a graded harmonic weight and no extra nested corrections. Two structural features are worth recording, since they are not assumptions but outputs: every non-constant term carries an A-letter of the highest weight in its monomial, and every monomial whose top letter is B, C or $H_n^{(r)}$ has coefficient 0 — the fit was run in a basis that allowed all of these.

The single most consequential feature of (6.2) is that the letter $A_r(k) = H^{(r)}_{n+k} - H^{(r)}_k$ has upper argument up to $2n$. *This one fact is the source of the* d_{2n} *in* (1.6) *and of the entire middle-row anomaly of §13*: $A_3(k)$ acquires a second p-pole as soon as $\lfloor (n+k)/p \rfloor \geq p$, which for $n = a < p$ happens iff $a + k \geq p$.

6.3. **The top row: the family, and (T1-top).** At weight 5 the situation is subtler, and the history is instructive. A first search, in a basis of 149 symmetric weight-5 monomials with at most three factors, was inconsistent, and this was read as evidence that the depth-2 period $\zeta(5) + 2\zeta(3)\zeta(2)$ forces genuinely nested letters. That reading is *false*. Removing only the factor-count cap — same alphabet, no nested letters — makes the system consistent.

Proposition 6.2 (Existence)**.** *[VERIFIED] In the alphabet* $\{A_r, B_r\}$ *(at* k *and* l*)* $\cup\{C_r\}\cup\{N_r\}$, $r = 1, \ldots, 5$*, with at most* 5 *factors in each single-variable slot and at most* 2 *in the coupling and constant slots, the fitting system* $P_n = \sum_{k,l} T(n,k,l)\,w_5(n,k,l)$ *has* 448 *basis monomials and rank* 313*; it is* consistent*, with all* 687 *independent excess equations satisfied at* $N = 1000$*. The solution space is a* 135*-dimensional affine family.*

The ablation that pins this is decisive. At $N = 1200$ equations, caps $(\mathrm{mf}_k, \mathrm{mf}_c, \mathrm{mf}_n)$:

$(3,2,2)$ inconsistent, $(3,2,5)$ inconsistent, $(3,5,2)$ inconsistent,

$(5,2,2)$ CONSISTENT, $(5,5,5)$ CONSISTENT.

The missing ingredient is monomials with four or five factors in one variable — e.g. $A_1(k)^4A_1(l)$, $A_1(k)^3B_1(k)^2$, $A_1(k)^5$. Likewise the alphabet ablation (at $N = 1000$) shows $\{A_r, B_r, C_r, N_r\}$ is exactly right: dropping B or dropping C is inconsistent, and $H^{(r)}_{2n}$ is not needed. *So the* *alphabet of w_5 is exactly the alphabet of $\hat{w}_3$; the sole difference between the weights is the number of factors* — $\hat{w}_3$ uses at most two, w_5 needs up to five.

The family is then cut down by the depth conditions of §7 to a 124-dimensional subfamily, and inside it we fix explicit representatives.

Definition 6.3 (The representatives)**.**

- w_5^{allp}: 178 terms, coefficient denominators supported on $\{2,3\}$, hence p-integral for *every* $p \geq 5$. Obtained by a CRT combination of two depth-minimal representatives with disjoint bad primes (Lemma 6.4).
- w_5^{I}: 207 terms, denominators on $\{2,3\}$; in addition to the depth conditions it satisfies the full cell-wise cap on every pole pattern except one band, which is what makes the base case of §11 work.

Both are the CRT outputs, not the intermediates they are built from; in the accompanying archive they are the files `w5_allp.json` (178 terms) and `w5_exIII_allp.json` (207 terms). The intermediate from which the second is built, `w5_exIII.json` (equivalently `w5_I.json`), has 155 terms and denominators $\{2,3,71\}$, so it is *not* p-integral at $p = 71$ and must not be substituted for w_5^{I}.

Lemma 6.4 (CRT combination)**.** *[PROVED] Let x_1, x_2 lie in the depth-conditioned family; suppose x_1 is p-integral for all $p \geq 5$ except P_1 (with maximal denominator exponent e_1) and x_2 except P_2 (exponent e_2), $P_1 \neq P_2$. Put $d = x_2 - x_1$ and choose $t \in \mathbb{Z}$ with $t \equiv 1 \pmod{P_1^{e_1+1}}$, $t \equiv 0 \pmod{P_2^{e_2+1}}$. Then $x = x_1 + td$ lies in the family and is p-integral for every $p \geq 5$.*

Proof. At P_1, $x = x_2 - (1-t)d$ with $v_{P_1}((1-t)d) \geq (e_1+1) - e_1 = 1$; at P_2, $x = x_1 + td$ with $v_{P_2}(td) \geq (e_2+1) - e_2 = 1$; at every other $p \geq 5$ both x_1 and d are p-integral and $t \in \mathbb{Z}$. □

We can now state the first of the two hypotheses of Theorem 1.3 (the second, (DEPTH), is stated in §7).

(T1-top). $P_n = \displaystyle\sum_{k,l=0}^{n} T(n,k,l)\, w_5(n,k,l)$ for all $n \geq 0$, where w_5 is w_5^{allp} or w_5^{I} as appropriate.

Remark 6.5 (which representative)**.** The induction of §9 consumes w_5 *only* through the depth bound (DEPTH-gen), so it works for any point of the depth-conditioned family; and the base case of §11 uses w_5^{I}. Hence the whole $p \geq 5$ theorem can be run end-to-end on the *single* representative w_5^{I}, and its certificate target is then the *one* identity $P_n = \sum_{k,l} T \cdot w_5^{\mathrm{I}}$. Alternatively one may certify w_5^{allp} and add the homogeneous identity $\sum_{k,l} T \cdot (w_5^{\mathrm{I}} - w_5^{\mathrm{allp}}) = 0$; these are genuinely different obligations (see Proposition 14.2), and which is cheaper is a computational question, not a mathematical one.

6.4. **New closed forms for Apéry's $\zeta(3)$ numerators.** The residue construction behind (6.2) also produces pure harmonic-monomial closed forms one weight down, replacing the classical non-generalising weight $\sum_{m \leq k} (-1)^{m-1}/(2m^3\binom{n}{m}\binom{n+m}{m})$.

Proposition 6.6. *[PROVED] For Apéry's $\zeta(3)$ numerators* [Ap79]*,*

$$b_n = \sum_{j=0}^{n} \binom{n}{j}^2 \binom{n+j}{n}^2 \Big[H_j^{(3)} + \big(2H_j - H_{n+j} - H_{n-j}\big) H_j^{(2)} \Big].$$

Proof. Put $F(t) = \prod_{m=1}^{n}(t-m) / \prod_{m=0}^{n}(t+m)$, so that $F(t)^2 = \sum_{j=0}^{n} \big[\beta_{2,j}(t+j)^{-2} + \beta_{1,j}(t+j)^{-1}\big]$ with $\beta_{2,j} = \binom{n}{j}^2\binom{n+j}{n}^2$ and $\beta_{1,j} = \beta_{2,j} \cdot \frac{d}{dt} \log[(t+j)^2 F(t)^2]\big|_{t=-j} = 2\beta_{2,j}(2H_j - H_{n+j} -$

H_{n-j}). Since $\deg\mathrm{num}-\deg\mathrm{den}=-2$, the residues sum to zero: $\sum_j \beta_{1,j}=0$. Now $-(F^2)'(t)=\sum_j\big[2\beta_{2,j}(t+j)^{-3}+\beta_{1,j}(t+j)^{-2}\big]$, so summing over $t\geq 1$ and using $\sum_{t\geq1}(t+j)^{-m}=\zeta(m)-H_j^{(m)}$,

$$\sum_{t\geq 1}-(F^2)'(t)=2\Big(\sum_j\beta_{2,j}\Big)\zeta(3)+\Big(\sum_j\beta_{1,j}\Big)\zeta(2)-\Big[2\sum_j\beta_{2,j}H_j^{(3)}+\sum_j\beta_{1,j}H_j^{(2)}\Big].$$

The $\zeta(2)$ coefficient vanishes, $\sum_j\beta_{2,j}=a_n$ is Apéry's identity, and the linear form produced is $2(a_n\zeta(3)-b_n)$. □

Proposition 6.7. *[PROVED] over $\mathbb{Q}$ Writing $A_1(k)=H_{n+k}-H_k$, $B_1(k)=H_{n-k}-H_k$, $B_2(k)=H_{n-k}^{(2)}-H_k^{(2)}$, the same numerators b_n also have the closed form*

$$b_n=\sum_{k=0}^{n}\binom{n}{k}^2\binom{n+k}{k}^2\Big[H_n^{(3)}+\tfrac12\big(A_1(k)^2B_1(k)+A_1(k)B_1(k)^2+A_1(k)B_2(k)\big)\Big],$$

i.e. with the weight $w_3=H_n^{(3)}+\frac12A_1\big(B_1(A_1+B_1)+B_2\big)$, and this weight is cell-wise p-integral*: $d_3(n,k)\leq v_pT_A(n,k)$ with 0 violations for $p\in\{5,7,11,13,17,19,23\}$, all $n<p$ and all k.*

Remark 6.8. Propositions 6.6 and 6.7 are two closed forms for *the same* sequence b_n; they are not a pair of forms for b_n and a_n. (The denominators a_n need no harmonic weight: $a_n=\sum_k\binom{n}{k}^2\binom{n+k}{k}^2$ is Apéry's own closed form.) That the two weights differ while their T_A-sums agree is an instance of the non-uniqueness that Proposition 14.2 quantifies one weight up: the fitting system has a large kernel of genuine summation identities, so a representative is never determined by the identity it satisfies.

Proposition 6.7 matters here for a negative reason: it shows that Apéry's own base case is cell-wise in the *purely harmonic* alphabet, so the $\zeta(3)$ precedent does *not* argue for binomial-reciprocal letters at weight 5. The mechanism it does name — spend the weight on the non-polar B slot — is exactly what the weight-5 alphabet turns out not to have room for, by exactly one functional (§12).

Evidence 6.9. Proposition 6.6: [VERIFIED] exact agreement with the standard b_n for $n=0,\dots,5$: 0, 6, 351/4, 62531/36, 11424695/288, 35441662103/36000. Proposition 6.7: exact $\mathbb{Q}$ solve over 14 levels plus one u^3 condition (rank 15), then [VERIFIED] on 24 levels and 7 primes, 0 mismatches; and its right-hand side agrees with the standard b_n exactly for $n=0,\dots,14$.

7. The pole calculus and the depth conditions

Throughout this section fix $p\geq5$ and $n\geq1$, put $L=\lfloor\log_p n\rfloor$, $M:=L+1$, $P:=p^M$ (so $n<P$), fix a cell $0\leq k,l\leq n$, and write $u:=p^{-1}$. Set

$$v_5(n,k,l):=w_5(n,k,l)-H_n^{(5)},\qquad d_5(n,k,l):=\max\big(0,-v_pv_5(n,k,l)\big).$$

7.1. Level expansion.

Lemma 7.1 (Level expansion)**.** *For all $N\geq0$ and $m\geq1$, $H_N^{(m)}=\sum_{e\geq0}u^{em}\,\Sigma_e^{(m)}(N)$ where $\Sigma_e^{(m)}(N):=\sum_{j\leq\lfloor N/p^e\rfloor,\ p\nmid j}j^{-m}\in\mathbb{Z}_p$, and $\Sigma_e^{(m)}(N)=0$ for $p^e>N$.*

Proof. Partition $\{1,\dots,N\}$ by $e=v_p(i)$: $i=p^ej$ with $p\nmid j$, $j\leq N/p^e$, and $i^{-m}=p^{-em}j^{-m}$. □

Two consequences are used constantly: $\Sigma_e^{(m)}(N)=\Sigma_{e-1}^{(m)}(\lfloor N/p\rfloor)$ for $e\geq1$, and $v_p(H_N^{(m)})\geq-m\lfloor\log_p N\rfloor$.

Since $n+k, n+l < 2P$, $k+l < 2P$, $n+k+l < 3P$ and $n-k, k, l, n < P$, and since $p \geq 5$ gives $3P < pP = p^{M+1}$, *every* letter has an expansion $X_m = \sum_{e=0}^{M} u^{em} \mathsf{X}_{m,e}$ with $\mathsf{X}_{m,e} \in \mathbb{Z}_p$, and a weight-5 monomial has u-order at most $5M$. Hence

$$v_5(n,k,l) = \sum_{j=0}^{5M} K_j\, u^j, \qquad K_j \in \mathbb{Z}_p, \qquad \text{so} \qquad d_5 \leq \max\{j : K_j \neq 0\}. \tag{7.1}$$

7.2. **Top-level residues and the pattern census.**

Definition 7.2. Put

$$\alpha := [\![n+k \geq P]\!], \qquad \gamma := [\![n+l \geq P]\!], \qquad \varepsilon := \lfloor (k+l)/P \rfloor \in \{0,1\},$$
$$\kappa := [\![n+k+l \geq (\varepsilon+1)P]\!], \qquad \theta := \varepsilon + 1,$$

(θ set to 1 when $\kappa = 0$), and $s := \alpha + \gamma + \kappa$. The tuple $\pi = (\alpha, \gamma, \kappa, \theta)$ is the *pole pattern* of the cell.

Lemma 7.3 (Top-level residues). *The $e = M$ coefficient of a letter is: α for $A_m(k)$, γ for $A_m(l)$, 0 for every B_m and for N_m, and $\kappa\,\theta^{-m}$ for C_m.*

Proof. $k, l, n, n-k < P$ gives $\Sigma_M(\cdot) = 0$ for those arguments. $\Sigma_M^{(m)}(n+k)$ is a sum over $j \leq \lfloor (n+k)/P \rfloor \leq 1$ with $p \nmid j$, i.e. equals α. For C_m: with $\Theta := \lfloor (n+k+l)/P \rfloor \leq 2$ and $\varepsilon = \lfloor (k+l)/P \rfloor$, every $j \leq 2 < p$, so the coefficient is $\sum_{j=\varepsilon+1}^{\Theta} j^{-m}$; and $n < P$ forces $\varepsilon \leq \Theta \leq \varepsilon + 1$, so $\Theta - \varepsilon = \kappa$ and the coefficient is $\kappa\theta^{-m}$. ☐

Note θ^{-m} is a p-unit for every $p \geq 5$, so the whole calculus is p-independent.

Lemma 7.4 (Kummer floor). $v_p\binom{n+k}{n} \geq \alpha$, $v_p\binom{n+l}{n} \geq \gamma$, $v_p\binom{n+k+l}{n} \geq \kappa$*; hence*

$$vT := v_p T(n,k,l) \ \geq \ \alpha + \gamma + \kappa = s. \tag{7.2}$$

Proof. Kummer: $v_p\binom{n+k}{n}$ is the number of carries in $n+k$. Both $n, k < P = p^M$, so a carry out of position $M-1$ occurs iff $n+k \geq P$, i.e. iff $\alpha = 1$; same for γ. For κ: write $k + l = \varepsilon P + \rho$, $0 \leq \rho < P$; the carry out of position $M-1$ in $n + (k+l)$ occurs iff $n + \rho \geq P$, i.e. iff $n+k+l \geq (\varepsilon+1)P$. ☐

Lemma 7.5 (The census, at every level). *The pattern π takes exactly the seven values*

$$(0,0,0,1),\ (0,0,1,1),\ (1,0,1,1),\ (0,1,1,1),\ (1,1,0,1),\ (1,1,1,1),\ (1,1,1,2),$$

with $s = 0,1,2,2,2,3,3$ respectively.

Proof. (i) $\kappa = 0 \Rightarrow \alpha = \gamma$. Suppose $\alpha = 1$, $\kappa = 0$. If $\varepsilon = 0$ then $\kappa = [\![n+k+l \geq P]\!] = 0$ gives $n+k+l < P$, contradicting $n+k \geq P$. So $\varepsilon = 1$, i.e. $k+l \geq P$; then $n+l \geq k+l \geq P$ (using $k \leq n$), so $\gamma = 1$. Symmetrically for $\gamma = 1$, $\kappa = 0$. So the $\kappa = 0$ patterns are $(0,0,0)$ and $(1,1,0)$. (ii) $\theta = 2$ (i.e. $k+l \geq P$) forces $n+k \geq k+l \geq P$ (using $l \leq n$) and $n+l \geq k+l \geq P$, i.e. $\alpha = \gamma = 1$. (iii) With $\kappa = 1$, $\varepsilon = 0$ all four (α, γ) occur. Total $2+4+1 = 7$. ☐

Remark 7.6. Lemma 7.5 is level-free and does *not* go through Lemma 5.1: at $M = 1$ it *reproves* "$a + b \geq p \Rightarrow vT \geq 2$" as a corollary rather than using it. The absence of the patterns $(1,0,0,\cdot)$ and $(0,1,0,\cdot)$ is precisely the census form of that statement.

7.3. **The depth conditions and their consistency.** The Lemma-F budget of §8 supplies precision $p^{2+\min(vT,2)}$, and the ledger needs p^{1+d_5}; so the pattern-determined cap we must enforce is

$$J(\pi) := 1 + \min(s, 2) \qquad \text{(and } J = 0 \text{ at the trivial pattern } \pi = (0,0,0,1)\text{)}. \tag{7.3}$$

Definition 7.7 ((DEPTH))**.** For each reachable pattern π and each $j > J(\pi)$, require $K_j = 0$ *identically in the* $\mathbb{Z}_p$*-symbols* $\mathsf{X}_{m,e}$.

Treating the symbols as independent indeterminates is a strengthening; it is $\mathbb{Q}$-linear and p-independent in the coefficients of w_5, and — as the next theorem shows — it costs nothing.

Theorem 7.8 (Consistency of (DEPTH))**.** *[CERT] Over two auxiliary primes $q = 33554393$ and $q = 33554467$, with $N = 600$ fitting equations:*

raw condition rows	68
rank(*conditions alone*)	42
rank(*fitting system alone*)	313
rank(*joint system*)	324
rank(*augmented joint system*)	324

so the joint system is consistent *at both primes, and the depth-conditioned family has dimension* $448 - 324 = 124$.

Thus of the 42 independent depth conditions, 31 are already implied by the decomposition identity itself and only 11 are new.

Corollary 7.9 (Depth-minimal family)**.** *[PROVED] modulo the certificate of Theorem 7.8. The decomposition family contains a 124-dimensional affine subfamily on which, for* every *prime $p \ge 5$ and every cell $0 \le b, c \le a < p$,*

$$d_5(a,b,c) \;\le\; 1 + \min\bigl(v_p T(a,b,c), 2\bigr),$$

provided the chosen representative's coefficients are p-integral — which Lemma 6.4 arranges for every $p \ge 5$ at once.

Evidence 7.10. [VERIFIED] 0 failures. Three independent exact tests on each saved representative: (V1) the exact-$\mathbb{Q}$ ladder identity $P_n = \sum T w_5$; (V2) the depth sweep $d_5 \le 1 + \min(vT, 2)$ over *all* cells; (V3) the cell-by-cell test $v_p(\mathcal{T}(b,c) - (Q_n/Q_a)T(a,b,c)) \ge 1 + d_5(b,c)$ computed mod p^{10} from the true fibre tables. For w_5^{allp}: (V1) $n \le 34$, 0; (V2) $\max d_5 = 3$, 0 violations, min slack 0, up to $p = 31$; (V3) 0 failures at $p = 5, 7, 11, 13$. The comparison is the point:

representative	$p=5$	$p=7$	$p=11$	$p=13$
130-term (pre-depth)	1/270	18/707	3385/5379	1285/10634
106-term sparsest canonical	34/270	91/707	3503/5379	—
depth-minimal	**0**	**0**	**0**	**0**

i.e. 0/16 990 cells fail, where the sparsest representatives failed thousands. *Sharpness:* min slack is 0 at every prime — the value $d_5 = 1 + \min(vT, 2)$ is attained. So (DEPTH) sits exactly at the edge of what Lemma 8.8 supplies.

Remark 7.11 (sparsest is not depth-minimal)**.** Canonicalisation by sparsity and canonicalisation by depth are different, and it is depth that is load-bearing. The sparsest representative found (106 terms) has $\max d_5 = 5$ and violates the cap on the $vT \ge 2$ cells; a 130-term one has $\max d_5 = 4$. Neither works. Depth is not visible in any sparsity or weight statistic — and neither is the second load-bearing criterion, p-integrality of the coefficients: a stray prime in a denominator destroys the bound at that one prime and nowhere else.

7.4. **Level-lifting: the conditions are level-independent.** This is the technical heart of the induction. Fix a pattern π. For a letter-multiset S (letters counted with multiplicity) write $\mathrm{wt}(S) = \sum_{X\in S} \mathrm{weight}(X)$ and

$$\Phi_\pi(S) := \sum_{\mathcal{M}\supseteq S} c_{\mathcal{M}} \prod_{X\in\mathcal{M}\setminus S} \rho(X;\pi), \tag{7.4}$$

where $c_{\mathcal{M}}$ is the w_5-coefficient of the monomial $\mathcal{M}$ and $\rho(X;\pi) \in \{\alpha, \gamma, \kappa\theta^{-m}, 0\}$ is the top-level residue of Lemma 7.3. These are $\mathbb{Q}$-linear functionals of the 448 coefficients, independent of p, n and M.

Proposition 7.12 (LIFT)**.** *[PROVED] Let $J \geq 0$ and suppose*

$$\Phi_\pi(S) = 0 \quad \textit{for every letter-multiset } S \textit{ with } \mathrm{wt}(S) \leq 4 - J. \tag{7.5}$$

Then for every $M \geq 1$ *and every cell (n,k,l) with $\lfloor \log_p n\rfloor = M-1$ and pattern π,*

$$d_5(n,k,l) \;\leq\; 5(M-1)+J \;=\; 5L+J.$$

Moreover at $M=1$ the conditions (7.5) are exactly *the conditions "$K_j = 0$ identically in the symbols for $j > J$" of Definition 7.7.*

Proof. Expand each letter by Lemma 7.1 and collect. Writing $j = 5M-\delta$ and $f_i := M - e_i \geq 0$ (so $\sum_i f_i m_i = \delta$),

$$K_{5M-\delta} = \sum_{\mathcal{M}} c_{\mathcal{M}} \sum_{\substack{f:\ \sum f_i m_i = \delta \\ f_i \leq M}} \Big(\prod_{i:\, f_i\geq 1} \mathsf{X}_{i,M-f_i}\Big)\Big(\prod_{i:\, f_i=0} \rho(X_i;\pi)\Big).$$

Regard the $\mathsf{X}_{i,e}$ ($0 \leq e \leq M-1$, one indeterminate per (letter, level)) as independent symbols — this is the same strengthening already made at $M=1$. Two contributions carry the same symbol monomial iff they have the same multiset $\{(X_i, M-f_i) : f_i \geq 1\}$; that multiset determines the dropped multiset S and the drop profile $f|_S$. Summing over $\mathcal{M}$ at fixed $(S, f|_S)$ produces $c(S,f)\,\Phi_\pi(S)$, where $c(S,f)$ is the multinomial count of ways to distribute the copies of a repeated letter among the prescribed levels — a *nonzero rational independent of* $\mathcal{M}$: for a letter occurring $N \geq q$ times in $\mathcal{M}$ the count is $\binom{N}{q} q!/\prod_u q_u!$, and its ratio to the $M=1$ count $\binom{N}{q}$ is $q!/\prod_u q_u!$, free of N. Hence

$$\begin{aligned} &K_{5M-\delta} = 0 \text{ identically in the symbols} \\ \iff\;& \Phi_\pi(S) = 0 \text{ for every } S \text{ realisable by an admissible } f \text{ with } \textstyle\sum_i f_i m_i = \delta. \end{aligned}$$

The minimal δ realising a given support S is $\delta = \mathrm{wt}(S)$ (all $f_i = 1$), admissible for every $M \geq 1$. Therefore $\{K_j = 0\ \forall j > 5(M-1)+J\} \Longleftarrow \{\Phi_\pi(S) = 0\ \forall S,\ \mathrm{wt}(S) \leq 4-J\}$, because $j > 5(M-1)+J \iff \delta = 5M - j < 5-J \iff \mathrm{wt}(S) \leq \delta \leq 4-J$. With (7.1) this gives $d_5 \leq 5(M-1)+J$. At $M=1$, $\delta = 5 - (u\text{-order})$, so $u > J \iff \mathrm{wt}(S) \leq 4-J$: the two condition sets coincide row for row. □

Corollary 7.13 ((DEPTH-gen))**.** *[PROVED] With w_5 any point of the depth-conditioned 124-dimensional family, for* every *prime $p \geq 5$,* every $n \geq 1$ *and* every *cell,*

$$d_5(n,k,l) \;\leq\; 5L+1+\min(\alpha+\gamma+\kappa, 2) \;\leq\; 5L+1+\min\big(v_p T(n,k,l), 2\big).$$

Proof. The caps are $J(\pi) = 1+\min(s,2)$ for $s \geq 1$; for $\pi = (0,0,0,1)$ every top residue vanishes, so $\Phi_\pi(S) = 0$ automatically whenever $\mathcal{M}\setminus S \neq \emptyset$, i.e. whenever $\mathrm{wt}(S) \leq 4$, and the corollary holds there with $J=0$. Apply Proposition 7.12. The second inequality is Lemma 7.4. □

Evidence 7.14. [VERIFIED] 0 violations in 150 955 cells: $p \in \{5, 7, 11, 13\}$, $n \leq 40/55/45/45$ (digit levels $L = 0, 1, 2$), all (k, l). In the same sweep $v_pT \geq \alpha + \gamma + \kappa$ has 0 counterexamples, and the bound is *sharp* (min slack 0 at every prime, attained already at $L = 0$).

Corollary 7.15 (Theorem 1.6)**.** *[PROVED] Assume* (T1-top) *and* (DEPTH)*. Then* $\mathrm{ord}_p(P_n) \geq -5\lfloor \log_p n\rfloor - 1$ *for every prime* $p \geq 5$ *and every* $n \geq 1$*; equivalently* $d_n^5 \cdot \prod_{5 \leq p \leq 3n} p \cdot P_n \in \mathbb{Z}[1/6]$*. For* $p > 3n$ *all letter arguments are* $< p$*, so* $\mathrm{ord}_p(P_n) \geq 0$ *outright.*

Proof. Cell by cell, $v_p(Tv_5) \geq vT - d_5 \geq vT - 5L - 1 - \min(vT, 2) \geq -5L - 1$ by Corollary 7.13; and $v_p(H_n^{(5)} Q_n) \geq -5L$ because $Q_n \in \mathbb{Z}$ and by Lemma 7.1. □

7.5. **A hard negative: the deficit is irremovable.** The cell-wise bound at $L = 0$ is $v_p(Tv_5) \geq -1$, and the value -1 is *attained*, at $(n, k, l) = \left(\frac{p+1}{2}, 0, \frac{p-1}{2}\right)$ for every p (so $p = 5 : (3, 0, 2)$, $p = 7 : (4, 0, 3)$, $p = 11 : (6, 0, 5)$, $p = 13 : (7, 0, 6)$, $\ldots$), where $vT = 2$, $d_5 = 3$ and the pattern is $(0, 1, 1, 1)$. The cell-wise bound would follow from $d_5 \leq v_pT$, i.e. from strengthened conditions. It does not exist.

Theorem 7.16. *[CERT] at two auxiliary primes* $q = 33554393$*,* $q = 33554467$*,* $N = 600$*:*

system	*caps* $J(\pi)$	*rows*	rank(*cond*)	rank(*joint*)	*consistent?*
(DEPTH)	$1 + \min(s, 2)$	68	42	324	*yes, nullity* 124
(DEPTH$^+$)	s	239	123	342	***NO***
(DEPTH^{++})	2 *on the three* $s = 2$ *patterns*	149	81	342	***NO***

The third row is the sharp localisation: it is *exactly the three* $s = 2$ *patterns — the ones carrying the* -1 *deficit — that cannot be tightened*; the $s \leq 1$ and $s = 3$ patterns are already at the cell-wise optimum. So within the p-independent linear framework the deficit is irreducible, and the base case requires a genuine cancellation *between* cells. It is moreover global in both indices: [VERIFIED] ($p \leq 13$) the partial sums $\sum_c T(a, b, c)v_5(a, b, c)$ at fixed b still have $v_p = -1$; only the full double sum is integral. No row identity suffices.

Theorem 7.17 (Alphabet enlargement does not help)**.** *[VERIFIED] Let* $R^{(a)}$ *be the Apéry-type letters and* $Y_{a,b}, V_{a,b}$ *the nested interval letters,*

$$R^{(a)}(n, k) = \sum_{m \leq k} \frac{(-1)^{m-1}}{m^a \binom{n}{m}\binom{n+m}{m}},$$

$$Y_{a,b}(n, k) = \sum_{k < m_2 < m_1 \leq n+k} m_1^{-a} m_2^{-b}, \qquad V_{a,b}(n, k + l) = \sum_{k+l < m_2 < m_1 \leq n+k+l} m_1^{-a} m_2^{-b}.$$

Then $R^{(a)}$ *has pole order* ≤ 1 *at every weight* $a = 1, \ldots, 5$ *with indicator exactly* α*, and* $Y_{a,b}$ *has pole order exactly* $\max(a, b) < a + b$ *with indicator* α *(*$V_{a,b}$ *likewise with indicator* κ*) — i.e. both families supply precisely the missing order. Nevertheless, with the full Apéry alphabet* $R^{(1..5)}$ *at both slots (*1210 *coefficients,* rank(fit) $313 \to 960$*, depth-conditioned value space* $\dim U$ $261 \to 641$*) the strengthened system is* still inconsistent*, at both the* (DEPTH^{++}) *and the* (DEPTH$^+$) *cap, at* $N = 1300$ *(row-saturated). The same for* Y/V*.*

The structural reason is worth recording, since it applies to *any* enlargement.

Proposition 7.18 (Block diagonality)**.** *[PROVED] Let the alphabet be enlarged by new letters whose* u^0 *part is a* new $\mathbb{Z}_p$ *symbol (true for* R, D, Y, V, Z*). Split the basis into* old *(monomials using no new letter) and* new*. Then the depth-condition matrix is block diagonal,* $C = \mathrm{diag}(C_{\text{old}}, C_{\text{new}})$*, because every* u*-expansion term of a new monomial carries at least one new* *symbol in its symbol key and no old term does. Hence* $\operatorname{rank} C = \operatorname{rank} C_{\text{old}} + \operatorname{rank} C_{\text{new}}$*, and with* $W_{\text{old}} := M_{\text{old}} \ker C_{\text{old}}$*,* $W_{\text{new}} := M_{\text{new}} \ker C_{\text{new}}$ *in value-sequence space,*

$$\text{the enlarged system is consistent} \iff P \in W_{\text{old}} + W_{\text{new}}.$$

So enlargement does not relax the old conditions at all: the new letters must supply the missing value-direction from scratch. And the obstruction is *global*: the smallest prefix of levels $n = 1, \dots, \Lambda$ on which the strengthened system is already inconsistent is $\Lambda = 262$, one more than $\dim U = 261$, exactly the generic threshold. There is no short window of levels and hence no small-n congruence carrying the obstruction.

Remark 7.19 (the honest caveat)**.** The depth conditions used throughout are the *symbol-independent* form: for every reachable pattern π and every u-power $j > \operatorname{cap}(\pi)$, K_j must vanish identically in the $\mathbb{Z}_p$ symbols. This is *sufficient* for the cell-wise depth bound at every prime simultaneously, and it is what makes the computation possible at all. It is *not necessary*: at a fixed prime the symbols range over the finitely many values the cells supply. So Theorem 7.17 refutes the p-independent symbol-independent form of the enlargement route; the prime-by-prime form is a different, non-linear question which we have not settled. Two remarks bound its plausibility: the symbol-independent conditions are *sharp* at the base cap (min slack 0), so nothing is being wasted; and as $p \to \infty$ the cells fill out and the symbols become generic, so the two forms agree asymptotically. Any gap lives at small primes.

8. The fibre congruence

Throughout this section $p \geq 5$, $n = ap + r$, $0 \leq r < p$, $k = bp + s$, $l = cp + t$ with $0 \leq b, c \leq a$, and

$$\mathcal{T}(b, c) := \sum_{s,t=0}^{p-1} T(n,\, bp + s,\, cp + t) \in \mathbb{Z}.$$

8.1. **The block factorisation.** Write T as a *balanced factorial ratio* — this is the form that makes everything work:

$$T(n, k, l) = \frac{(n + k)!\,(n + l)!\,(n + k + l)!\,n!}{k!^3\,l!^3\,(n - k)!^2\,(n - l)!^2\,(k + l)!}, \tag{8.1}$$

which is checked by expanding the five binomials; the argument sums balance, $(n+k)+(n+l)+(n+k+l)+n = 3k+3l+2(n-k)+2(n-l)+(k+l) = 4n+2k+2l$. Call the 15 factorial slots $(n+k), (n+l), (n+k+l), n$ (numerator) and $k, k, k, l, l, l, (n-k), (n-k), (n-l), (n-l), (k+l)$ (denominator). For $m \geq 0$ write $m = m_1 p + m_0$ with $0 \leq m_0 < p$, and $(m!)_p := \prod_{j \leq m,\ p \nmid j} j$, so that $m! = p^{m_1} m_1!\,(m!)_p$.

Lemma 8.1 (Wolstenholme block)**.** *[PROVED] For $p \geq 5$ and every $i \geq 0$,*

$$B_i := \prod_{u=1}^{p-1} (ip + u) \ \equiv\ (p - 1)! \pmod{p^3}.$$

Proof. $B_i = \sum_{j \geq 0} (ip)^j e_{p-1-j}(1, \dots, p - 1)$ with $e_{p-1} = (p - 1)!$, $e_{p-2} = (p - 1)!\,H_{p-1}$ and $e_{p-3} = (p-1)!\,(H_{p-1}^2 - H_{p-1}^{(2)})/2$. Wolstenholme's congruence ($p \geq 5$) gives $H_{p-1} \equiv 0 \pmod{p^2}$ and $H_{p-1}^{(2)} \equiv 0 \pmod{p}$. Hence the $j = 1$ term lies in $p^3\mathbb{Z}_p$, the $j = 2$ term is $p^2 i^2 (p-1)!(H_{p-1}^2 - H_{p-1}^{(2)})/2 \in p^3\mathbb{Z}_p$, and the terms $j \geq 3$ lie in $p^3\mathbb{Z}_p$. □

Consequently, writing

$$G(m_1,m_0) := \Big[\prod_{i<m_1} B_i \Big/ ((p-1)!)^{m_1}\Big] \prod_{u=1}^{m_0}\Big(1+\frac{m_1 p}{u}\Big), \tag{8.2}$$

one has the *exact* identity $(m!)_p = ((p-1)!)^{m_1}\, m_0!\, G(m_1,m_0)$ together with

$$G(m_1,m_0) = 1 + p\,m_1 H_{m_0} + p^2 m_1^2 \frac{H_{m_0}^2 - H_{m_0}^{(2)}}{2} + O(p^3), \qquad G \in 1+p\mathbb{Z}_p$$

(all H_{m_0} here are p-integral because $m_0 < p$).

Lemma 8.2 (Exact fibre block factorisation)**.** *[PROVED] Let $p \ge 5$, $n = ap + r$ with $0 \le r < p$, $k = bp + s$, $l = cp + t$ with $0 \le b, c \le a$ and $0 \le s \le r$, $0 \le t \le r$. Put*

$$e_1 = [\![r+s \ge p]\!], \quad e_2 = [\![r+t \ge p]\!], \quad e_3 = \lfloor (r+s+t)/p \rfloor \in \{0,1,2\}, \quad e_4 = [\![s+t \ge p]\!].$$

Then, exactly*,*

$$T(n,k,l) = T(a,b,c)\,T(r,s,t)\,\Pi\,\hat{G}, \qquad \Pi = \frac{\binom{a+b+e_1}{e_1}\binom{a+c+e_2}{e_2}\binom{a+b+c+e_3}{e_3}}{\binom{b+c+e_4}{e_4}}, \qquad \hat{G} \in 1+p\mathbb{Z}_p, \tag{8.3}$$

where $\hat{G} = \prod_{\mathrm{slots}}\big[G(m_1,m_0)/G(e_{\mathrm{slot}},m_0)\big]^{\pm 1}$, with (m_1,m_0) the true base-p digits of each slot and $e_{\mathrm{slot}} \in \{e_1,e_2,e_3,e_4\}$ on the four shifted slots, 0 on the other eleven. The hypothesis $a < p$ is not *used.*

Proof. Under $s \le r$, $t \le r$ the true digit pairs of the 15 slots are

$$(n+k) \to (a+b+e_1,\ r+s-e_1 p), \qquad (n+l) \to (a+c+e_2,\ r+t-e_2 p),$$
$$(n+k+l) \to (a+b+c+e_3,\ r+s+t-e_3 p),$$

$n \to (a,r)$, $k \to (b,s)$, $l \to (c,t)$, $(n-k) \to (a-b, r-s)$, $(n-l) \to (a-c, r-t)$, $(k+l) \to (b+c+e_4,\ s+t-e_4 p)$. Substituting $m! = p^{m_1} m_1!\,((p-1)!)^{m_1} m_0!\,G(m_1,m_0)$ in all 15 slots gives

$$T(n,k,l) = p^E ((p-1)!)^E \Big[\prod (m_1!)^{\pm 1}\Big]\Big[\prod (m_0!)^{\pm 1}\Big]\Big[\prod G(m_1,m_0)^{\pm 1}\Big],$$

with $E = \sum \pm m_1 = e_1+e_2+e_3-e_4$. For each shifted slot write $m_1! = (m_1-e)! \cdot m_1!/(m_1-e)!$ and $m_0! = (m_0+ep)!\big/\big(p^e e!\,((p-1)!)^e\, G(e,m_0)\big)$ (the second by the same substitution applied to m_0+ep, whose digits are (e,m_0)). The unshifted level-a digits are exactly $a+b$, $a+c$, $a+b+c$, a; b,b,b, c,c,c, $a-b, a-b$, $a-c, a-c$, $b+c$, i.e. $\prod (m_1-e)!^{\pm 1} = T(a,b,c)$; and the unshifted level-r digits are $r+s$, $r+t$, $r+s+t$, r; s,s,s, t,t,t, $r-s, r-s$, $r-t, r-t$, $s+t$, i.e. $\prod (m_0+ep)!^{\pm 1} = T(r,s,t)$. Collecting, the powers $p^{\pm e}$ and $((p-1)!)^{\pm e}$ cancel $p^E((p-1)!)^E$ exactly, $\prod [m_1!/((m_1-e)!e!)]^{\pm 1} = \Pi$, and the leftover G's are $\hat{G}$. Finally $\hat{G} \in 1+p\mathbb{Z}_p$ because each $G(m_1,m_0) \in 1+p\mathbb{Z}_p$ by Lemma 8.1, for every $m_1 \ge 0$. ☐

The admissible carry patterns involve only r,s,t and are

$$(e_1,e_2,e_3,e_4) \in \{(0,0,0,0),(0,0,1,0),(1,0,1,0),(0,1,1,0),(1,1,1,0),(1,1,1,1),(1,1,2,1)\} \tag{8.4}$$

($e_4 = 1 \Rightarrow e_1 = e_2 = 1$ since $t \le r$ gives $r+s \ge s+t \ge p$; $e_1 = 1 \Rightarrow e_3 \ge 1$; $e_3 = 2 \Rightarrow e_4 = 1$).

Evidence 8.3. [VERIFIED] 8 247 294 in-regime cells, 0 failures ($p \in \{5,7,11\}$, all $a \le 3p+1$, all r, all $0 \le b, c \le a$, all $s,t \le r$): $\hat{G}$ recomputed as $T(n,k,l)/(T(a,b,c)T(r,s,t)\Pi)$ over $\mathbb{Q}$ and tested for $\hat{G} - 1 \in p\mathbb{Z}_p$.

8.2. **The residue identity.**

Lemma 8.4 (Φ)**.** [PROVED] *For all integers $r \geq 0$, $t \geq 0$,*

$$\sum_{s=0}^{r} T(r,s,t)\,\Phi_b(s,t) = 0, \qquad \Phi_b(s,t) := H_{r+s}+H_{r+s+t}-3H_s+2H_{r-s}-H_{s+t} = A_1(s)+2B_1(s)+C_1$$

in the level-r letters. By the $k \leftrightarrow l$ symmetry of T, also $\sum_{t=0}^{r} T(r,s,t)\Phi_c(s,t) = 0$ for each s, with $\Phi_c(s,t) = A_1(t) + 2B_1(t) + C_1$.

Proof. $\Phi_b = \partial_s \log T(r,s,t)$ in the Γ-continuation, and the identity is a residue sum. Concretely, put

$$G(x) := \prod_{i=1}^{r}(x+i)\prod_{i=1}^{r}(x+t+i)\Big/\prod_{j=0}^{r}(x-j)^2.$$

G is rational with deg(den) $-$ deg(num) $= (2r+2) - 2r = 2$, so $G(x) = O(x^{-2})$ and the sum of all its residues vanishes. Its poles are exactly the double poles at $x = 0, 1, \dots, r$ (the numerator zeros are at $x = -1, \dots, -r$ and $x = -t-1, \dots, -t-r$, all negative, so no cancellation). At $x = s$ write $G = N/((x-s)^2E)$ with $E(x) = \prod_{j\neq s}(x-j)^2$; then

$$\begin{aligned}\mathrm{Res}_{x=s}\, G &= \frac{N}{E}(s)\Big[\frac{N'}{N}(s) - \frac{E'}{E}(s)\Big] \\ &= \frac{N}{E}(s)\big[(H_{r+s}-H_s) + (H_{r+s+t}-H_{s+t}) - 2H_s + 2H_{r-s}\big] = \frac{N}{E}(s)\,\Phi_b(s,t),\end{aligned}$$

and $\frac{N}{E}(s) = \frac{(r+s)!(r+s+t)!}{s!^3(s+t)!(r-s)!^2} = T(r,s,t)/K_t$ with $K_t = \frac{(r+t)!\,r!}{t!^3(r-t)!^2}$ independent of s. Summing the residues gives the claim. □

Remark 8.5. Φ_b, Φ_c are exactly the level-r weight-1 letter combinations $A_1 + 2B_1 + C_1$ — the same alphabet as $\hat{w}_3$ and w_5. Lemma 8.4 is the weight-1 shadow of the same residue mechanism that produced the closed forms of §6.4; the weight-2 level is Lemma 12.9. [VERIFIED] exact, 0 exceptions, $r \leq 11$, all t, plus the aggregated $\sum_{s,t}$ version for $r \leq 10$ over $\mathbb{Q}$.

8.3. **The two carry lemmas and Lemma F.** Write $vT := v_pT(a,b,c) = \alpha+\gamma+\kappa$ with $\alpha = [\![a+b \geq p]\!]$, $\gamma = [\![a+c \geq p]\!]$, $\kappa = v_p\binom{a+b+c}{a} \in \{0,1\}$, each a single Kummer carry because $a, b, c < p$; so $vT \leq 3$.

Lemma 8.6 (Off-fibre-regime vanishing)**.** [PROVED] *If $s > r$ or $t > r$ then $v_p\big(T(n, bp+s, cp+t)\big) \geq 2 + \min(vT, 2)$.*

Proof. By $k \leftrightarrow l$ symmetry assume $s > r$. If $b = a$ then $k = ap + s > ap + r = n$ and $T = 0$; so $b < a$. The subtraction $n - k$ borrows at position 0, so $v_p\binom{n}{k} \geq 1$ and the *squared* factor contributes 2. It remains to produce $\min(vT, 2)$ further powers. If $\alpha = 1$, the position-1 digit sum of $n + k$ is $a + b + [\![r+s \geq p]\!] \geq a + b \geq p$, a carry, so $v_p\binom{n+k}{n} \geq 1$; identically if $\gamma = 1$. If $\kappa = 1$, write $k + l = \beta p + \sigma$ with $\beta = b + c + e_4$, $\sigma = s + t - e_4p$: if $b + c \geq p$ then $\kappa = 1$ means $a + b + c \geq 2p$ and the position-1 sum of $n + (k + l)$ is $a + (\beta - p) + \text{carry} \geq a + b + c - p \geq p$, a carry; if $b + c < p$ and $\beta < p$, the position-1 sum is $a + \beta + \text{carry} \geq a + b + c \geq p$, a carry. The only gap is $\beta = p$, i.e. $b + c = p - 1$ and $e_4 = 1$. Now count. If $vT \leq 1$, at most one of α, γ, κ is 1 and the above supplies it (in the gap case $\kappa = 1$, $\alpha = \gamma = 0$ one gets $b + c = p - 1$, $b, c < p - a$, hence $p - 1 < 2(p - a)$ and $p - 1 \leq 2a$, forcing $a = (p-1)/2$ and $b = c = a$, contradicting $b < a$). If $vT \geq 2$, at least two of α, γ, κ equal 1; if they are α, γ we are done. Otherwise the gap $b + c = p - 1$, $e_4 = 1$ may occur: if $\alpha = 1, \gamma = 0$ then $\gamma = 0$ gives $c \leq p - a - 1$, hence $b = p - 1 - c \geq a$, contradicting $b < a$ — vacuous. If $\gamma = 1, \alpha = 0$ then symmetrically $c = a$;

if $t > r$ then $\binom{n}{l}^2$ gives 2 more and we are done, so $t \le r$, and $e_4 = 1$ gives $s \ge p - t \ge p - r$, i.e. $r + s \ge p$; then $n + k$ carries at position 0 and (since $a + b + 1 = a + (p - 1 - a) + 1 = p$) at position 1, so $v_p\binom{n+k}{n} \ge 2$ and the total is $\ge 2 + 1 + 2 = 5 \ge 4$. If $\alpha = \gamma = 1$ then $v_p\binom{n+k}{n} + v_p\binom{n+l}{n} \ge 2$. $\square$

Lemma 8.7 (First-order fibre expansion). *[PROVED] For $0 \le s \le r$, $0 \le t \le r$,*

$$T(n, bp + s, cp + t) \equiv T(a, b, c)T(r, s, t)\big(1 + p[a\Phi_a + b\Phi_b + c\Phi_c]\big) \pmod{p^{2+vT}},$$

$$\Phi_a(s, t) = H_{r+s} + H_{r+t} + H_{r+s+t} + H_r - 2H_{r-s} - 2H_{r-t},$$

except in the single carry pattern $(e_1, e_2, e_3, e_4) = (1, 1, 1, 1)$ with $\alpha = \gamma = 1$, where the congruence holds mod p^{1+vT} (and then $vT = 3$, so $1 + vT = 4 = 2 + \min(vT, 2)$). In all cases the congruence holds mod $p^{2+\min(vT,2)}$.

Proof. Set $\Delta_1 := a\Phi_a + b\Phi_b + c\Phi_c$; expanding the definitions, $\Delta_1 = \sum_{\text{slots}} \pm\mu_1 H_{\mu_0}$ where (μ_1, μ_0) are the *naïve* (level-a, level-r) digits, i.e. $\mu_0 = m_0 + e_{\text{slot}}p$. Each H_{μ_0} has $\mu_0 < 3p$, hence $H_{\mu_0} = (\mathbb{Z}_p) + p^{-1}H_{\lfloor \mu_0/p \rfloor}$ with $\lfloor \mu_0/p \rfloor \le 2 < p$, so $p\Delta_1 \in \mathbb{Z}_p$; moreover $H_{\mu_0} - H_{m_0} = p^{-1}H_{e_{\text{slot}}} + (\mathbb{Z}_p)$, whence $1 + p\Delta_1 = 1 + pX + \sum_{\text{shifted}} \pm\mu_1 H_{e_{\text{slot}}} + p(\mathbb{Z}_p)$ with $X = \sum_{\text{slots}} \pm\mu_1 H_{m_0}$. By Lemma 8.2, $T(n, k, l)/(T(a, b, c)T(r, s, t)) = \Pi\hat{G} = \Pi(1 + pX + O(p^2))$, so the difference equals $T(a, b, c)T(r, s, t)\big[\Pi - 1 - \sum \pm\mu_1 H_e + O(p)\big]$. Let $J := v_pT(r, s, t)$ and K the valuation of the bracket; the conclusion is $v_p(\text{diff}) \ge vT + J + K$. Running the seven patterns (8.4):

(e_1, e_2, e_3, e_4)	Π	$1 + \sum \pm\mu_1 H_e$	$K \ge$	$J \ge$
$(0, 0, 0, 0)$	1	1	2	0
$(0, 0, 1, 0)$	$a{+}b{+}c{+}1$	$1 + (a{+}b{+}c)$ — equal	1	1
$(1, 0, 1, 0), (0, 1, 1, 0)$	$(a{+}b{+}1)(a{+}b{+}c{+}1)$	$1 + (a{+}b) + (a{+}b{+}c)$	0	2
$(1, 1, 1, 0)$	3 factors	—	0	3
$(1, 1, 1, 1)$	$\frac{(a+b+1)(a+c+1)(a+b+c+1)}{b+c+1}$	—	-1	2
$(1, 1, 2, 1)$	with $\binom{a+b+c+2}{2}$	—	-1	3

$K \ge 0$ whenever $e_4 = 0$ (then $\Pi \in \mathbb{Z}$), and $K \ge -1$ always since $b + c + 1 \le 2a + 1 < 2p$. So $J + K \ge 2$, i.e. $v_p(\text{diff}) \ge vT + 2$, in every pattern except $(1, 1, 1, 1)$, and there $K = -1$ forces $v_p\binom{b+c+1}{1} = 1$, i.e. $b + c = p - 1$, with $v_p(a + c + 1) = v_p(a + b + 1) = 0$. But $b + c = p - 1$ gives $\kappa = 1$, and: $\alpha = \gamma = 0$ forces (as in Lemma 8.6) $a = (p - 1)/2$, $b = c = a$, so $a + b + 1 = p$, a contradiction; exactly one of α, γ equal to 1 forces $b = a$ resp. $c = a$, hence $a + c + 1 = p$ resp. $a + b + 1 = p$, again a contradiction. So $K = -1$ implies $\alpha = \gamma = 1$, hence $vT = 2 + \kappa = 3$ and $v_p(\text{diff}) \ge 3 + 2 - 1 = 4 = 2 + \min(vT, 2)$. $\square$

Theorem 8.8 (Lemma F). *[PROVED] Let $p \ge 5$, $n = ap + r$, $1 \le a < p$, $0 \le r < p$, $p \nmid Q_a$ and $0 \le b, c \le a$. Then*

$$\mathcal{T}(b, c) \;\equiv\; \frac{Q_n}{Q_a}\, T(a, b, c) \pmod{p^{\,2+\min(v_pT(a,b,c),\,2)}}.$$

Proof. Split the fibre at $s \le r$, $t \le r$. By Lemma 8.6 the complementary terms are $\equiv 0$ mod $p^{2+\min(vT,2)}$, and there $T(r, s, t) = 0$. By Lemma 8.7, for $s, t \le r$, $T(n, bp + s, cp + t) \equiv T(a, b, c)T(r, s, t)(1 + p[a\Phi_a + b\Phi_b + c\Phi_c])$ mod $p^{2+\min(vT,2)}$. Every product $T(r, s, t)\Phi_x(s, t)$ is *p-integral*: a pole of Φ_x needs $r + s \ge p$ (so $\binom{r+s}{r}$ carries), $r + t \ge p$, $s + t \ge p$ (so $r + s \ge s + t \ge p$), or $r + s + t \ge p$ with $r + s, r + t < p$ (so $s + t < p$ by Lemma 3.4, hence $\binom{r+s+t}{r}$ carries); in each case $v_pT(r, s, t) \ge 1$, and every pole has order exactly 1 because all arguments are $< 3p$.

Summing over $0 \le s,t \le p-1$ and using $T(r,s,t)=0$ for $s>r$ or $t>r$,

$$\mathcal{T}(b,c) \equiv T(a,b,c)\Big[\sum_{s,t=0}^{r} T(r,s,t) + p\,a\,\Psi_a + p\,b\sum_{s,t} T\Phi_b + p\,c\sum_{s,t} T\Phi_c\Big] \pmod{p^{2+\min(vT,2)}},$$

with $\Psi_a := \sum_{s,t=0}^{r} T(r,s,t)\Phi_a(s,t) \in \mathbb{Z}_p$. Now $\sum_{s,t=0}^{r} T(r,s,t) = Q_r$ by (1.5) at $n=r$, and by Lemma 8.4 the last two sums vanish. *The whole (b,c)-dependence of the first-order term therefore cancels*, leaving $\mathcal{T}(b,c) \equiv \mu\,T(a,b,c) \bmod p^{2+\min(vT(b,c),2)}$ with $\mu := Q_r + p\,a\,\Psi_a \in \mathbb{Z}_p$ independent of (b,c). Summing over $0 \le b,c \le a$ and using $\sum_{b,c}\mathcal{T}(b,c) = Q_n$, $\sum_{b,c} T(a,b,c) = Q_a$ and $2+\min(vT,2) \ge 2$ gives $Q_n \equiv \mu Q_a \pmod{p^2}$, hence $\Lambda := Q_n/Q_a \equiv \mu \pmod{p^2}$ as $p \nmid Q_a$. Finally $\mathcal{T}(b,c) - \Lambda T(a,b,c) = T(a,b,c)(\mu-\Lambda) + [\mathcal{T}(b,c) - \mu T(a,b,c)]$, whose two terms have $v_p \ge vT+2$ and $\ge 2+\min(vT,2)$ respectively. □

Corollary 8.9 (A mod p^2 supercongruence for the bottom row)**.** *[PROVED] For $p \ge 5$, $1 \le a < p$, $0 \le r < p$ with $p \nmid Q_a$,*

$$Q_{ap+r} \equiv Q_a\big(Q_r + p\,a\,\Psi_a\big) \pmod{p^2}, \qquad \Psi_a = \sum_{s,t=0}^{r} T(r,s,t)\,\Phi_a(s,t),$$

with $\Phi_a = A_1(s) + A_1(t) + C_1 - 2B_1(s) - 2B_1(t) - H_s - H_t + H_{s+t} + H_r$ in the level-r letters.

This is a genuine mod p^2 refinement of Theorem 1.4, and Ψ_a is the level-r weight-1 harmonic functional of the Brown–Zudilin summand. [VERIFIED] 0 mismatches at $p = 5,7,11,13$, all a,r with $p \nmid Q_a$, $n \le 360$.

Evidence 8.10 (sharpness of Lemma F). [VERIFIED] 0 failures, and *sharp*: over 20 998 cells at $p = 5,7,11,13$ the pairs $(vT, v_p(\mathrm{diff}))$ realised are exactly $(0,2),(1,3),(2,4),(3,4)$, i.e. the bound $2+\min(vT,2)$ is attained in every stratum (min slack 0). Consequently "prove Lemma F one order deeper" is *impossible*: the congruence mod $p^{3+\min(vT,2)}$ is false. Lemmas 8.6 and 8.7 are each [VERIFIED] at $p=5,7$ over all (a,b,c,r,s,t), stratified by carry pattern (26 strata; min slack ≥ 0 in every one and $=0$ in seven).

Remark 8.11 (what Lemma F did *not* need)**.** Not needed: Jacobsthal's $\binom{ap}{bp} \equiv \binom{a}{b} \pmod{p^3}$, and Granville's general prime-power binomial congruences. The only external input is Wolstenholme, used once, in Lemma 8.1 — and that is exactly why $p \ge 5$ is the right hypothesis. The real content is (i) the exact block factorisation, where T's being a *balanced* factorial ratio is used, and (ii) Lemma 8.4. The proof is weight-agnostic: Lemma 8.8 is a statement about T alone.

8.4. **Two carry inequalities.** The next three sections use two elementary base-p facts repeatedly; we isolate them once. Write $\mathrm{car}(x,y,\epsilon)$ for the number of carries in the base-p addition $x+y+\epsilon$, with $\epsilon \in \{0,1\}$ the carry into position 0; write $\mathrm{s}(N)$ for the base-p digit sum of N, and $z(N)$ for the number of trailing digits of N equal to $p-1$.

Lemma 8.12 (Carry inequalities)**.** *[PROVED] With $n = ap+r$, $k = bp+s$, $l = cp+t$ and $e_1 = [\![r+s \ge p]\!]$, $e_2 = [\![r+t \ge p]\!]$ as in Lemma 8.2:*

(C1) $\mathrm{car}(x,y,1) = \mathrm{car}(x,y,0) + z(x+y) \;\ge\; \mathrm{car}(x,y,0)$ *for all $x,y \ge 0$;*

(C2) $v_p\binom{n+k}{n} = e_1 + \mathrm{car}(a,b,e_1) \ge v_p\binom{a+b}{a}$*, and likewise for $(n+l,n)$ with e_2; and* $v_p\binom{n}{k} = [\![s>r]\!] + \mathrm{car}(b,\, a-b-[\![s>r]\!],\, [\![s>r]\!]) \ge v_p\binom{a}{b} + [\![s>r]\!]$*, and likewise for (n,l).*

Proof. (C1): $\mathrm{car}(x,y,\epsilon) = (\mathrm{s}(x)+\mathrm{s}(y)+\epsilon-\mathrm{s}(x+y+\epsilon))/(p-1)$, and $\mathrm{s}(N+1) = \mathrm{s}(N)+1-(p-1)z(N)$.

(C2): by Kummer's theorem $v_p\binom{n+k}{n}$ counts the carries of $n+k$ in base p; splitting that addition at position 0 contributes e_1 there and $\operatorname{car}(a,b,e_1)$ above, and (C1) gives $\operatorname{car}(a,b,e_1) \geq \operatorname{car}(a,b,0) = v_p\binom{a+b}{a}$. For the second, split the subtraction $n-k$ at position 0: it borrows exactly when $s > r$, the higher-position carries of $k+(n-k)$ sum to a, and $\mathrm{s}(a-b-1)+1 \geq \mathrm{s}(a-b)$. □

8.5. **The multi-digit weakening.** The induction consumes one order less precision, and at that precision the statement holds for *all* a, with a shorter proof and no auxiliary hypotheses at all.

Lemma 8.13 (Lemma F-gen)**.** *[PROVED] Let $p \geq 5$, $a \geq 1$, $0 \leq r < p$, $n = ap+r$, $0 \leq b, c \leq a$. Let α, γ, κ be the level-a indicators of Definition 7.2 evaluated at (a,b,c) with $P_a = p^{\lfloor \log_p a \rfloor + 1}$, and $s_a := \alpha + \gamma + \kappa$. Then*

$$v_p\big(\mathcal{T}(b,c) - Q_r\, T(a,b,c)\big) \ \geq \ 1 + \min(s_a, 2) \ =: \ G \ (\leq 3).$$

Three features distinguish this from Theorem 8.8: it holds for *all* a, not only $a < p$; the comparison constant is Q_r itself, with no $\mu = Q_r + pa\Psi_a$ correction and no $\Lambda = Q_n/Q_a$, hence *no $p \nmid Q_a$ hypothesis* and no exceptional-prime compensation; and it is one order weaker, which is what makes the general-a proof short. It is exactly the precision the induction consumes.

Proof. Both parts use the carry inequalities of Lemma 8.12.

(A) Off-regime terms $s > r$ or $t > r$. By symmetry assume $s > r$. If $b = a$ then $k > n$ and $T = 0$. So $b < a$; the subtraction $n-k$ borrows at position 0, so $v_p\binom{n}{k} \geq 1$ and the squared factor gives $v_p T \geq 2$. That settles $G \leq 2$, i.e. $s_a \leq 1$. If $s_a \geq 2$ then at least two of α, γ, κ are 1; $\alpha = \gamma = 0$ would force $s_a = \kappa \leq 1$, so $\alpha = 1$ or $\gamma = 1$, and then Lemma 8.12(C2) gives one further power: $v_p T \geq 3 = G$.

(B) In-regime terms $s \leq r$, $t \leq r$. Since $Q_r = \sum_{s,t \leq r} T(r,s,t)$ exactly, (8.3) gives $\sum_{s,t\leq r} T(n,k,l) - Q_r T(a,b,c) = T(a,b,c) \sum_{s,t \leq r} T(r,s,t)[\Pi \hat{G} - 1]$, so it suffices termwise that $vT + J + v_p(\Pi\hat{G} - 1) \geq G$ with $J := v_p T(r,s,t)$. Note $v_p(\Pi\hat{G}-1) \geq \min(v_p(\Pi - 1), v_p(\Pi)+1) \geq \min(v_p(\Pi), 0)$, and $G \leq 3$, $vT \geq s_a$. Running the seven patterns (8.4):

(e_1,e_2,e_3,e_4)	Π	$J \geq$	$v_p(\Pi\hat{G}-1) \geq$	total $\geq$
$(0,0,0,0)$	1	0	1	$s_a + 1$ ✓
$(0,0,1,0)$	$a{+}b{+}c{+}1 \in \mathbb{Z}$	1	0	$s_a + 1$ ✓
$(1,0,1,0), (0,1,1,0)$	$\in \mathbb{Z}$	2	0	$s_a + 2$ ✓
$(1,1,1,0)$	$\in \mathbb{Z}$	3	0	3 ✓
$(1,1,1,1)$	$\frac{(a+b+1)(a+c+1)(a+b+c+1)}{b+c+1}$	2	see below	✓
$(1,1,2,1)$	$\frac{(a+b+1)(a+c+1)\binom{a+b+c+2}{2}}{b+c+1}$	3	see below	✓

(the J column: $e_1 = 1 \Rightarrow \binom{r+s}{r}$ carries; $e_2 = 1 \Rightarrow \binom{r+t}{r}$ carries; if $e_4 = 0$ and $e_3 \geq 1$ then $\binom{r+s+t}{r}$ carries at position 0; if $e_4 = 1$ and $e_3 = 2$ then $r + \sigma \geq p$ with $\sigma = s+t-p$, so $\binom{r+s+t}{r}$ carries).

For the two $e_4 = 1$ patterns put $m := v_p(b+c+1)$ and $\tau := v_p(a)$. Then $a+b+c+1 = a+(b+c+1)$ has $v_p \geq \min(\tau, m)$, so $v_p(\Pi) \geq \min(\tau,m) - m$ and therefore $v_p(\Pi\hat{G}-1) \geq \min(\tau,m)-m$ (which is ≤ 0, so the bound is valid whether or not $v_p(\Pi) \geq 0$).

Case $\tau \geq m$. Then $v_p(\Pi\hat{G} - 1) \geq 0$ and the totals are $vT + 2 \geq 2$ resp. $vT + 3 \geq 3$; since $G \leq 3$ and, for $(1,1,1,1)$, $vT + 2 \geq s_a + 2 \geq \min(s_a, 2) + 1 = G$.

Case $\tau < m$ (so $m \geq 1$). Then $b + c \equiv -1 \pmod{p^m}$, i.e. the digits of $b+c$ in positions $0, \ldots, m-1$ are all $p-1$, while a has digits 0 in positions $0, \ldots, \tau - 1$ and $a_\tau \geq 1$. In the addition $a+(b+c)$: no carry at positions $0,\dots,\tau-1$ $(0+(p-1)=p-1)$, a carry at position τ $(a_\tau+(p-1)\ge p)$, and a carry at each of $\tau+1,\dots,m-1$ $(a_i+(p-1)+1\ge p)$. Hence

$$E := v_p\tbinom{a+b+c}{a} \ge m-\tau, \qquad \text{so} \qquad vT \ge (m-\tau)+A+A'+2D+2D',$$

with $A=v_p\binom{a+b}{a}\ge\alpha$, $A'=v_p\binom{a+c}{a}\ge\gamma$, $D=v_p\binom{a}{b}$, $D'=v_p\binom{a}{c}$. For $(1,1,1,1)$ the total is $\ge vT+2+(\tau-m)\ge 2+A+A'+2D+2D'$: if $s_a\le 1$ then $G\le 2$; if $s_a\ge 2$ then as in (A) $\alpha=1$ or $\gamma=1$, so $A+A'\ge 1$ and the total is $\ge 3=G$. For $(1,1,2,1)$ the total is $\ge vT+3+(\tau-m)\ge 3\ge G$. ☐

Remark 8.14. Lemma 8.4 is *not* used. The single-digit Lemma 8.8 needs it because it aims at relative precision p^2, so its first-order term must be (b,c)-independent; at relative precision p^1 the first-order term is simply absorbed, and the whole first-order analysis — Lemmas 8.7, 8.6, 8.4 and the Ψ_a supercongruence — disappears. That is the structural reason the multi-digit statement is *easier* than the single-digit one.

Evidence 8.15. [VERIFIED] 0 failures in 111 963 cells: $p=5,7,11,13$, multi-digit $a\le 30/22/15/13$, all r, all (b,c); sharp (min slack 0).

9. The digit-scaled induction

9.1. The inductive statement.

Definition 9.1 ((IND))**.** Let $p\ge 5$. For $L\ge 0$ let $\mathcal{I}(L)$ be the assertion

$$\text{for every } m\ge 1 \text{ with } \lfloor\log_p m\rfloor\le L: \quad v_p(W_m)\ \ge\ -5\lfloor\log_p m\rfloor, \qquad W_m=P_m-H_m^{(5)}Q_m.$$

Note $v_p(H_m^{(5)}Q_m)\ge -5\lfloor\log_p m\rfloor$ always ($Q_m\in\mathbb{Z}$, Lemma 7.1), so $\mathcal{I}(L)$ is *equivalent* to $v_p(P_m)\ge -5\lfloor\log_p m\rfloor$ on the same range.

Remark 9.2 (why this form)**.** The naïve per-digit ratio statement $p^5P_n/Q_n\equiv P_a/Q_a \pmod p$ fails multi-digit: the depth of the single-step ratio is $5-2s$, negative for $s\ge 3$. (IND) is scaled: it never divides by Q, never renormalises, and its per-level target is an inequality on $v_p(W)$, not a congruence.

9.2. **The step and its ledger.** Let $L\ge 1$, $\lfloor\log_p n\rfloor=L$, $n=ap+r$ with $0\le r<p$, $a=\lfloor n/p\rfloor$, so $\lfloor\log_p a\rfloor=L-1$. Group the cells of W_n by $(b,c)=(\lfloor k/p\rfloor,\lfloor l/p\rfloor)$, $0\le b,c\le a$, and split

$$p^5W_n-Q_rW_a=(\mathrm{I})+(\mathrm{II}),\qquad \begin{aligned}(\mathrm{II}) &:= \sum_{b,c} v_5(a,b,c)\big[\mathcal{T}(b,c)-Q_rT(a,b,c)\big],\\ (\mathrm{I}) &:= \sum_{k,l} T(n,k,l)\,\mathcal{E}(n,k,l),\quad \mathcal{E}:=p^5v_5(n,k,l)-v_5(a,b,c).\end{aligned} \tag{9.1}$$

If $v_p(\mathrm{I}),v_p(\mathrm{II})\ge -5(L-1)$ then, using $\mathcal{I}(L-1)$ for $v_p(Q_rW_a)\ge v_p(W_a)\ge -5(L-1)$,

$$v_p(p^5W_n)\ge -5(L-1) \quad\Longrightarrow\quad v_p(W_n)\ge -5-5(L-1)=-5L,$$

which is $\mathcal{I}(L)$.

Proposition 9.3 (Ledger for the fibre term)**.** *[PROVED] Per cell, per digit level:*

provided: $v_pv_5(a,b,c)$	$\ge -5(L-1)-1-\min(s_a,2)$	*Corollary 7.13*
provided: $v_p(\mathcal{T}-Q_rT)$	$\ge 1+\min(s_a,2)$	*Lemma 8.13*
consumed: target	$\ge -5(L-1)$	*(IND) step*
slack	***exactly*** 0	*both inputs sharp*

So the two lemmas match to the digit: nothing is wasted and nothing is missing in (II). Note what is *not* needed: $\Lambda = Q_n/Q_a$, the hypothesis $p \nmid Q_a$, the supercongruence of Corollary 8.9, and Lemma 8.4. The exceptional primes (where $v_p(Q_n)$ can reach 7 at $p = 7$) are irrelevant to (IND).

9.3. **The descent term, in-regime.** By Lemma 7.1, $p^m H_N^{(m)} = H_{\lfloor N/p \rfloor}^{(m)} + p^m \Sigma_0^{(m)}(N)$. Hence, letter by letter,

$$\begin{aligned}
p^m A_m^{(n)}(k) &= A_m^{(a)}(b) + e_1(a+b+1)^{-m} + p^m \sigma^A, \qquad\qquad \sigma^A := \Sigma_0(n+k) - \Sigma_0(k) \in \mathbb{Z}_p,\\
p^m B_m^{(n)}(k) &= B_m^{(a)}(b) - [\![s > r]\!](a-b)^{-m} + p^m \sigma^B,\\
p^m C_m^{(n)} &= C_m^{(a)}(b,c) + [e_3\text{-terms}] - [e_4\text{-terms}] + p^m \sigma^C,\\
p^m N_m^{(n)} &= N_m^{(a)} + p^m \sigma^N.
\end{aligned} \tag{9.2}$$

Definition 9.4 (regimes)**.** A cell is *in-regime* if $s \le r$, $t \le r$ and $e_1 = e_2 = e_3 = e_4 = 0$; otherwise *off-regime.*

In-regime all the mismatch terms vanish and $p^m X_m^{(n)} = X_m^{(a)} + p^m \sigma^X$ exactly. Expanding the product,

$$\mathcal{E} = \sum_{S \neq \emptyset} p^{\mathrm{wt}(S)} \sigma_S \Psi_S, \qquad \sigma_S = \prod_{X \in S} \sigma^X \in \mathbb{Z}_p, \qquad \Psi_S := \sum_{\mathcal{M} \supseteq S} c_{\mathcal{M}} \prod_{X \in \mathcal{M} \setminus S} X^{(a)},$$

a level-a form of weight $5 - \mathrm{wt}(S)$.

Proposition 9.5 (In-regime descent costs nothing)**.** *[PROVED] For every in-regime cell and every digit level, $v_p(T\mathcal{E}) \ge -5(L-1)$.*

Proof. By the argument of Proposition 7.12 applied to Ψ_S — its u-order is $(5 - \mathrm{wt}\, S)M_a - \delta'$ and its top non-vanishing δ' obeys $\mathrm{wt}(S) + \delta' \ge 5 - J(\pi)$ because $\Phi_\pi(S \cup S') = 0$ for $\mathrm{wt}(S \cup S') \le 4 - J(\pi)$ — one gets

$$v_p\big(p^{\mathrm{wt}\, S} \sigma_S \Psi_S\big) \ \ge\ 5 - J(\pi) - (5 - \mathrm{wt}\, S)M_a$$

whenever $\mathrm{wt}(S) \le 5 - J(\pi)$; if $\mathrm{wt}(S) > 5 - J(\pi)$ then $\mathrm{wt}(S) \ge 6 - J(\pi) \ge 3$, so $\mathrm{wt}(S)M_a \ge 3 \ge J(\pi)$ and the requirement below holds trivially. In the fully in-regime case Lemma 8.12(C2) is an equality at every slot, so $v_p T(n,k,l) = v_p T(a,b,c) = vT$. The requirement $v_p(T\mathcal{E}) \ge -5(M_a - 1)$ becomes $vT \ge J(\pi) - \mathrm{wt}(S)M_a$, and with $J(\pi) = 1 + \min(s_a, 2) \le 3$, $\mathrm{wt}(S) \ge 1$, $vT \ge s_a$:

M_a	$\mathrm{wt}(S)$	needed $vT \ge$	
1	1	$\min(s_a, 2)$	have $vT \ge s_a$ ✓
1	2	$\min(s_a, 2) - 1$	✓
1	≥ 3	≤ 0	✓
≥ 2	≥ 1	≤ 1, $= 1$ only when $J = 3$ i.e. $s_a \ge 2$	have $vT \ge 2$ ✓

□

10. The descent term off-regime

Fix $p \ge 5$ and $n \ge p$; put $L = \lfloor \log_p n \rfloor \ge 1$, $M = L + 1$, $P = p^M$, $a = \lfloor n/p \rfloor$, $r = n - ap$, $P_a = p^L$ (so $p^{L-1} \le a < P_a$ and $\lfloor \log_p a \rfloor = L - 1$). For a cell write $b = \lfloor k/p \rfloor$, $s = k - bp$, $c = \lfloor l/p \rfloor$, $t = l - cp$, and set $\zeta := e_3 - e_4$. Let $(\alpha, \gamma, \kappa, \theta)$ be the level-n pattern (with P) and $(\alpha_a, \gamma_a, \kappa_a, \theta_a)$ the level-a pattern (with P_a); $s_n := \alpha + \gamma + \kappa$, $s_a := \alpha_a + \gamma_a + \kappa_a$.

10.1. The digit dictionary.

Lemma 10.1 (Digit dictionary). *[PROVED]*

(1) $\lfloor (n+k)/p \rfloor = a+b+e_1$, $\lfloor (n+l)/p \rfloor = a+c+e_2$, $\lfloor (n+k+l)/p \rfloor = a+b+c+e_3$, $\lfloor (k+l)/p \rfloor = b+c+e_4$, $\lfloor (n-k)/p \rfloor = a-b-[\![s>r]\!]$, $\lfloor n/p \rfloor = a$.
(2) $0 \le \zeta = e_3 - e_4 \le 1$, *and* $\zeta = 1$ *iff the addition* $n+(k+l)$ *carries out of position* 0.
(3) $\alpha = [\![a+b+e_1 \ge P_a]\!] \ge \alpha_a$ *and* $\gamma = [\![a+c+e_2 \ge P_a]\!] \ge \gamma_a$.
(4) $\varepsilon := \lfloor (k+l)/P \rfloor = \lfloor (b+c+e_4)/P_a \rfloor = \varepsilon_a + e_4[\![b+c = P_a - 1]\!]$, *and* $\kappa = [\![a+b+c+e_3 \ge (\varepsilon+1)P_a]\!]$.
(5) $\kappa \ge \kappa_a$ unless $e_4 = 1$ *and* $b+c = P_a - 1$*; hence in all cases* $s_n \ge s_a - [\![e_4 = 1]\!]$.
(6) *In-regime,* $s_n = s_a$ *and* $v_pT(n,k,l) = v_pT(a,b,c)$.

Proof. (1) is the definition of the four indicators ($n = ap+r$, $k = bp+s$, $l = cp+t$, and $n-k = (a-b-[\![s>r]\!])p + ((r-s) \bmod p)$). (2) $e_4 \le e_3$ because $r+s+t \ge s+t$, and $e_3 \le e_4+1$ because $r<p$. The digit of $k+l$ in position 0 is $s+t-e_4p$, so $n+(k+l)$ carries out of position 0 iff $r+s+t-e_4p \ge p$ iff $e_3 \ge e_4+1$. (3) $n+k \ge P = p\,P_a$ iff $\lfloor (n+k)/p \rfloor \ge P_a$ iff $a+b+e_1 \ge P_a$, and $a+b+e_1 \ge a+b$. (4) $b+c \le 2a \le 2P_a - 2$, so the interval $[1, b+c+1]$ shifted appropriately contains exactly one multiple of P_a, namely P_a itself; hence $\lfloor (b+c+1)/P_a \rfloor - \lfloor (b+c)/P_a \rfloor = [\![b+c = P_a - 1]\!]$. The formula for κ is $n+k+l \ge (\varepsilon+1)P \iff \lfloor (n+k+l)/p \rfloor \ge (\varepsilon+1)P_a$. (5) If $\varepsilon = \varepsilon_a$ then $\kappa = [\![a+b+c+e_3 \ge (\varepsilon_a+1)P_a]\!] \ge [\![a+b+c \ge (\varepsilon_a+1)P_a]\!] = \kappa_a$. Otherwise ε exceeds ε_a, which by (4) forces $e_4 = 1$ and $b+c = P_a - 1$; and always $\kappa \ge \kappa_a - 1$ since both are 0/1. Adding (3) gives the display. (6) In-regime all four e_i vanish, so (3),(4) give $\alpha = \alpha_a$, $\gamma = \gamma_a$, $\varepsilon = \varepsilon_a$, $\kappa = \kappa_a$. □

Item (5) is sharp: $e_4 = 1$ with $b+c = P_a - 1$ really does drop κ from 1 to 0. Example: $p = 5$, $n = 101$, $(k,l) = (63,63)$, so $L = 2$, $a = 20$, $r = 1$, $b = c = 12$, $s = t = 3$, $b+c = 24 = P_a - 1$, $(e_3, e_4) = (1,1)$, level-a pattern $(1,1,1,1)$ with $s_a = 3$, level-n pattern $(1,1,0,1)$ with $s_n = 2$; there $B = 4$ and $v_pT = 10 \ge 1 + \max(s_n, s_a) = 4$.

10.2. The descent Kummer lemma.

Define the *bottom-carry count*

$$B := e_1 + e_2 + 2[\![s>r]\!] + 2[\![t>r]\!] + \zeta. \tag{10.1}$$

Lemma 10.2 (Descent Kummer). *[PROVED] Let* $p \ge 5$*,* $n \ge p$*,* $L = \lfloor \log_p n \rfloor \ge 1$*, and let* (k,l) *be off-regime. Then*

$$\text{(DK1)}\quad B \ge 1, \text{ and if } e_4 = 1 \text{ then } B \ge 2;$$

$$\text{(DK2)}\quad v_pT(n,k,l) \ge s_n + B; \qquad \text{(DK3)}\quad v_pT(n,k,l) \ge 1 + \max(s_n, s_a).$$

Proof. (DK1). Three exhaustive cases. If $s>r$ or $t>r$ then $B \ge 2$ outright, from the term $2[\![s>r]\!]$ resp. $2[\![t>r]\!]$. If $s \le r$, $t \le r$ and $e_4 = 1$ then $r+s \ge s+t \ge p$ and $r+t \ge s+t \ge p$, so $e_1 = e_2 = 1$ and again $B \ge 2$. If $s \le r$, $t \le r$ and $e_4 = 0$ then $\zeta = e_3$, and off-regime forces $e_1 + e_2 + e_3 \ge 1$; each summand contributes 1 to B (for e_3 through $\zeta = e_3$), so $B \ge 1$. Since $e_4 = 1$ occurs only in the first two cases, $e_4 = 1 \Rightarrow B \ge 2$.

(DK2). Write $T = \binom{n+k}{n}\binom{n}{k}^2\binom{n+l}{n}\binom{n}{l}^2\binom{n+k+l}{n}$ and let $V_1, \dots, V_5$ be the v_p of the five binomials, so $v_pT = V_1 + V_2 + 2V_3 + 2V_4 + V_5$. By Kummer each V_i is a count of carries in one addition. In each addition we locate the *bottom* carry (out of base-p position 0) and, where the level-n pattern demands one, the *top* carry (out of position $M-1 = L$); the two positions are distinct precisely because $L \ge 1$ — this is the one and only place the hypothesis is used.

- V_1 counts the carries of $n+k$. Position 0: carries iff $r+s \ge p$, i.e. iff $e_1 = 1$. Position L: since $n, k < P$, it carries iff $n+k \ge P$, i.e. iff $\alpha = 1$ (Lemma 7.4). Hence $V_1 \ge \alpha + e_1$; likewise $V_2 \ge \gamma + e_2$.

- V_3 counts the carries of $k+(n-k)$. In position 0 the digits are s and $(r-s) \bmod p$, summing to $r + p[\![s > r]\!]$, so it carries iff $s > r$. Hence $V_3 \geq [\![s > r]\!]$, likewise $V_4 \geq [\![t > r]\!]$.
- V_5 counts the carries of $n + (k + l)$. Position 0: carries iff $\zeta = 1$ (Lemma 10.1(2)). Position L: writing $k + l = \varepsilon P + \rho$ with $0 \leq \rho < P$, the low M digits of the addition sum to $n + \rho$, so it carries out of position $M - 1 = L$ iff $n + \rho \geq P$, i.e. iff $\kappa = 1$. Hence $V_5 \geq \kappa + \zeta$.

Summing with multiplicities $(1, 1, 2, 2, 1)$, $v_p T \geq (\alpha + \gamma + \kappa) + (e_1 + e_2 + 2[\![s > r]\!] + 2[\![t > r]\!] + \zeta) = s_n + B$.

(DK3). By (DK2) and (DK1), $v_p T \geq s_n + 1$. For the level-a half, Lemma 10.1(5) gives $s_n \geq s_a - [\![e_4 = 1]\!]$ and (DK1) gives $B \geq 1 + [\![e_4 = 1]\!]$; hence $v_p T \geq s_n + B \geq (s_a - [\![e_4 = 1]\!]) + (1 + [\![e_4 = 1]\!]) = s_a + 1$. □

The mechanism is one line: *off-regime means a carry in base-p position* 0*, the depth-pattern indicators live in position $L \geq 1$, and Kummer counts both.*

Theorem 10.3 (Off-regime descent)**.** *[PROVED] Let $p \geq 5$, $n \geq p$, $L = \lfloor \log_p n \rfloor \geq 1$, $a = \lfloor n/p \rfloor$, and let (k, l) with $0 \leq k, l \leq n$ be off-regime. Then, with $\mathcal{E} = p^5 v_5(n, k, l) - v_5(a, b, c)$,*

$$v_p\big(T(n, k, l)\, \mathcal{E}(n, k, l)\big) \;\geq\; -5(L - 1).$$

Proof. Corollary 7.13 at level n ($\lfloor \log_p n \rfloor = L$, pattern sum s_n) gives $v_p v_5(n, k, l) \geq -5L - 1 - \min(s_n, 2)$, hence $v_p(p^5 v_5(n, k, l)) \geq -5(L - 1) - 1 - \min(s_n, 2)$. The same corollary at level a ($\lfloor \log_p a \rfloor = L - 1$, pattern sum s_a) gives $v_p v_5(a, b, c) \geq -5(L - 1) - 1 - \min(s_a, 2)$. Therefore

$$v_p(\mathcal{E}) \;\geq\; -5(L - 1) - 1 - \max\big(\min(s_n, 2), \min(s_a, 2)\big) \;\geq\; -5(L - 1) - 1 - \max(s_n, s_a),$$

while Lemma 10.2 gives $v_p T(n, k, l) \geq 1 + \max(s_n, s_a)$. Add. □

Corollary 10.4 (The descent term)**.** *[PROVED] For every $p \geq 5$ and every n with $L = \lfloor \log_p n \rfloor \geq 1$, $v_p((\mathrm{I})) \geq -5(L - 1)$, cell by cell: off-regime by Theorem 10.3, in-regime by Proposition 9.5.*

Corollary 10.5 (The induction step)**.** *[PROVED] $\mathcal{I}(L - 1) \Rightarrow \mathcal{I}(L)$ for every $L \geq 1$.*

Proof. In (9.1), term (II) is bounded by Proposition 9.3 and term (I) by Corollary 10.4. □

Proof of Theorem 1.3*.* $\mathcal{I}(0)$ is Theorem 1.5, proved in §11; the step is Corollary 10.5. Hence $v_p(W_m) \geq -5\lfloor \log_p m \rfloor$ for all $m \geq 1$, and therefore $\mathrm{ord}_p(P_m) \geq -5\lfloor \log_p m \rfloor$. □

Remark 10.6 (scope)**.** Theorem 10.3 is uniform in the level: nothing in §10 distinguishes $a < p$ from $a \geq p$, and the hypothesis $L \geq 1$ enters exactly once, in (DK2), to know that digit positions 0 and L are distinct. At $L = 0$ there is no descent to make. It uses no $p \nmid Q_a$, no Λ, no supercongruence, no Lemma 8.4, no Lemma 5.3, and — in itself — no Wolstenholme: $p \geq 5$ is inherited only from Corollary 7.13. And it consumes w_5 only through Corollary 7.13, so it holds verbatim for any point of the depth-conditioned family, in particular for w_5^{I}.

10.3. **Why the expected route fails.** It is worth recording that the natural letter-wise route to Theorem 10.3 does not exist, since the failure is instructive. That route reads off from (9.2) the mismatch $e_1(a + b + 1)^{-m}$ and observes that a mismatch *forces carries*:

$$v_p\tbinom{n+k}{n} = v_p\tbinom{a+b}{a} + e_1(1 + \lambda_1), \qquad \lambda_1 := v_p(a + b + 1) \tag{10.2}$$

(correct, and a consequence of Lemma 8.12(C1) with $z(a + b) = \lambda_1$). *But at weight* 5 *the trade loses.* Expanding $\mathcal{E}$ letter-wise as $\sum_{S \neq \emptyset} (\prod_{X \in S} D_X) \Psi_S$ with D_X the per-letter mismatch, the single term $S = \{A_5(k)\}$ contributes $c\,(a + b + 1)^{-5}$ of valuation $-5\lambda_1$, where c is the

w_5-coefficient of the symmetrised $A_5(k)+A_5(l)$ monomial ($=3830046703/48$ for w_5^{allp}, a p-unit), against a total Kummer gain of only $1+\lambda_1$ in the whole of T. Cell-wise:

p	n	(k,l)	(a,b)	λ_1	v_pT	letter-wise $vT-5\lambda_1$	target $-5(L-1)$
5	19	$(6,0)$	$(3,1)$	1	4	$\mathbf{-1}$	0
5	19	$(6,19)$	$(3,1)$	1	4	$\mathbf{-1}$	0
5	99	$(26,30)$	$(19,5)$	2	8	-2	-5
7	244	$(100,50)$	$(34,14)$	2	8	-2	-5

At $p=5$, $n=19$, $(k,l)=(6,0)$ the letter-wise ledger is short by exactly one power, and no refinement of the carry bookkeeping can repair it because (10.2) is an *equality.* (The true value there is $v_p(T\mathcal{E})=1$, comfortably above the target.) What actually happens is that the pole is not present in $\mathcal{E}$ at all: the mismatch poles of the individual letters cancel among themselves *inside* v_5, because the (DEPTH) conditions annihilate its top u-coefficients at level n. Hence the correct instrument is Corollary 7.13, not a letter-wise ledger — and the only thing left to check is that T pays for the *two* depth caps, which is Lemma 10.2. This is the weight-5 shadow of the fact that the $a<p$ analysis could afford the trade only up to weight 3: Lemma 5.3 supplies $v_pT\geq 4$, which covers A_3 but not A_5.

Evidence 10.7. All exact; p-adic valuations computed with tracked precision, the evaluator cross-checked against exact `Fraction` evaluation of the 178-term w_5^{allp} on 1290 cells with 0 mismatches of valuation or unit.

sweep	range	cells	failures
Lem. 10.2 (DK1)–(DK3), Lem. 10.1(2),(5)	931 levels, $p\leq 31$, $L=1,2,3$	142 820 576	**0**
the same, at $L=4$ (and $L=3$)	13 levels, $p\leq 11$, $n\leq 3124$	45 533 157	**0**
slot-wise inequalities behind (DK2)	$p\leq 31$, $L=1,2,3$	11 096 075	**0**
Thm 10.3, exact p-adic $\mathcal{E}$	$p\leq 13$, $L=1$	174 527	**0**
Thm 10.3, exact p-adic $\mathcal{E}$	$p\leq 13$, $L=2$	530 900	**0**
Thm 10.3, exact p-adic $\mathcal{E}$	$p\leq 7$, $L=3$	934 656	**0**
whole term (I), both regimes, $+\ v_p(\sum T\mathcal{E})$	$p\leq 13$, $L=2,3$	15 553 + 345 334	**0**
Cor. 7.13 and Thm 10.3 for w_5^{I}	$p\leq 13$, $L\leq 2$	15 228 + 43 213	**0**

The first two rows are decisive: together they check the criterion on *every* cell of 944 different levels — 188 353 733 off-regime cells, 0 failures, no sampling inside a level. *Control:* in-regime, $B=0$ and $s_n=s_a$ at every one of the 13 082 671 in-regime cells of the sweep (as Lemma 10.1(6) predicts) and the criterion $v_pT\geq\max(J(\pi_n),J(\pi_a))$ *fails* at 134 964 of them — by design. So off-regime is not a convenience of the proof: it is exactly the hypothesis that buys the extra power of p. *Sharpness:* the slack is 0 at 41 284 cells, so neither Lemma 10.2 nor Theorem 10.3 can be weakened; and the true quantity $v_p(T\mathcal{E})+5(L-1)$ attains 0 (e.g. $p=13$, $n=153$, $(k,l)=(1,41)$: $v_pT=3$, $v_p\mathcal{E}=-3$, target 0).

Remark 10.8 (a formalisation target)**.** Lemmas 10.1 and 10.2 are pure base-p combinatorics over $\mathbb{N}$ — no harmonic numbers, no representative, no analysis. They are the most formalisation-ready statements in the chain after Theorem 1.4 (already [LEAN]), and formalising them would make the Kummer half of the induction step machine-checked, leaving Corollary 7.13 as the analytic input.

11. The nucleus: the base case

By §7.5 the base case is not cell-wise for any p-independent representative, and by §12 below no summand-side identity can supply it. What does work is a split of the range $n<p$ at the midpoint. Throughout this section $w_5=w_5^{\mathrm{I}}$ and $v_5=w_5^{\mathrm{I}}-H_n^{(5)}$.

11.1. **One band.**

Theorem 11.1 (Partial closure)**.** *[PROVED] given* (T1-top) *for w_5^{I} and the [CERT] linear-algebra certificate. For every prime $p \geq 5$ and every $n < p$, every cell outside the band*

$$\mathrm{III} = \{(k,l) : \ k,l \geq q := p-n, \quad p \leq k+l < p+q\} \tag{11.1}$$

contributes a p-integral summand to $P_n = \sum_{k,l} T(n,k,l) w_5^{\mathrm{I}}(n,k,l)$. Consequently

$$\text{(BASE)} \quad \iff \quad v_p\Big(\sum_{(k,l)\in\mathrm{III}} T(n,k,l)\, v_5(n,k,l)\Big) \ \geq \ 0.$$

The conditions cutting out w_5^{I} are the full cell-wise cap $d_5 \leq v_p T$ on every pole pattern *except* the one giving III, where the cap of (7.3) is kept. They are p-independent, so their sufficiency is the pole calculus of §7; the joint system has rank(cond) = 111, rank(joint) = 342 and is consistent, against rank(cond) = 123, rank(joint) = 342 and *inconsistent* with no exemption. So *the entire residue of the base case is one order of pole on one pattern group*, and the harmonic alphabet already supplies everything else.

This is a strict sharpening: the previously known reduction was $2\Sigma_{\mathrm{I}} + \Sigma_{\mathrm{III}} \equiv 0$ over two regions, and the attained deficit cell $\left(\frac{p+1}{2}, 0, \frac{p-1}{2}\right)$ has $k = 0 < q$, so it lies in region I: *the representative w_5^{I} genuinely moves the deficit off the cell where it was found.*

Evidence 11.2. [VERIFIED], exact `Fraction` arithmetic: (V1) $P_n = \sum T\, w_5^{\mathrm{I}}$ exactly over $\mathbb{Q}$ for $n = 1, \ldots, 20$ (and $n = 1, \ldots, 16$ for the all-primes form), 0 mismatches; (V2) $v_p(Tv_5) \geq 0$ at every cell outside III over $p = 5, 7, 11, 13, 17, 19, 23$: 10 092 cells (1 025 exempt), 0 violations; and 5 769 cells, 0 violations, for the all-primes form; (V3) the reduction itself, 0 failures.

11.2. **Below the midpoint the band is empty.**

Lemma 11.3. *[PROVED] If $n \leq (p-1)/2$ then $\mathrm{III} = \emptyset$, hence $v_p(P_n) \geq 0$.*

Proof. III requires $k + l \geq p$ with $k, l \leq n$, so $2n \geq p$; but $2n \leq p - 1$. □

So half of the base case is free, and the deficit can first appear only at $n = (p+1)/2$ — exactly where §7.5 located the attained deficit cell. The two localisations agree for a structural reason: the midpoint singularity of L_{BZ} ($2n + 5 \equiv 0$) and the midpoint appearance of III ($2n \geq p$) are the same phenomenon seen from the recurrence side and from the summand side.

11.3. **At the midpoint the band is three cells.** Let $n = (p+1)/2$, $q = p - n = (p-1)/2$.

Lemma 11.4. *[PROVED]* $\mathrm{III} = \{(q, q+1), (q+1, q), (q+1, q+1)\} = \{(n-1, n), (n, n-1), (n, n)\}$.

Proof. $k, l \geq q$ and $k, l \leq n = q + 1$ force $k, l \in \{q, q+1\}$. Then $k + l \in \{p-1, p, p+1\}$; the condition $k + l \geq p$ excludes (q, q), and $k + l < p + q$ holds for the other three since $p + 1 < p + (p-1)/2$ for $p \geq 5$. □

Lemma 11.5. *[PROVED] At $n = (p+1)/2$, $p \geq 5$: $v_p(T) = 2$ at all three cells and*

$$T(n, n-1, n)/p^2 \equiv T(n, n, n-1)/p^2 \equiv 2, \qquad T(n,n,n)/p^2 \equiv 24 \pmod{p}.$$

Proof. Write $\epsilon := (-1)^{(p+1)/2}$, $q = (p-1)/2$; Wilson's theorem gives $(q!)^2 \equiv \epsilon$.

Cell $B = (n,n)$. $T = \binom{p+1}{n}^2\binom{3n}{n}$. Kummer: $n + n = p + 1$ has exactly one carry in base p, so $v_p\binom{p+1}{n} = 1$; $n + (p+1)$ has none, so $v_p\binom{3n}{n} = 0$; thus $v_p(T) = 2$. As $p + 1 - n = n$, $\binom{p+1}{n}/p = (p+1)(p-1)!/(n!)^2 \equiv -1/(n!)^2$; with $n! = (q+1)q!$ and $q + 1 \equiv 1/2$, $(n!)^2 \equiv \epsilon/4$, so $\binom{p+1}{n}/p \equiv -4\epsilon$ and $(\binom{p+1}{n}/p)^2 \equiv 16$. Lucas on $3n = p + (p+3)/2$, $n = (p+1)/2$ gives $\binom{3n}{n} \equiv \binom{(p+3)/2}{(p+1)/2} = (p+3)/2 \equiv 3/2$. Hence $T/p^2 \equiv 16 \cdot (3/2) = 24$.

Cell $A = (n-1, n) = (q, q+1)$. $\binom{n+k}{n} = \binom{p}{n}$ with $v_p = 1$ and $\binom{p}{n}/p = (p-1)!/(n!q!) \equiv -1/((q+1)(q!)^2) \equiv -2\epsilon$; $\binom{n}{k} = \binom{q+1}{q} = q+1 \equiv 1/2$; $\binom{n+l}{n} = \binom{p+1}{n}$ with $\binom{p+1}{n}/p \equiv -4\epsilon$; $\binom{n}{l} = 1$; $\binom{n+k+l}{n} = \binom{(3p+1)/2}{(p+1)/2} \equiv \binom{1}{0}\binom{(p+1)/2}{(p+1)/2} = 1$ by Lucas. So $v_p(T) = 2$ and $T/p^2 \equiv (-2\epsilon)(1/2)^2(-4\epsilon) = 2\epsilon^2 = 2$. □

At $n = (p+1)/2$ every letter argument is $< 2p$, so each letter contains at most one multiple of p and its u-expansion is exact and elementary. With $h_r := H_q^{(r)} \bmod p$, $s_r := \sum_{j=1}^{p-1} j^{-r} \bmod p$, $(q+1)^{-1} \equiv 2$, $(q+2)^{-1} \equiv 2/3$:

letter	cell $A = (q, q+1)$	cell $B = (q+1, q+1)$
$A_r(k)$	$u^r + s_r - h_r$	$u^r + s_r + 1 - h_r - 2^r$
$B_r(k)$	$1 - h_r$	$-h_r - 2^r$
$A_r(l)$	$u^r + s_r + 1 - h_r - 2^r$	$u^r + s_r + 1 - h_r - 2^r$
$B_r(l)$	$-h_r - 2^r$	$-h_r - 2^r$
C_r (no pole)	$h_r + 2^r$	$h_r + 2^r + (2/3)^r - 1$
N_r	$h_r + 2^r$	$h_r + 2^r$

Lemma 11.6. *[PROVED] (finite exact symbolic computation over $\mathbb{Q}[h_1, \ldots, h_5, s_1, \ldots, s_5]$). For every prime $p \geq 5$, substituting the table above into the* 207 *monomials of w_5^{I} and extracting the u^j coefficients gives*

$$K_4 = K_5 = 0 \quad \textit{at both cells}, \qquad K_3(A) = 3 - \tfrac{s_2}{2}, \qquad K_3(B) = -\tfrac{1}{2} - \tfrac{s_2}{2}.$$

Both are free of $h_1, \ldots, h_5$ and of s_1, s_3, s_4, s_5*: the entire harmonic content of the u^3 coefficient cancels.*

Only letters of weight ≤ 3 can reach K_3 — the pole part must have total weight 3 and the remainder weight 2 — which is why s_4, s_5 cannot appear and the lemma is valid down to $p = 5$. That $K_4 = K_5 = 0$ is the Lemma-F cap $d_5 \leq 3$ being met.

Theorem 11.7 (The nucleus, dissolved)**.** *[PROVED] For every prime $p \geq 5$, at $n = (p+1)/2$,*

$$\sum_{(k,l)\in \mathrm{III}} \frac{T}{p^2} K_3 \;\equiv\; 2\Big(3 - \frac{s_2}{2}\Big) + 2\Big(3 - \frac{s_2}{2}\Big) + 24\Big(-\frac{1}{2} - \frac{s_2}{2}\Big) \;=\; -14\, s_2 \;\equiv\; 0 \pmod{p},$$

because $s_2 = \sum_{j=1}^{p-1} j^{-2} \equiv \sum_{j=1}^{p-1} j^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0$ for $p \geq 5$. Since $K_0, K_1, K_2 \in \mathbb{Z}_p$ and $T/p^2 \in \mathbb{Z}$, this gives $v_p(\sum_{\mathrm{III}} T v_5) \geq 0$, hence by Theorem 11.1 $v_p(P_{(p+1)/2}) \geq 0$.

The whole nucleus consumes exactly one classical fact, $\sum_{j<p} j^{-2} \equiv 0$, i.e. the same $p \geq 5$ hypothesis Wolstenholme imposes elsewhere. No Fermat quotient, no Bernoulli number.

Remark 11.8 (the statement is not vacuous)**.** The same extraction on w_5^{allp} gives an h_1-dependent K_3 with non-vanishing corner combination. The content of Theorem 11.7 lives precisely in the extra depth conditions defining w_5^{I} — exactly as the wall theorem of §12 predicts: the value cannot be moved by summand-side identities, only by pinning the representative. *The wall is respected, and the door it points at is the representative.*

Evidence 11.9. [VERIFIED] 0 failures: $v_p(T) = 2$ and $T/p^2 \bmod p = 2, 2, 24$ at every prime $p \in \{5, \ldots, 31\}$; and, by exact `Fraction` arithmetic through an independent evaluator, at every prime $5 \leq p \leq 139$ the three corner residues $p\,T v_5 \bmod p$ are exactly $(6, 6, -12) \bmod p$ and sum to 0 — 32/32 primes clean. Cell-wise integrality below the midpoint: 1 588 cells over all levels $n < (p+1)/2$, $p \leq 23$, 0 violations, plus the top below-midpoint level $n = (p-1)/2$ at $p = 29, 31, 37$ (842 cells, $|\mathrm{III}| = 0$ as Lemma 11.3 predicts).

11.4. **Above the midpoint: the exceptional steps are apparent.** For $n > (p+1)/2$ we use forward induction along L_{BZ}. To produce level $m \le p-1$ we use the step $\nu = m-3 \ge (p-3)/2$; on that range $2\nu+5 \in [p+2, 2p-3]$ is odd and $\neq p$, and $\nu+3 \le p-1$. *Hence the only possible exceptional steps above the midpoint are roots of a_0 modulo p.*

Lemma 11.10 (Apparency). *[PROVED] by exact polynomial division. Put $U(\nu) := (c_0, c_1, c_2)(\nu)$ and $V(\nu) := (c_1, c_2, c_3)(\nu-1)$, both acting on $(Y_\nu, Y_{\nu+1}, Y_{\nu+2})$. All three 2×2 minors of $[U(\nu); V(\nu)]$ are divisible by $a_0(\nu)$ in $\mathbb{Q}[\nu]$ (each minor has degree 18; the remainders on division by a_0 are exactly 0). Hence if $p \mid a_0(\nu)$, $\nu \ge 1$, $V(\nu) \not\equiv 0 \pmod p$, and Y solves L_{BZ} with $Y_{\nu-1}, \ldots, Y_{\nu+2} \in \mathbb{Z}_p$, then $U(\nu)\cdot(Y_\nu, Y_{\nu+1}, Y_{\nu+2}) \equiv 0 \pmod{p^{v_p(a_0(\nu))}}$, so $Y_{\nu+3} \in \mathbb{Z}_p$.*

Proof. Because $c_0(\nu-1) = \nu^5(\nu+1)a_0(\nu)$, the row at $\nu-1$ degenerates in the same place as the row at ν; it gives $V(\nu)\cdot(Y_\nu, Y_{\nu+1}, Y_{\nu+2}) = -\nu^5(\nu+1)a_0(\nu)Y_{\nu-1} \equiv 0 \pmod{p^v}$ with $v := v_p(a_0(\nu))$. Each minor is $a_0(\nu)\cdot(\text{integer})$, so $\equiv 0 \pmod{p^v}$. Pick i with V_i a p-unit and set $\lambda := U_i/V_i \in \mathbb{Z}_p$; then $U_j - \lambda V_j = (U_jV_i - U_iV_j)/V_i \equiv 0 \pmod{p^v}$, so $U\cdot Y \equiv \lambda(V\cdot Y) \equiv 0 \pmod{p^v}$. On this range $v_p(c_3(\nu)) = v$. ☐

Lemma 11.10 is the shadow of an explicit desingularisation. a_0 is irreducible over $\mathbb{Q}$, so $K := \mathbb{Q}[\nu]/(a_0)$ is a field and $U = \lambda V$ in K with $\lambda = c_0(\nu)/c_1(\nu-1)$ of degree ≤ 2:

$$\lambda(\nu) = \frac{392627556035671426586\,\nu^2 + 1282015597875460006266\,\nu + 1052781309790247665282}{D},$$

$$D = 3641620092914355321 = 3\cdot 7\cdot 11\cdot 29\cdot p_0, \qquad p_0 := 543606522303979.$$

Every coefficient of $\mathrm{row}(\nu) - \lambda\,\mathrm{row}(\nu-1)$ — an order-4 relation on $Y_{\nu-1}, \ldots, Y_{\nu+3}$ — is divisible by $a_0(\nu)$ in $\mathbb{Q}[\nu]$; dividing through and clearing D leaves:

Proposition 11.11 (Desingularisation). *[PROVED], [VERIFIED] There is an order-4 left multiple $\tilde{L}:\ d_0Y_{\nu-1} + d_1Y_\nu + d_2Y_{\nu+1} + d_3Y_{\nu+2} + d_4Y_{\nu+3} = 0$ of L_{BZ} with $d_i \in \mathbb{Z}[\nu]$, $\deg d_0, \ldots, d_3 = 8$ and*

$$d_4 = 2D(\nu+3)^5(2\nu+5) = 7283240185828710642\,(\nu+3)^5(2\nu+5):$$

a_0 is gone from the leading coefficient. ($\tilde{L}$ annihilates Q, P and $\hat{P}$ exactly over $\mathbb{Q}$ for $\nu = 1, \ldots, 40$, 0 residuals. Note d_1 is not identically zero: the Y_ν coefficient vanishes only modulo a_0, and $d_1 = (c_0(\nu) - \lambda c_1(\nu-1))/a_0(\nu)$ is a genuine degree-8 polynomial.)

Consequently, *on the desingularised ladder the only singular step in $0 \le n \le p-4$ is the midpoint $n_0 = (p-5)/2$* (the alternative $\nu+3 \equiv 0$ gives $\nu = p-3$, out of range), for every $p \notin \{2, 3, 7, 11, 29, p_0\}$. The recurrence half and the region half of the proof dovetail with no residue: $\tilde{L}$'s single singular step is precisely the single level that Theorem 11.7 supplies.

Lemma 11.12 (The fully degenerate steps). *[PROVED] (explicit finite set). Let $p \ge 5$ and let ν be a root of a_0 modulo p with $1 \le \nu \le p-4$. Then $V(\nu) \equiv 0$ forces p to divide all three resultants $\mathrm{Res}_\nu(a_0, c_i(\nu-1))$, whose gcd is $2^6\cdot 3^3\cdot 7^3\cdot 11\cdot 29^3\cdot 37^2\cdot 557^3\cdot p_0$, and a root-by-root check over every prime in that gcd gives exactly three occurrences:*

$$(p,\nu) = (7,2), \qquad (11,6), \qquad (p_0,\, 416574044722681).$$

In all three cases $\mathrm{row}(\nu-1)/p$ is a valid $\mathbb{Z}$-functional with a unit $Y_{\nu+2}$ coefficient; combined with the regular row at $\nu-2$ it gives two independent functionals, and the ranks over $\mathbb{F}_p$ are: $(11,6)$ and $(p_0,\cdot)$ give $\mathrm{rank} = 2$ both with and without $U(\nu)$ (apparent); $(7,2)$ gives 2 and 3, so $p = 7$ is a finite check (levels $5, 6$; [VERIFIED]).

Remark 11.13. The range hypothesis $\nu \le p-4$ is not cosmetic. Dropping it admits exactly one further pair, $(p,\nu) = (29,27)$: there $a_0(27) \equiv 0$ and $c_1(26) \equiv c_2(26) \equiv c_3(26) \equiv 0 \pmod{29}$, the last because $\nu + 2 = 29$. It is harmless for us — the forward induction of §11.5 only ever uses steps $\nu \le p-4$, and at $p = 29$ the above-midpoint range is $[(p-3)/2,\, p-4] = [13, 25]$ — but it is a genuine fourth solution of the unrestricted system. (At $p = 2, 3$ the pair $(p,1)$ is likewise fully degenerate; both are excluded by $p \ge 5$.)

Two of the three listed cases arise from consecutive roots of a_0, and there are exactly two such primes, not one: $\mathrm{Res}_\nu\big(a_0(\nu), a_0(\nu-1)\big) = -2^2 \cdot 11 \cdot 37^2 \cdot 557^3 \cdot p_0$, and of its prime divisors only $p = 11$ (roots $5, 6, 9$; the pair $(5,6)$) and $p = p_0$ (roots 416574044722680, 416574044722681, 423023739121937) actually carry a consecutive pair — a_0 has no root at all modulo 37, a single root 49 modulo 557, and a single root 1 modulo 2. The case $(11,6)$ is thus produced by the same mechanism as $(p_0,\cdot)$, while $(7,2)$ and the out-of-range $(29,27)$ are not.

Evidence 11.14. [VERIFIED] Sweep over all 428 primes $5 \le p < 3000$: 296 267 steps above the midpoint, 223 exceptional, *every one* an a_0-root, 0 exceptions; 2 fully degenerate, both predicted by the resultant computation.

11.5. **Assembly, and a corollary that was expected to be an input.**

Proof of Theorem 1.5. For $n \le (p-1)/2$, Lemma 11.3. For $n = (p+1)/2$, Theorem 11.7. For $n > (p+1)/2$, forward induction from the four consecutive integral levels below: each step $\nu \ge (p-3)/2$ has $p \nmid (\nu+3)(2\nu+5)$, so is exceptional only if $p \mid a_0(\nu)$, and then it is apparent by Lemma 11.10 (Lemma 11.12 for the three fully degenerate cases; $p = 7$ by the finite check). ☐

Proposition 11.15 (The universal midpoint row)**.** *[PROVED] $n_0 := (p-5)/2$ is the unique n with $2n+5 = p$, so $n_0 \equiv -5/2 \pmod p$, and since $\deg c_i = 9$,*

$$2^9 c_i(n_0) \equiv 2^9 c_i(-5/2) = 28\, R_i \pmod p, \qquad (R_0, R_1, R_2, R_3) = (11907, -334374, -19292, 0);$$

equivalently, as an identity in $\mathbb{Z}[m]$, $128\, c_i(m) = 7R_i + (2m+5)H_i(m)$. The content $28 = 2^2 \cdot 7$ is a p-unit for every $p \ge 5$ except $p = 7$, where the whole row vanishes mod p and the recurrence obligation at n_0 is vacuous. The row (R_0, R_1, R_2) is primitive ($11907 = 3^5 \cdot 7^2$, $334374 = 2 \cdot 3 \cdot 23 \cdot 2423$, $19292 = 2^2 \cdot 7 \cdot 13 \cdot 53$).

Corollary 11.16 ((REC-$\star$))**.** *[PROVED] For every prime $p \ge 5$, with $n_0 = (p-5)/2$,*

$$11907\, P_{n_0} - 334374\, P_{n_0+1} - 19292\, P_{n_0+2} \;\equiv\; 0 \pmod p.$$

Proof. $c_0(n_0)P_{n_0} + c_1(n_0)P_{n_0+1} + c_2(n_0)P_{n_0+2} = -c_3(n_0)P_{(p+1)/2}$ has $v_p \ge v_p(c_3(n_0)) \ge 1$ by Theorem 1.5; multiply by $2^9/28$ and use Proposition 11.15. At $p = 7$ use the finite check. ☐

The order of implication is inverted relative to expectation: (REC-$\star$) was the anticipated *input* to the base case; it is its *output*. And its universal normalised row is *identical* to the row governing the $a = 1$ midpoint band one digit up in the earlier campaign of this program [P-M] — the base level and its $a = 1$ lift are governed by the same universal row.

Evidence 11.17. [VERIFIED] 0 failures, exact `Fraction` ladders, all 44 primes $5 \le p \le 199$: $v_p\big(11907P_{n_0} - 334374P_{n_0+1} - 19292P_{n_0+2}\big) = 1$ *exactly* at every one, so the congruence is tight everywhere. The corresponding quantity for Q is 1 (except $p = 29, 131$ where it is 2) — automatic, since $Q \in \mathbb{Z}$ solves L_{BZ} — and for $\hat{P}$ it is 0 at 42 of the 44 primes, which is the control showing the congruence is real content. A previously reported anomaly at $p = 13$ was an artefact of the *un-normalised* row: writing $c_i(n_0) = uR_i + pe_i$ with $u = 28 \cdot 2^{-9}$, the raw combination is $u\varphi_R(P) + p\sum e_i P_{n_0+i}$, whose valuation is ≥ 1 always and jumps to 2 exactly when a second-order term vanishes mod p, as it happens to at $p = 13$. In the normalised row $p = 13$ is tight like every other prime.

Remark 11.18 (two harmless primes)**.** $v_p(c_3(n_0)) = 1 + v_p(a_0(n_0))$ (as $n_0 + 3 = (p+1)/2 \in [1, p-1]$ and 2 are units), and $8a_0(-5/2) = -241144 = -2^3 \cdot 43 \cdot 701$, so $p \mid a_0(n_0)$ exactly for $p \in \{43, 701\}$, both with multiplicity 1. There the recurrence route would need $v_p \geq 2$; the proof above never uses the midpoint recurrence step, so $\{43, 701\}$ cost nothing.

12. Two structural theorems

This section is not needed for Theorem 1.3; it explains why the proof has the shape it has, and we believe the statements are of independent interest.

12.1. **The base case is one identity.** Let $n < p$. At $L = 0$ one has $v_pT = \alpha + \gamma + \kappa$ *exactly* (by Kummer: $v_p\binom{n+k}{n} = \alpha$, $v_p\binom{n}{l} = v_p\binom{n}{k} = 0$, $v_p\binom{n+k+l}{n} = \kappa$, all arguments being $< 2p$ and $n, k, l < p$). Write $v_5 = \sum_j K_j w^j$ as in (7.1) and $S_j := \sum_{k,l} T(n,k,l)K_j(n,k,l)$.

Proposition 12.1. *[PROVED] For $w_5 = w_5^{\mathrm{allp}}$, $K_2 = 0$ identically on every pattern with $s \leq 1$ — strictly stronger than the cap $J = 1 + \min(s, 2)$, which permits $K_2 \neq 0$ at $s = 1$. Hence at $L = 0$ every cell with $K_2 \neq 0$ has $v_pT = s \geq 2$, so $(T/p)K_2 \equiv 0 \pmod{p}$ cell by cell.*

Proposition 12.2. *[PROVED] For $p \geq 5$ and $n < p$, the base case is* equivalent *to the single congruence*

$$(\mathrm{V3})_0 \qquad \sum_{k,l=0}^{n} \frac{T(n,k,l)}{p^2} K_3(n,k,l) \;\equiv\; 0 \pmod{p},$$

a full *double sum: $K_3 = 0$ off the five patterns with $s \geq 2$, and on those $v_pT = s \geq 2$, so every summand is p-integral.*

Proof. $v_p(W_n) \geq 0 \iff v_p(P_n) \geq 0$ ($n < p$ makes $H_n^{(5)}Q_n$ integral). $W_n = \sum_j p^{-j}S_j$ with $S_0 \in \mathbb{Z}_p$; $K_1 \neq 0 \Rightarrow s \geq 1 \Rightarrow v_pT \geq 1$, so $v_p(S_1) \geq 1$; $K_2 \neq 0 \Rightarrow s \geq 2$ by Proposition 12.1, so $v_p(S_2) \geq 2$; and $K_4 = K_5 = 0$. Hence $v_p(W_n) \geq 0 \iff v_p(S_3) \geq 3$. $\square$

With $q = p - n$, $\tilde{n} = 2n - p$, the $s = 2$ locus is exactly

$$\mathrm{I}: \ (\alpha, \gamma, \kappa) = (0, 1, 1), \ k < q, \ l \geq q; \qquad \mathrm{II}: \ (1, 0, 1) \text{ (the mirror)};$$
$$\mathrm{III}: \ (1, 1, 0), \ k, l \geq q, \ p \leq k + l < p + q,$$

and $\Sigma_{\mathrm{I}} = \Sigma_{\mathrm{II}}$, so $(\mathrm{V3})_0 \iff 2\Sigma_{\mathrm{I}} + \Sigma_{\mathrm{III}} \equiv 0$. Theorem 11.1 improves this to one region.

Evidence 12.3. [VERIFIED] 0 failures. The residue density $R(k,l) := pTv_5 \bmod p$ is supported *exactly* on $\mathrm{I} \cup \mathrm{II} \cup \mathrm{III}$, is $k \leftrightarrow l$ symmetric, and $\sum_{k,l} R = 0$ for all 66 levels $n < p$, $p \in \{5, 7, 11, 13, 17, 19\}$; row sums are *not* 0. Graded checks at $L = 0$ (reconstruction of v_5 from the K_j, integrality of each K_j, the caps, and $v_p(S_j) \geq j$): 1 516 cells, $p \leq 13$, 0 failures, with $v_p(S_3) = 3$ attained at most n. At $p = 199$ the identity holds at all 99 levels $n = 100, \ldots, 198$ and at all 15 levels tested at $p = 31$ — by far the strongest single verification, in exact $\mathbb{F}_p$ arithmetic. Graded $L = 1$ checks at $p = 5$, $n \leq 14$: 0 failures.

Remark 12.4 (it is irreducibly a congruence)**.** $(\mathrm{V3})_0$ can be completely de-p-adicised. Using Wilson's theorem, the reflections $H_{p-1-j} \equiv H_j$ and $H_{p-1-j}^{(2)} \equiv -H_j^{(2)}$ ($p \geq 5$), and the carry binomial $\binom{p+m}{n}/p \equiv (-1)^{n-m+1}/(n\binom{n-1}{m})$ for $0 \leq m < n < p$, every ingredient becomes an explicit rational function of (n, q) alone, and $(\mathrm{V3})_0$ becomes $F(n, p-n) \equiv 0 \pmod{p}$ for an explicit finite double sum F in which p does not occur. [VERIFIED] for all 24 pairs (n, q) with $2 \leq n \leq 12$, $q = p - n$, $p = n + q$ prime ≥ 5. *But $F(n, q) \neq 0$ as a rational number* for every tested pair: the reduction consumes Wilson and the reflections, so $(\mathrm{V3})_0$ is genuinely a mod-p congruence, not the specialisation of a rational identity. In particular creative telescoping has nothing to telescope here.

12.2. **The wall.**

Theorem 12.5 (The θ wall)**.** *[PROVED] Let $\mathcal{D} \subset \mathbb{Q}^{448}$ be the space of weight-5 forms satisfying the* (DEPTH) *conditions (*$\dim \mathcal{D} = 406$*), and for $w \in \mathcal{D}$ put $V_w(n) := \sum_{k,l} T(n,k,l)w(n,k,l)$.* (DEPTH) *gives $v_p(V_w(n)) \geq -1$ for $n < p$, and the entire content of the cancellation is the linear functional*

$$\theta_{p,n} : \mathcal{D} \to \mathbb{F}_p, \qquad \theta_{p,n}(w) := \big(p\, V_w(n)\big) \bmod p.$$

Then $\theta_{p,n}(w)$ depends on w only through the single rational number $V_w(n)$. Consequently:

1. *every exact identity $\sum_{k,l} T \cdot M = 0$ lies in $\ker V$ and therefore satisfies $\theta_{p,n}(M) = 0$ trivially; adding any such M to w_5 changes neither V nor θ;*
2. $(\mathrm{V3})_0$ *for w_5 is* literally *the assertion $v_p(P_n) \geq 0$ and carries no information beyond it.*

Proof. Immediate from the definition: V is linear and $\sum T\,M = 0$ means $M \in \ker V$. ∎

Proposition 12.6. *[VERIFIED] $\theta \not\equiv 0$ on $\mathcal{D}$: three independent random $w \in \mathcal{D}$ (each with ≈ 420 nonzero coefficients, drawn by rref-completion of the $\mathbb{Q}$ condition system) give* $\min_{n<p} v_p(\sum_{k,l} T(n,k,l)w(n,k,l)) = -1$ *at $p = 5, 7, 11, 13$ — all 12 (trial, prime) pairs — with cell-wise minimum also -1.*

Corollary 12.7 (the wall)**.** *The* (DEPTH) *calculus* plus arbitrarily much summand-side identity work *cannot prove the base case. The property "$v_p(V_w(n)) \geq 0$ for all $p \geq 5$, $n < p$" is a* p-dependent *linear condition on w for each (p, n), is false at a generic point of $\mathcal{D}$, and can only be enforced by conditions that pin the value sequence V_w — i.e. by the fitting system itself.*

This is the structural explanation of Theorem 7.16: those systems were trying to buy, with p-independent linear conditions, something that is not p-independent-linear. It also explains why the proof of §11 had to import a property of the *number* P_n — which it does twice, through the decomposition identity and through L_{BZ}.

Remark 12.8 (a reflection route, refuted)**.** One might hope to pair the deficit cells by an involution. This fails concretely. The attained cell $\big(\frac{p+1}{2}, 0, \frac{p-1}{2}\big)$ is the *corner* of region I, not a reflection centre. In coordinates $(u, v) = (\lambda, k + \lambda)$ on I and $(u, v) = (\tilde{n} - \lambda', \kappa')$ on III, both regions are width-q bands $0 \leq v - u \leq q - 1$ and $\mathrm{III} = \mathrm{I} \cap \{v \leq \tilde{n}\}$; so the index sets *are* related by $\lambda' \mapsto \tilde{n} - \lambda'$, but the summands are not — the two T/p^2 weights carry $\binom{n}{v-u}^2 \binom{n}{\tilde{n}-u}^2 / \binom{n-1}{u}$ and $-\binom{n}{\tilde{n}-v}^2 \binom{n}{u}^2 / \binom{n-1}{\tilde{n}-u}$, which no substitution matches. Numerically, $|\mathrm{I} \cup \mathrm{II}| \neq |\mathrm{III}|$ (12 vs 3 at $p = 7$, $n = 4$; 16 vs 5 at $p = 7$, $n = 5$) and the residue multisets differ ($\{1, 2, 3, 5, 1, 6\} \times 2$ vs $\{1, 1, 4\}$ at $p = 7$, $n = 4$). And by Theorem 12.5 even a successful pairing would have been a kernel identity and could not have proved the base case.

12.3. **The weight-2 residue identities.** Lemma 8.4 is the weight-1 member of a family. Here is the weight-2 level; it is new, and by Theorem 12.5 it cannot prove the base case — but it is exactly the kind of kernel element by which the w_5 family is translated, and it is reusable.

Lemma 12.9 (Φ_2)**.** *[PROVED] Fix $n \ge 0$ and $0 \le l \le n$; write $\Phi := A_1(k) + 2B_1(k) + C_1$ in the level-n letters. Then*

$$\text{(P1)} \qquad \sum_{k=0}^{n} T(n,k,l)\big[\Phi\, A_1(k) - A_2(k)\big] = 0,$$

$$\text{(P2)} \qquad \sum_{k=0}^{n} T(n,k,l)\big[\Phi\, C_1 - C_2\big] = 0,$$

$$\text{(P3)} \qquad \sum_{k=0}^{n} T(n,k,l)\big[\Phi\, B_1(k) - \tfrac{1}{2}(\Phi^2 - A_2(k) - C_2)\big] = 0.$$

(P1) + 2(P3) + (P2) is the trivial $\Phi^2 - \Phi^2 = 0$, so exactly two of the three are independent; with (P0) : $\sum_k T\Phi = 0$ they span a 3-dimensional space of exact k-row identities. The $k \leftrightarrow l$ mirrors hold by symmetry of T.

Proof. Let $G(x) := \prod_{i=1}^{n}(x+i)\prod_{i=1}^{n}(x+l+i)\big/\prod_{j=0}^{n}(x-j)^2$, so $\deg\mathrm{den} - \deg\mathrm{num} = 2$ and $\mathrm{Res}_{x=k}\, G = (T(n,k,l)/K_l)\Phi(k,l)$ with K_l independent of k (Lemma 8.4).

(P1). For $1 \le i \le n$ put $G_i := G(x)/(x+i)$. Since $(x+i)$ divides the numerator of G, G_i has no pole at $x = -i$; its only poles are the double poles at $x = 0, \ldots, n$, and $\deg\mathrm{den}-\deg\mathrm{num} = 3 \ge 2$, so all residues sum to 0. Writing $(x-k)^2 G = N/E$ with $(N/E)(k) = T(n,k,l)/K_l$ and $(N/E)'(k) = (T/K_l)\Phi(k,l)$, we get $\mathrm{Res}_{x=k}\, G_i = (T/K_l)[\Phi/(k+i) - (k+i)^{-2}]$. Sum over k, then over $i = 1, \ldots, n$, using $\sum_i (k+i)^{-1} = A_1(k)$ and $\sum_i (k+i)^{-2} = A_2(k)$.

(P2). Identical with $G(x)/(x+l+i)$ (again a numerator factor), using $\sum_i (k+l+i)^{-1} = C_1$ and $\sum_i (k+l+i)^{-2} = C_2$.

(P3). Take $G_j := G(x)/(x-j)$, $0 \le j \le n$; now $x = j$ is a *triple* pole and $\deg\mathrm{den}-\deg\mathrm{num} = 3$. For $k \ne j$, $\mathrm{Res}_{x=k}\, G_j = (T_k/K_l)[\Phi(k)/(k-j) - (k-j)^{-2}]$. At $x = j$, with $\Lambda := \log(N/E)$, $\mathrm{Res} = \frac{1}{2}(N/E)''(j) = \frac{1}{2}(T_j/K_l)[\Phi(j)^2 + \Lambda''(j)]$ and $\Lambda''(j) = -A_2(j) - C_2 + 2(H_j^{(2)} + H_{n-j}^{(2)})$. Sum over $j = 0, \ldots, n$ and use $\sum_{j\ne k}(k-j)^{-1} = -B_1(k)$, $\sum_{j \ne k}(k-j)^{-2} = H_k^{(2)} + H_{n-k}^{(2)}$. The two occurrences of the *non-alphabet* symbol $H_k^{(2)} + H_{n-k}^{(2)}$ cancel exactly (-1 from the double poles, $+1$ from the triple pole), leaving (P3). ∎

The cancellation of $H_j^{(2)} + H_{n-j}^{(2)}$ in (P3) is why a *closed* weight-2 identity exists inside the $\{A, B, C, N\}$ alphabet at all; the same construction with $G(x)/((x+i)(x+i'))$ produces the weight-3 level. [VERIFIED] exact over $\mathbb{Q}$, 0 failures: all 120 pairs (n,l) with $0 \le l \le n \le 14$, each of (P0)–(P3) exactly 0.

13. The middle row

The weight-3 companion is proved by the same machinery, and it is instructive because it *fails* in a way the top row does not. Throughout, $n = ap + r$ with $p \ge 5$, $1 \le a < p$, $0 \le r < p$, and

$$v(n,k,l) := \hat{w}_3(n,k,l) - H_n^{(3)}, \qquad \hat{W}_n = \hat{P}_n - H_n^{(3)}Q_n = \sum_{k,l} T(n,k,l)\, v(n,k,l)$$

by Theorem 6.1; the constant letter is exactly the H-layer removed by Theorem 4.2, and v has no constant letter. Level-a letters are written $A_m^{(a)}(b) = H_{a+b}^{(m)} - H_b^{(m)}$, $B_m^{(a)}(b) = H_{a-b}^{(m)} - H_b^{(m)}$, $C_m^{(a)}(b,c) = H_{a+b+c}^{(m)} - H_{b+c}^{(m)}$.

Lemma 13.1 (Letter descent)**.** *[PROVED] In-regime (Definition 9.4) one has $\lfloor (n+k)/p\rfloor = a+b$, $\lfloor (n-k)/p\rfloor = a-b$, $\lfloor (k+l)/p\rfloor = b+c$, $\lfloor (n+k+l)/p\rfloor = a+b+c$, and hence by Lemma 4.1 applied twice*

$$p^m A_m(k) = A_m^{(a)}(b) + p^m\mathbb{Z}_p, \qquad p^m B_m(k) = B_m^{(a)}(b) + p^m\mathbb{Z}_p, \qquad p^m C_m = C_m^{(a)}(b,c) + p^m\mathbb{Z}_p.$$

Consequently, for every in-regime (k,l), $p^3\, T(n,k,l)\, v(n,k,l) \equiv T(n,k,l)\, v(a,b,c) \pmod p$. Off-regime the same congruence holds, using Lemma 5.3.

Proof. For the displayed identities, e.g. $A_m(k) = [U_m(n+k) - U_m(k)] + p^{-m}[H_{a+b}^{(m)} - H_b^{(m)}]$ by Lemma 4.1, and the bracket is $A_m^{(a)}(b)$; multiply by p^m. (For B one needs $s \le r$ to get $\lfloor (n-k)/p\rfloor = a-b$; for C one needs $s+t<p$ and $r+s+t<p$.) Now let $M = L_1L_2$ (or L_1) be one of the five monomial types of v, of total weight 3; then $p^3M = \prod(L_i^{(a)} + p^{m_i}z_i)$ with $z_i \in \mathbb{Z}_p$, so $p^3M - M^{(a)} = \sum_{\emptyset\ne S}\prod_{i\in S} p^{m_i}z_i \prod_{i\notin S} L_i^{(a)}$. The level-$a$ letters obey $B_m^{(a)} \in \mathbb{Z}_p$ (both arguments $< p$), $C_m^{(a)} \in p^{-m}\mathbb{Z}_p$, and $A_m^{(a)}(b) \in \mathbb{Z}_p$ unless $a+b \ge p$, in which case $A_m^{(a)}(b) \in p^{-m}\mathbb{Z}_p$. Checking the five types: for $A_3(k)$ the error is p^3z; for $A_2(k)X_1$ with $X_1 \in \{A_1(k), B_1(k), C_1, A_1(l)\}$ the errors are $p^2z_1X_1^{(a)}$ (and $X_1^{(a)} \in p^{-1}\mathbb{Z}_p$, so this is in $p\mathbb{Z}_p$), $p\,z_2A_2^{(a)}(b)$, and $p^3z_1z_2$. The only dangerous term is $p\,z_2A_2^{(a)}(b)$, which lies in $p^{-1}\mathbb{Z}_p$ *only if* $a+b\ge p$ — and then Lemma 5.1 gives $v_p(T(n,k,l)) \ge 2$, so after multiplying by T it lies in $p\mathbb{Z}_p$. Off-regime the digit identities acquire a $+1$ and the mismatch of $p^mA_m(k)$ against $A_m^{(a)}(b)$ is $\varepsilon_1(a+b+1)^{-m}$ with $\varepsilon_1 = [\![r+s\ge p]\!]$, a p-unit except when $a+b+1=p$; there Lemma 5.3 supplies $v_p(T)\ge 4$, which the A_3 monomial needs, and ≥ 3 suffices for the A_2X_1 monomials. ☐

Proof of Theorem 1.7. By Theorem 4.2, $p^3\hat{P}_n - \hat{P}_aQ_r \equiv p^3\hat{W}_n - \hat{W}_aQ_r \pmod p$. By Theorem 6.1 and Lemma 13.1,

$$p^3\hat{W}_n = \sum_{k,l} p^3T(n,k,l)v(n,k,l) \equiv \sum_{k,l} T(n,k,l)v(a,b,c) = \sum_{b,c} v(a,b,c)\,\mathcal{T}(b,c) \pmod p.$$

Put $d(b,c) := \max(0, -v_p(v(a,b,c))) \in \{0,1,2,3\}$. Every monomial of v contains an A_2 or an A_3 letter, so $d(b,c)\ge 2$ forces $a+b\ge p$ or $a+c\ge p$; and then $v_p(T(a,b,c))\ge 2$, whence $d(b,c) \le 1+\min(v_pT(a,b,c),2)$. Therefore Theorem 8.8 applies and $v(a,b,c)\big[\mathcal{T}(b,c) - (Q_n/Q_a)T(a,b,c)\big]$ has $v_p \ge -d(b,c)+1+d(b,c) = 1$, so

$$p^3\hat{W}_n \equiv \frac{Q_n}{Q_a}\sum_{b,c} v(a,b,c)T(a,b,c) = \frac{Q_n}{Q_a}\hat{W}_a \pmod p.$$

Hence $p^3\hat{W}_n - \hat{W}_aQ_r \equiv \lambda\hat{W}_a$ with $\lambda := Q_n/Q_a - Q_r$ and $v_p(\lambda)\ge 1$ by Theorem 1.4. Finally $\hat{W}_a = \hat{P}_a - H_a^{(3)}Q_a$ with $H_a^{(3)} \in \mathbb{Z}_p$ (as $a<p$) and $Q_a \in \mathbb{Z}$, so $v_p(\hat{W}_a) \ge \min(0, v_p(\hat{P}_a))$ and $v_p(\lambda\hat{W}_a) \ge 1 + \min(0, v_p(\hat{P}_a))$. ☐

Corollary 13.2. *[PROVED] — no Lemma 8.8, no off-regime descent, no integrality hypothesis. For $1\le a<p/3$, every level-a letter is p-integral ($a+b\le 2a<p$ kills the A-poles, $a+b+c\le 3a<p$ kills the C-poles, B never has one), hence $v(a,b,c)\in\mathbb{Z}_p$, $d(b,c)=0$, and Lemma 8.8 degenerates to Lemma 3.5. Therefore*

$$p^3\hat{P}_{ap+r} \equiv \hat{P}_a\,Q_r \pmod p \qquad \textit{for all } 1\le a<p/3,\ 0\le r<p,$$

and $\hat{P}_a \in \mathbb{Z}_p$ automatically in that range.

13.1. **Why the product form fails, exactly.**

Proposition 13.3 (Size of the middle row)**.** *[PROVED], conditional on Theorem 6.1. For $n < p^2$ and $p \geq 5$, $v_p(\hat{P}_n) \geq -4$; and for $n < p$, $v_p(\hat{P}_n) \geq -1$.*

Proof sketch. Inspect $\hat{w}_3$'s letters. $H_n^{(3)} = (\mathbb{Z}_p) + p^{-3} H_a^{(3)}$ with $H_a^{(3)} \in \mathbb{Z}_p$: pole order 3. $A_3(k) = (\mathbb{Z}_p) + p^{-3}(H_\beta^{(3)} - H_b^{(3)})$ with $\beta = \lfloor (n+k)/p \rfloor$ and $b = \lfloor k/p \rfloor < p$; since $\beta \leq 2a+1 < 2p$, $H_\beta^{(3)} = (\mathbb{Z}_p) + p^{-3}[\![\beta \geq p]\!]$, so $A_3(k) = p^{-6}[\![\beta \geq p]\!] + p^{-3}(\mathbb{Z}_p) + (\mathbb{Z}_p)$ — only the indicator term reaches depth 6, and by Lemma 5.2 every (k,l) contributing to it has $v_p T \geq 2$: pole order ≤ 4 after weighting. For $A_2(k)X_1$ the deepest term is $A_2(k)C_1$, which reaches p^{-4} already with $\beta < p$, whenever $\lfloor (n+k+l)/p \rfloor \geq p > \lfloor (k+l)/p \rfloor$; that case also gives -4. Taking the minimum gives -4. For $n = a < p$ the only offending letter is $A_3(k)$ with $a + k \geq p$ (no p-divisible $m \leq a$ exists, so $H_a^{(3)}$, B_r and C_r are all p-integral), giving $-3 + 2 = -1$ by Lemma 5.1. □

Proposition 13.3 explains the anomaly: the naïve product congruence $p^3 \hat{P}_{ap+r} \equiv \hat{P}_a Q_r \pmod{p}$ is *false*, and it fails precisely by the term $\lambda \hat{W}_a$ of the proof above — λ supplies one power of p, and $\hat{W}_a$ eats it back when $v_p(\hat{P}_a) = -1$. Equivalently, the middle Frobenius slot needs p^4, not p^3, if one insists on an unnormalised product congruence.

The top row has no such defect: [VERIFIED], 0 exceptions for $p \in \{5, 7, \ldots, 31\}$, $v_p(P_a) \geq 0$ *and* $v_p(W_a) \geq 0$ for every $1 \leq a < p$, whereas $v_p(\hat{P}_a) = -1$ for most $a \in (p/2, p)$. That is why (LB_5) and (W5) are clean and need no integrality hypothesis.

Evidence 13.4. [VERIFIED] 0 failures, all $1 \leq a < p$, $0 \leq r < p$, $n = ap + r \leq 360$, $p \in \{5, 7, 11, 13, 17, 19\}$: $v_p(p^3 \hat{P}_n - \hat{P}_a Q_r) \geq 1 + \min(0, v_p(\hat{P}_a))$ with both floors attained — the minimum is 1 on the cells with $v_p(\hat{P}_a) = 0$ and 0 on those with $v_p(\hat{P}_a) = -1$. The plain product form fails on 3/5/24/55/84/128 of the 5/7/33/65/119/152 cells with $v_p(\hat{P}_a) = -1$ at those primes and on *none* of the cells with $\hat{P}_a \in \mathbb{Z}_p$. Also [VERIFIED]: $v_p(\hat{P}_a) \geq 0 \Rightarrow v_p(\hat{P}_{ap+r}) \geq -3$ for every r, and every cell with $v_p(\hat{P}_n) = -4$ has $v_p(\hat{P}_a) = -1$; and $\min_{n<p^2} v_p(\hat{P}_n) = -4$ for $p \in \{5, \ldots, 23\}$, $\min_{a<p} v_p(\hat{P}_a) = -1$ for every $p \in \{5, \ldots, 31\}$ — both bounds of Proposition 13.3 attained. The *master* form $p^3 \hat{P}_n Q_a \equiv \hat{P}_a Q_n \pmod{p}$ holds on all cells; it does not imply the product form when $p \mid Q_a$ or $\hat{P}_a \notin \mathbb{Z}_p$.

14. Certification status of the decomposition identities

This section states, as of the time of writing, exactly what is and is not certified. It is written so that a single sentence flips when a certificate lands.

14.1. **Status.** The table records the state at the date of this preprint. Both [VERIFIED] entries are the subject of ongoing machine computation; if either is upgraded, the only change to this paper is the corresponding row and the sentence in Remark 14.6.

object	class	basis
$Q_n = \sum_{k,l} T(n,k,l)$	[CERTIFIED]	two-step CT; telescoper $= L_{\mathrm{BZ}}$ exactly; re-verified
$L_{\mathrm{BZ}} \cdot T = \Delta_k(\rho T) + \Delta_l(\sigma T)$	[CERTIFIED]	explicit ρ, σ; checked to 0 twice
(DEPTH) linear system	[CERT]	rank(joint) = rank(aug) = 324, two primes
w_5^{I} cell-wise conditions	[CERT]	three primes, identical pivot sets
Theorem 6.1 $\hat{P}_n = \sum T \hat{w}_3$	[VERIFIED]	see §14.3
(T1-top) $P_n = \sum T w_5$	[VERIFIED]	see §14.2

14.2. **(T1-top).**

Evidence 14.1. [VERIFIED] For $w_5 = w_5^{\mathrm{allp}}$, evaluating $\sum_{k,l} T(n,k,l)w_5(n,k,l)$ directly from the saved representative — a code path different from the linear algebra that produced it — modulo $q = 33554393$ and $q = 33554467$ for $n = 0,\dots,750$:

against the exact ladder P_n, every $n \le 360$	**0 mismatches, both primes**
L_{BZ} residual, $n = 0,\dots,747$	**0 at all 748 values, both primes**
minimal recurrence (search over order ≤ 12, deg ≤ 30)	**(order 3, degree 9), nullity 1**

The fitting system imposed equations only for $n \le 600$, so roughly 147 of these are genuine excess checks; and the minimal recurrence being exactly L_{BZ} pins what a certificate must produce. Separately, exact over $\mathbb{Q}$: $n \le 34$ for w_5^{allp}, $n \le 20$ for w_5^{I}, 0 mismatches, by standalone re-implementations reading only the stored coefficients and the ladder; plus *all* 287 excess equations satisfied modulo two primes at $N = 600$ (that is, 600 equations against rank 313), and, in the earlier run at $N = 1000$, all 687.

Proposition 14.2 (No cheap representative). *[PROVED] The* 448 *basis monomials are linearly independent* as functions *on the cells: the cell-level matrix (rows the* 1331 *cells of* $n = 17,\dots,22$*, columns the* 448 *symmetrised monomials) has rank* 448 *modulo* $q = 33554393$*, hence rank* 448 *over* $\mathbb{Q}$*. Consequently* $\ker(\textit{design}) \cap \{\textit{pointwise zero}\} = 0$*: every one of the* 135 *kernel directions is a genuine* summation *identity* $\sum_{k,l} T \cdot z = 0$*, of exactly the same difficulty as the target.* Certifying one representative does not certify another for free*, and the representative one certifies must be the representative one uses.*

Theorem 14.3 (Structural obstruction at weight 5). *[CERT] negative. The weight-*5 *fitting system* $P_n = \sum_{k,l} T \cdot w$ *is* inconsistent *when the support is restricted to letter monomials of degree* ≤ 3*, and the obstruction is the* fit identity alone *— not the p-integrality conditions — so no pole-cap regime can rescue it. Verified in three alphabets at* $N = 600$ *equations,* $q = 33554393$ *(and re-run at* $q = 33554467$ *for the two depth-*1 *alphabets, identical verdicts):*

alphabet	*degree cap*	*cols*, rank(fit)	*fit alone*
A_r, B_r, C_r, N_r	≤ 2	44, 41	*INCONSISTENT*
	≤ 3	176, 151	*INCONSISTENT*
	≤ 4	338, 269	*INCONSISTENT*
	≤ 5	448, 313	*consistent*
$+R_r(k)$ *(Apéry)*	≤ 3	369, 329	*INCONSISTENT*
$+Y_{ab}, V_{ab}, Z_{ab}$ *(nested)*	≤ 3	488, 327	*INCONSISTENT*

(The third alphabet was run at $q = 33554393$ *only.) The harness reproduces both a known positive (*rank(fit) $= 313$*; the* w_5^{I} *system with* 212 *condition rows,* rank(joint) $= 342$*, consistent) and a known negative. Every negative entry is immune to the "vacuous consistency" caveat: at degree* ≤ 3 *the ranks are* 151/329/327 *against* $N = 600$*, i.e. the systems are overdetermined by* 449/271/273 *independent equations and fail.*

Remark 14.4 (what the label means here). Inconsistency over $\mathbb{F}_q$ does not formally imply inconsistency over $\mathbb{Q}$: reduction can lower both $\mathrm{rank}(A)$ and $\mathrm{rank}([A \mid b])$, and a rational solution can hide q in a denominator. Two auxiliary primes with identical verdicts make this overwhelming evidence, which is what [CERT] means in §1.5; it is not a proof. What would upgrade it to [PROVED] is a finite, exactly checkable object: a rational annihilator y with $y^\top A = 0$ and $y^\top b \neq 0$, verified over $\mathbb{Q}$. We have not produced one. Nothing else in this paper depends on Theorem 14.3; it is an explanation of where the certification cost sits, not a step in any proof.

The consequence for certification is concrete. Weight-lowering by the certified ρ, σ turns the target into a creative telescoping whose cost is governed by the *degree* of the monomials of w, because a squarefree degree-d monomial contributes $2^d - 1$ sub-monomials under the shift. Measured supports of $E(w)/T$: **6** for $\hat{w}_3$ (whose folded form has degree ≤ 2), 208 for w_5^{allp}, 220 for w_5^{I}. Degree histograms of the known representatives (w_5^{I}: $1{:}3, 2{:}19, 3{:}59, 4{:}73, 5{:}53$; w_5^{allp}: $1{:}2, 2{:}14, 3{:}52, 4{:}66, 5{:}44$; $\hat{w}_3$ folded: $1{:}4, 2{:}2$) sit exactly where Theorem 14.3 says they must. *So $\hat{w}_3$'s degree-≤ 2 folded form — the single structural fact that makes the weight-3 certificate route exist — has no weight-5 analogue.* We record this as a mathematical fact about w_5, not a resource shortfall: the cheapest certified route to (T1-top) is a creative telescoping on a ∂-finite module of rank 100 to 220, against rank 3 for Theorem 6.1, and nothing short of a new idea about the shape of w_5 changes this. Adding depth would not have helped even had the degree test passed: a single nested letter is degree 1 but its ∂-module is not rank 2 (shifting it produces $A_a(k), A_b(k), H_k^{(a)}, H_k^{(b)}$), so the 2^d support model becomes roughly 4^d.

14.3. **Theorem B.**

Evidence 14.5. [VERIFIED] exact over $\mathbb{Q}$ for $n \leq 40$ (and independently re-derived and re-checked for $n \leq 30$, and re-implemented directly from the displayed formula (6.2) for $n \leq 16$), 0 discrepancies; 340 excess equations satisfied modulo q in a 93-element basis at $N = 400$ (with a negative control: the same basis against the target P is inconsistent, as it must be); and $\hat{P}_n - \sum_{k,l} T\hat{w}_3 = 0$ exactly for $n = 0, \ldots, 300$ — 301 values.

Theorem 6.1 is a *compute* node, not a structural one. It has been reduced by proof to $\sum_{k,l} E(v) = 0$ for an explicit $E(v)$; the boundary and telescoping steps are proved from the pole structure of ρ, σ (using $\rho(n,0,l) = \sigma(n,k,0) = 0$ and the double zeros of T), the regularised boundary lemma is [VERIFIED] exactly for $n_0 = 1, \ldots, 6$, and $E(v)/T$ is a single hypergeometric term times a rank-3 factor ($E(v)/T = c_0 + \beta A_2(k) + \alpha\Psi$ after the proved relations $\gamma = 3\alpha$, $\delta = \frac{3}{2}\alpha$, $\varepsilon = \frac{1}{2}\alpha$). Splitting $E(v) = \sum_\tau F_\tau$ over the five shift terms before telescoping — a split that is itself [CERTIFIED], symbolically and without loading any package — drops the weights from 132 917 leaves to $\{84, 86, 66, 12471, 2255\}$ at unchanged rank ≤ 3; splitting further by letter reduces every piece to a rational multiple of T, i.e. to rank 1. Since $\hat{P}_n - \sum T\hat{w}_3 = 0$ is [VERIFIED] for $n = 0, \ldots, 300$, any operator of order ≤ 298 finishes the identity by matching initial values.

At the date of this preprint no complete M_τ has been produced. What has changed, and is worth recording because it inverts the earlier diagnosis, is where the cost sits. The pipeline has five stages — annihilator, first telescoping (eliminate l), Gröbner normalisation, ∂-finite product, second telescoping — and after the letter-split *four of the five are clear on every τ and every piece tried.* On the largest piece the annihilator returns in 34 s and the first telescoping in 33 s (the 12489-leaf term; the 13069-leaf unsplit object of which it is one piece had failed with an out-of-memory at 7.8 GB after 85 minutes), the Gröbner step in 1–5 s, and the ∂-finite product — until recently the second wall — in 500 s at no measurable memory, returning 4 generators on the largest cofactor in the problem. The *second* telescoping is now the sole remaining obstacle, and it returns no telescoper by any method: not under any support bound attempted (a full ladder $d = 0, \ldots, 5$ on two pieces), and not under unconstrained search with no ansatz box at all (421 s on the largest piece, 600 s on another). We therefore do not quote a kernel-hour estimate: earlier ones were withdrawn, and we prefer to state the position as measured.

That the residual difficulty is exactly one stage sharpens, but also qualifies, the description of Theorem 6.1 as a compute node. A direct measurement (`guessrec` over order ≤ 12,

$\deg \le 30$) finds that *no* single-letter component sum satisfies an operator in that box, while their combination satisfies the unique minimal one, of order 3 and degree 9 — namely L_{BZ}. So the second telescoping, applied to split pieces, has been searching an empty box: the split that made the first four stages cheap is the same split that puts the target out of range.

A subsequent measurement locates a second obstruction one stage earlier, and this one is structural rather than a matter of resources. The cost model above — support size, hence monomial degree — is only one of two axes: `Annihilator` needs the summand small both in the number of distinct harmonic letters it carries *and* in the size of the rational coefficients multiplying them, and refolding w can address only the first. Carried out, it does: a refolded $\tilde{v}$, equal to v modulo proved kernel identities of the Lemma 12.9 family together with the $k \leftrightarrow l$ symmetry, reduces E from 9 distinct letters to 7. The annihilator nevertheless becomes *more* expensive, not less — it aborts against an 8.5 GB memory cap after 536 s (peak 8.4 GB) with memory rising a straight 3.6 GB per minute, where the 13069-leaf 9-letter object above reached 7.8 GB only after 85 minutes. The reason is visible in E itself. Writing $E(w) = \sum_\tau F_\tau$ with $F_\tau = G_\tau\,(\tau.w - w)$ over the five shifts, the two one-variable shifts have $G_{kk} = -(\rho T)|_{k\to k+1}$ and $G_{ll} = -(\sigma T)|_{l\to l+1}$: the coefficients of $E(w)$ carry the very cofactors ρ, σ of $L_{\mathrm{BZ}} \cdot T = \Delta_k(\rho T) + \Delta_l(\sigma T)$ — the two largest rational functions in the problem — unless $\tau.w - w = 0$ for $\tau = kk$ and $\tau = ll$ both, i.e. unless w is a function of n alone, which the fit for $\hat{w}_3$ excludes. *So no choice of representative makes $E(w)$ cheap.* The obstruction lives in the certificate for Q_n that produced E, not in the weight w, and it is invariant under every refolding of w.

Splitting further cannot help either, so what remains is the one route that never forms E at all: telescope the summand $T\,w$ directly. Its coefficients are polynomial rather than $\Theta(\rho, \sigma)$, it needs no weight-lowering step and hence no boundary lemma for E, and its second telescoping searches the support $\{1, S_n, S_n^2, S_n^3\}$ — the one box in this problem known to contain its answer, since $L_{\mathrm{BZ}}\hat{P}_n = 0$ is [PROVED] and $\sum_{k,l} T\hat{w}_3 = \hat{P}_n$ is [VERIFIED] to $n = 300$. Every side condition such a certificate would need is discharged in advance: $\sum_{k,l} T\tilde{v} = \hat{P}_n$ exactly for $n \le 33$, $\hat{w}_3 - \tilde{v}$ is an explicit combination of proved summation identities, the far-edge boundary terms vanish (T has a double zero at each integer $k > n$ where $\tilde{v}$ has at worst a simple pole, and none in l), and 301 exact initial values are banked. The next move on Theorem 6.1 is therefore a single annihilator on a 91-leaf object, or else an algebraic one; in neither case is it a larger version of the computation already attempted.

Remark 14.6 (what would flip)**.** If a certificate for (T1-top) lands, Theorem 1.3 and Theorems 1.5, 1.6 rest on the single remaining input (DEPTH); they become *unconditional* only when (DEPTH) is in addition promoted from [CERT] to a proof, which is a separate and much smaller task — a rational consistency witness for a finite linear system, of the kind described after Theorem 14.3. If a certificate for Theorem 6.1 lands, Theorem 1.7 becomes unconditional outright, since it does not consume (DEPTH). No other statement in this paper changes; in particular a certificate for Theorem 6.1 changes nothing about Theorem 1.3, which does not use it.

15. The primes 2 and 3

The constant $12 = 2^2 \cdot 3$ is not covered by anything above: $p \ge 5$ is forced by the single use of Wolstenholme inside Lemma 8.1, and again by $\sum_{j<p} j^{-2} \equiv 0$ at the nucleus. The residual statement is:

> **(Sharp-12, $p \in \{2, 3\}$ part).** $\mathrm{ord}_2(P_n) \ge -2 - 5\lfloor \log_2 n \rfloor$ and $\mathrm{ord}_3(P_n) \ge -1 - 5\lfloor \log_3 n \rfloor$, i.e. the *constant* $12 = 2^2 \cdot 3$ and nothing worse.

[VERIFIED] 361/361, $n \leq 360$ (Evidence 1.1). We state its provenance precisely, including a wrinkle we have not resolved.

The mechanism proposed for the factor 2 in the earlier campaign of this program is an index-2 computation on the Betti side of the period matrix, via the bicomplex of [Du18]; its own recorded bottom line is that

> "the claim 'index 2 derived from geometry alone' is **not** yet earned; the honest status is 'index 2 is the unique value consistent with all closed sub-computations, with the sole open input being the integral bicomplex differential at two explicitly identified strata'."

Of the three sub-results there, two are proved and one is not: it is *proved* that the factor 2 is absent from the arrangement combinatorics on the other side (all Smith elementary divisors are 1, under every orientation convention); it is *proved* that only the prime 2 can arise from those incidence matrices; and the factor 2 itself is *not closed*, being localised to the integral weight spectral-sequence differential across the uncoloured generic facets on two explicitly identified strata — i.e. to an integral refinement of the $\mathbb{Q}$-only bicomplex of [Du18] that is not in the literature.

The honest consequence, which we flag rather than paper over, is this: *the second proved sub-result says the geometric route cannot produce a* 3. So the source of the factor 3 in $12 = 2^2 \cdot 3$ is at present unidentified, and (1.6) at $p = 3$ has no derivation at all — only the 361/361 verification. It is possible that the 3 has a different origin entirely (for instance in the normalisation constant $24 = 2^3 \cdot 3$ attached to $\zeta(2) = \pi^2/6$ in Brown and Zudilin's own heuristic), and we regard identifying it as the most concrete open question left by this paper.

The general-parameter conjecture of the earlier campaign, which our totally symmetric statement specialises, is

$$12 \cdot D_\nu(\boldsymbol{a}, n) \cdot I'(\boldsymbol{a} \cdot n) \in \mathbb{Z}\zeta(5) + \mathbb{Z}$$

for all admissible $\boldsymbol{a}$ and $n \geq 1$, where D_ν is [BZ]'s Φ_n-sharpened denominator with the exponents ν_p of [BZ, (nu_p)] extended from $p > \sqrt{m_1 n}$ to every prime [P-C]. Its evidence is 164 audited cells: excess at $p = 2$ is at most $+2$ (attained 46 times), at $p = 3$ at most $+1$ (attained 30 times). One verified cell at $n = 1$ shows excess $+1$ at $p = 7 = m_1$, the boundary prime of the multiset — a rare, orbit-coherent, non-persistent fluctuation which a re-audit at $n = 2$ flips to slack at both partner points. That effect is outside the totally symmetric family treated here, but it is a reminder that the general-parameter statement has a boundary-prime subtlety the symmetric one does not.

16. Summary of the dependency structure

node	class	where
Theorem 1.3 (Sharp-12, $p \geq 5$)	[PROVED] modulo (T1-top) + (DEPTH)	§10
Theorem 1.6 ($-5L-1$, all n)	[PROVED] modulo (T1-top) + (DEPTH)	Cor. 7.15
(IND) step $\mathcal{I}(L-1) \Rightarrow \mathcal{I}(L)$	[PROVED]	Cor. 10.5
fibre term (II)	[PROVED], slack 0	Prop. 9.3
(DEPTH-gen)	[PROVED] + [CERT]	Cor. 7.13
Prop. LIFT	[PROVED]	Prop. 7.12
Lemma F-gen	[PROVED]	Lem. 8.13
Lemma B (general a) $\to$ Lemma W	[PROVED]	Lem. 8.2, 8.1
descent (I), in-regime	[PROVED]	Prop. 9.5
descent (I), off-regime	[PROVED]	Thm 10.3
Lemma DK, Lemma D1	[PROVED]	Lem. 10.2, 10.1
base case $\mathcal{I}(0)$	[PROVED] modulo (T1-top) + (DEPTH)	Thm 1.5
one-band reduction	[PROVED] + [CERT]	Thm 11.1
corner sum $= -14s_2$	[PROVED]	Thm 11.7
apparency of the a_0 steps; $\tilde{L}$	[PROVED]	Lem. 11.10, Prop. 11.11
(T1-top) $P_n = \sum T w_5$	[VERIFIED] – **open**	§14.2
(DEPTH) depth linear system	[CERT] – **open**	Thm 7.8
Theorem 1.4 (Lucas, mod p)	[PROVED], [LEAN]	§3
the mod p^2 supercongruence	[PROVED] (not formalised)	Cor. 8.9
Theorem 4.2 (H-layer)	[PROVED]	§4
Theorem 1.7 (middle row)	[PROVED] modulo Theorem 6.1	§13
Theorem 6.1 $\hat{P}_n = \sum T \hat{w}_3$	[VERIFIED]	§14.3
Theorem 12.5 (θ wall)	[PROVED]	§12
Theorem 14.3 (degree obstruction)	[CERT] negative	§14.2
Sharp-12 at $p \in \{2, 3\}$	[VERIFIED]; 2-part reduced, 3-part open	§15

Two remarks close the summary. First, *the ledger has no slack anywhere*: (DEPTH) is attained, Lemma 8.8 is attained, Lemma 8.13 is attained, Lemma 10.2 is attained, and the induction consumes exactly what they provide, digit by digit. That is not bookkeeping tidiness; it is why the exponent in Theorem 1.3 is 5 and not 6. Second, *the whole chain has exactly two arithmetic inputs that constrain the prime* — Wolstenholme's congruence, used once inside Lemma 8.1, and $\sum_{j<p} j^{-2} \equiv 0 \pmod p$, used once at the nucleus. Both fail at $p = 3$, and both are exactly why the hypothesis of Theorem 1.3 is $p \geq 5$. The chain does use three further classical facts — Lucas's theorem, Kummer's theorem and Wilson's theorem (the last in Lemma 11.5 and in the de-p-adicisation remark of §12) — but all three hold at every prime and so cost nothing.

Appendix A. Reproduction

All numerical statements in this paper are exact. The Python computations use exact rational arithmetic, arbitrary-precision integers, $\mathbb{F}_q$ linear algebra, or tracked-precision p-adic arithmetic; the symbolic computations use exact rational-function arithmetic. Design matrices over $\mathbb{F}_q$ use a 13-bit `float64` split (two `dgemm` calls), exact for all sizes used and cross-checked against a direct implementation. The creative-telescoping work uses the package of [Ko10]; every certificate quoted as [CERTIFIED] was re-verified by exact arithmetic in a session that does not trust the search, at least once in a kernel that never loaded that package.

The principal artefacts, by section:

what it settles	where
consistency of (DEPTH): rank(joint) = rank(aug) = 324, two primes	Thm 7.8
(DEPTH^{+}), (DEPTH^{++}) inconsistent	Thm 7.16
w_5^{allp}: ladder identity, depth sweep, cell-by-cell test	Evidence 7.10
w_5^{I}: ladder identity, one-band integrality	Evidence 11.2
(DEPTH-gen) + Kummer floor, digit levels $L \le 2$	Evidence 7.14
Lemma F-gen at multi-digit a	Evidence 8.15
Lemma B for general a	Evidence 8.3
Lemma DK, its slot-wise ingredients, Theorem 10.3	Evidence 10.7
base case over every prime $5 \le p \le 367$	Evidence A.1
corner identity, apparency sweep, $\tilde{L}$	Evidence 11.9, 11.14
(REC-$\star$) normalised sweep	Evidence 11.17
the θ wall trials	Prop. 12.6
the degree-cap experiment	Thm 14.3
(Sharp-12) sweeps $n \le 360$ and $n \le 3000$	Evidence A.2

Evidence A.1 (the base case, directly). [VERIFIED] 11 884/11 884 cells, 0 failures: $\mathrm{ord}_p(P_n) \ge 0$ for every prime $5 \le p \le 367$ and every $1 \le n < \min(p, 361)$, with minimum value exactly 0.

Evidence A.2 (the main theorem, directly). [VERIFIED] 0 failures:

sweep	range	cells	$\min\big(\mathrm{ord}_p(P_n) + 5L\big)$
exact ladders	$n \le 360,\ p \in \{5, \dots, 31\}$	3 240	0 (at $p = 5,\ n = 1$)
same, as $\mathrm{ord}_p(p_n) \ge \kappa(n) - 5L$	$n \le 360,\ p \le 31$	3 240	—
certified order-3 recurrence, exact $\mathbb{Q}$	$n \le 3000,\ p \in \{5, \dots, 31\}$	27 000	0
recurrence vs. exact ladder	$n \le 360$	361	0 mismatches

where $p_n = \binom{2n}{n} P_n$ and $\kappa(n) = v_p\binom{2n}{n}$, so that $\mathrm{ord}_p(p_n) \ge \kappa(n) - 5L \iff \mathrm{ord}_p(P_n) \ge -5L$. The minimum slack is 0: the bound is attained.

Acknowledgments

The exploratory computations, the search for the weight-5 representatives, and a first assembly of several of the arguments above were carried out with the assistance of AI language models; every statement labelled [PROVED] has a human-readable proof given here, and every numerical claim was re-verified by exact arithmetic in an independent implementation.

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[P-C] Internal report `CONJECTURE.md` of the present programme, 16 July 2026 (unpublished): the general-parameter sharp-12 conjecture and its audit data.

[P-M] Internal report `PHASE2_A1_MIDPOINT_THEOREM.md` of the present programme, 20 July 2026 (unpublished): the complete generic midpoint theorem for the $a = 1$ band.